OUR CHANGING CLIMATE

SOCIAL SCIENCE

Climate Change in the Past and Present

Resource Guide

2025–2026

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Introduction

Can a changing climate transform humanity? Consider this fact: humans lived in an ice age for around 100,000 years at the end of the **Pleistocene**. While humans did migrate around the world, they left relatively little trace of their existence. Then, around 11,700 years ago, following a couple thousand years of thawing, that ice age ended, and the Earth's climate became roughly what we know it to be today. In just the past ten thousand years or so, humans have built great societies; developed new technology, including writing; multiplied exponentially; and even traveled to the moon. Was our changing climate a factor in this remarkable transformation in how humans live? It is difficult to say for sure. Just because two events happen one after another does not mean that the first event caused the second. But we can say this with confidence: humans have dramatically grown in number and capability while living in our climate since the end of the last ice age.

Today, our climate is changing again. Rather than heading back toward another ice age, however, the world is getting warmer. No one knows what life will be like for our descendants if the Earth's climate changes much more than it already has. We may be currently living in a moment that turns out to be a turning point for humanity, perhaps even the end of the brief era of human growth that began less than 12,000 years ago. But unlike our ancestors who had no control over their changing climate, we are now the cause of our changing climate. Our practice of burning fossil fuels is heating up the planet. One of the fundamental questions of our lifetimes is whether or not we will stop burning fossil fuels in time to keep the climate from changing much more than it already has.

This resource guide explores our changing climate from the perspective of the social sciences. The first section of the guide, "Conceptualizing Climate Change in the Past and the Present," introduces some of the tools that scientists and social scientists use to understand and talk about climate change. The next two sections are historical. Section II, "Humans in the **Holocene**," offers a global perspective on human life during the years from the end of the last ice age until the nineteenth century, with an emphasis on how people interacted with the climate. Section III, "The **Anthropocene**," explores how humans have contributed to global warming and how that warming has impacted human societies. The fourth and final section, "Responding to the **Climate Crisis**," explains how people have come to recognize that humans are causing climate change and examines some of the most notable responses to this knowledge.

NOTE TO STUDENTS: Throughout the resource guide, you will notice that some terms have been boldfaced and underlined. These terms are included in the glossary at the end of the resource guide. Also, students should be aware that early dates in history may vary depending on the source. The dates presented in this resource guide are those dates provided by the sources consulted by the author in writing this guide.

Section I

Conceptualizing Climate Change in the Past and the Present

SECTION I INTRODUCTION

Scholars and the public use the term “climate change” to describe a complex process of changes in the natural world. But what precisely do the words “climate change” mean? In Section I of this resource guide, we will establish some key concepts that will serve as the foundation for what “climate change” means in the rest of this guide.

The first set of key concepts comes from the field of **Earth System Science (ESS)**. ESS is a relatively new scientific approach to studying the natural world. Its distinctive approach views the Earth’s land, oceans, and **atmosphere** as a single system. Rather than studying the Earth’s different parts in isolation, ESS looks at the interactions between air, water, land, and living organisms. The concepts that we explore from ESS will provide a framework for understanding climate change as a natural phenomenon. Once that baseline understanding of climate change is established, we can add additional layers that consider the relationship between climate change and human society.

The second set of key concepts describes how scholars create and organize our knowledge of past climates. We will examine the sources that scholars use as clues for reconstructing climate and climate change. Scholars in different academic disciplines use different sources for studying the past, and they ask different questions to guide their investigations. It is important to pay attention to the methods used in a field to understand the strengths and limitations of the knowledge a given field produces. A multidisciplinary approach, which incorporates the findings of more than one academic discipline, offers a well-rounded picture of the history of climate change.

The third, and final, set of key concepts in Section I focuses on the idea of the Anthropocene. Many scholars consider the Anthropocene a new geological era in planetary history in which humans have become the driving force in planetary change. The term “Anthropocene” references the idea that current climatic conditions have been heavily influenced by human actions. It has been theorized by scholars that since around 1950, the world has entered a new era of climate history.

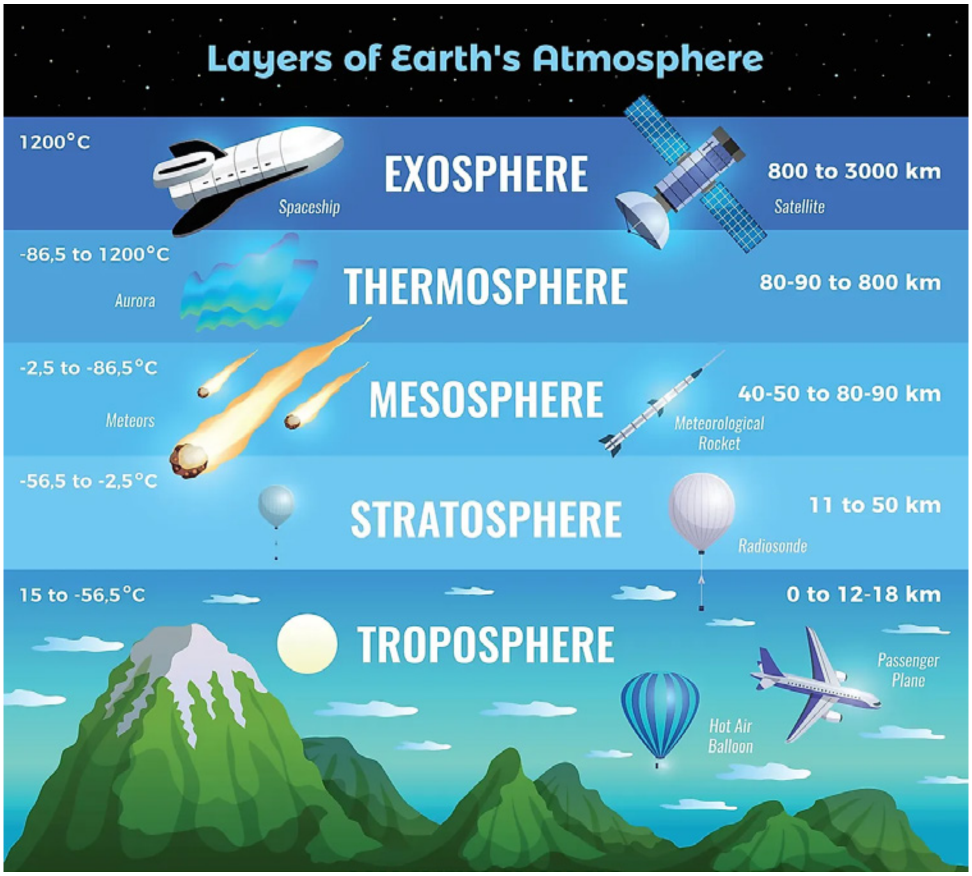
Depiction of the four subsystems (clockwise from top left: biosphere, hydrosphere, atmosphere, geosphere) of Earth System Science.

Source: California State University Northridge, [Earth Systems Interactions (csun.edu)](csun.edu).

Section I concludes with a reflection on how climate change and its history relate to the ways that scholars have traditionally told the story of global history. The content of Section I offers terms and ideas that allow for a robust conceptualization of climate change in the past and present.

ESSENTIAL CONCEPTS FROM EARTH SYSTEM SCIENCE (ESS)

What keeps the Earth’s climate in equilibrium? Why does the Earth’s climate sometimes change so dramatically that scholars designate a period of climate change? The factors that influence the climate



Earth's atmospheric layers.

Source: WorldAtlas

are varied. Moreover, the interactions between those factors complicate the situation exponentially. Different influences mix and mingle together in a myriad of ways, producing complex outcomes that are difficult to predict or even to understand precisely. The relatively young field of Earth System Science puts the complex factors that shape climate and climate change into an orderly system.[1] While ESS offers a useful framework for scholars who study climate, the basics of ESS also provide an easy-to-understand way for non-specialists to obtain an accurate, if simplified, picture of how climate works.

As the name Earth System Science suggests, ESS conceives of the world as a single system. The system has different parts, however, and those are called **subsystems**. On a very basic level, there are four subsystems. The four subsystems are the **geosphere** (earth and rock), **hydrosphere** (water and ice), atmosphere (air), and **biosphere** (living organisms). These subsystems interact to shape the weather and climate. The balance of the subsystems—and the stability of the weather and climate—are sometimes altered from external forces. Those forces are called **forcings**. When forcings alter the climate, it often causes reactions called **positive feedbacks** or **negative feedbacks**. These concepts will be explained in more detail shortly.

The Earth's Subsystems

The Earth's subsystems interact with each other to influence the weather and climate. These interactions can occur at different sized **scales** geographically. For instance, on a very large scale, an entire ocean could warm, causing changes to levels of moisture in the air over a large area. Conversely, on a small scale, a single stream might dry up, changing a local ecosystem. In both examples, the hydrosphere is interacting with the other subsystems in various ways that impact the future relationship between the subsystems in those places.

Geosphere

The geosphere encompasses all the land, earth, and rock that make up the planet. Sometimes, scholars use the term "**Lithosphere**," a word that incorporates the Greek word for rock or stone, to describe the same features encompassed by the term "geosphere." On a **geological time scale**—encompassing the millions and billions of years it has taken for the Earth to move and change form—notable events in climate history include the shifting of the Earth's plates and the release and recapture of minerals and particles from within the Earth. The Earth's crust, a relatively thin layer of rock at the surface of the Earth, is where most of the interactions between the geosphere and other subsystems occur. Living organisms, which are part of the biosphere, influence the composition of the soil itself. The geosphere also impacts weather and climate, as occurs in locations where mountain ranges cause clouds to form. The eruptions of volcanoes that release gases and particles from within the Earth also drive climate change.

Hydrosphere

The hydrosphere is all the water on the Earth, in the ground, and in the atmosphere. It includes oceans and freshwater rivers and lakes as well as clouds and water vapor. The hydrosphere also includes ice, which some scholars count as its own subsystem called the **cryosphere**. Some of the most well-publicized aspects of current climate change focus on the hydrosphere, including the melting of the cryosphere at the planet's North and South Poles, the warming of oceans and sea level rise, and increasingly severe droughts and flooding around the world.

Atmosphere

The atmosphere consists of various gases. On this basis, scholars define atmospheric zones, or layers. They are, from lowest to highest altitude: the troposphere, stratosphere, mesosphere, thermosphere, and exosphere. One important role that the atmosphere plays in climate change involves the **greenhouse gas effect** in which the concentration of certain gases released from the Earth's other subsystems traps heat in the lower layers of the atmosphere. Certain gases in the atmosphere cause the greenhouse gas effect because they are transparent to the Sun's rays, allowing them to the reach the surface. When those rays are re-radiated by the Earth back into space, they lose energy and drop to the infrared level. But at that level, the greenhouse gases absorb them, thus trapping the heat.

Biosphere

The biosphere is all living things on, in, and around the Earth. Life on Earth influences the chemical and thermal constitution of the Earth's subsystems. The public is probably most familiar with the carbon cycle in which humans participate as we breathe oxygen that trees and other plants produce from their absorption of carbon dioxide. In the past couple of centuries, human use of fossil fuels has radically accelerated the Earth's natural carbon cycle, rapidly increasing the amount of carbon in the atmosphere above natural levels.[2]

Forcings

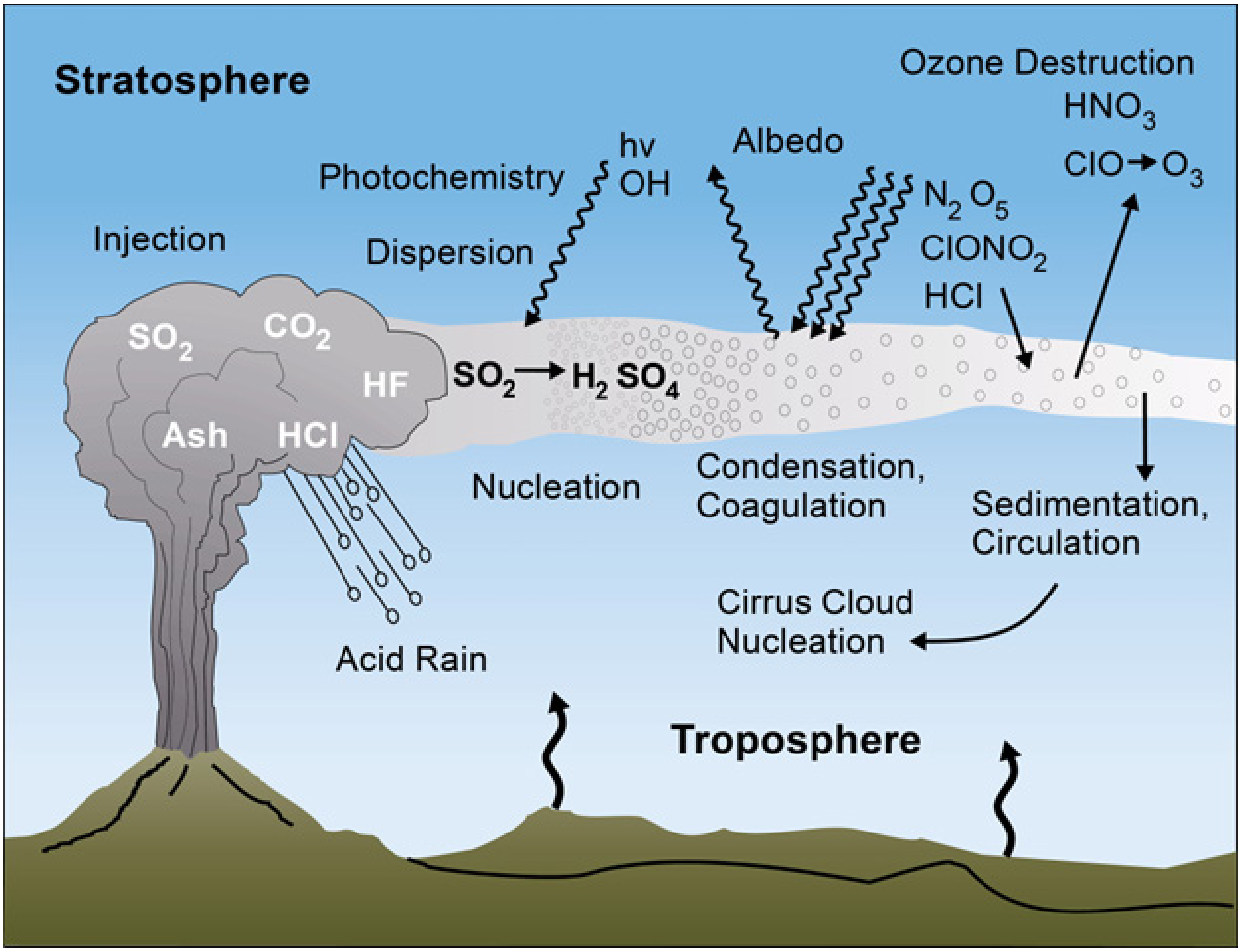
The Earth's climate system—that is, the different parts that combine to shape the climate around the globe—involves interactions between various components of the four subsystems just described. The Earth's climate system is also an open system. This means that it is not entirely self-contained. Solar energy radiated from the Sun is the Earth's vital source of external energy. That energy and the Earth's climate system mix to shape climatic conditions. Despite the variability of weather from day to day, over longer periods, climate is relatively stable. Climate, however, does change slowly over time. Scientists call the specific causes of climate change forcings. There are three particularly influential forcings that we will describe below: solar energy, volcanoes, and greenhouse gases.

Solar Energy

Energy from the Sun heats up the Earth. Everyone understands this reality from their own experience of feeling the Sun's warmth on a clear day. But what we cannot intuitively experience is that the amount of energy transferred from the Sun to the Earth is not completely consistent over time and space. The movement of the Earth and its relation to the Sun has been a driver of climate change for hundreds of thousands of years. For example, climatologists have identified that cooler temperatures in the Northern Hemisphere in the late 1600s and early 1700s corresponded to a period of fewer sunspots and low solar activity. The **Milankovitch cycles** reflect the fact that at intervals of around 100,000 years, 41,000 years, and 26,000 years, the Earth completes different cycles that influence which parts of the Earth receive more, or less, solar energy. These three cycles have interacted with each other to steer the climate of the Earth into and out of periodic ice ages.[4]

Volcanoes

If you were outside on a hot day, you might seek shelter from the sun by moving to a shady area. The ground that has not been hit by the sun will be cooler than somewhere nearby that has been absorbing solar



Volcanic gases in the atmosphere.

Source: United States Geological Survey, [Volcanoes Can Affect Climate | U.S. Geological Survey (usgs.gov)](usgs.gov).

energy throughout the day. When large volcanoes erupt, they emit a layer of dust and particles that can offer shade cover to large areas of the globe, creating cooler conditions over vast regions. When multiple large volcanoes erupt one after another, the cooling impact can be great enough to influence large regions of the world and even bring down the average global temperature.

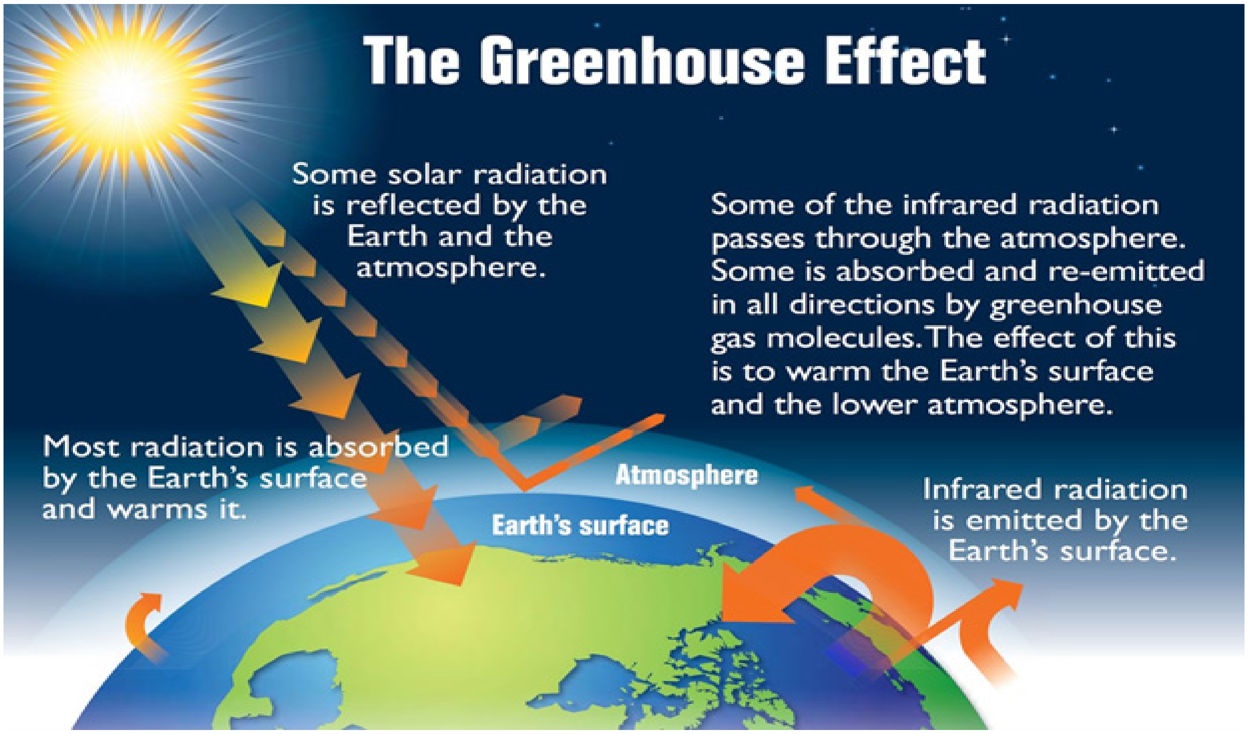
Greenhouse Gases

The term "greenhouse gases" refers to the design of buildings called greenhouses that capture heat from the Sun and are commonly used for agriculture. The public probably has experienced this heating dynamic more frequently in automobiles that are very hot on the inside after being parked in the sun with the windows up. This occurs when solar radiation enters through the glass windows and warms up the interior surfaces of the car, such as the dashboard. The windows also trap the heat that radiates off the dashboard and other surfaces within the car, creating a warming effect inside the entire car. Water vapor, carbon dioxide, and methane are some of the gases that produce a greenhouse effect around the surface of the Earth. Humans release greenhouse gases into the atmosphere through a variety of practices, contributing to increasing concentrations of greenhouse gases in the atmosphere and a more severe greenhouse effect.

Positive and Negative Feedbacks

Because there are so many interconnected parts to the Earth's climate system, when forcings begin to change the climate, the four subsystems are affected in many different ways. Reactions to climate change caused by forcings are called feedbacks. Some feedbacks have already been mentioned. One is the melting of sheets of ice around the North Pole. Forcings that have caused climatic warming in the Northern Hemisphere have resulted in this melting ice.

Scientists classify feedbacks as either positive or negative. In this context, "positive" and "negative"



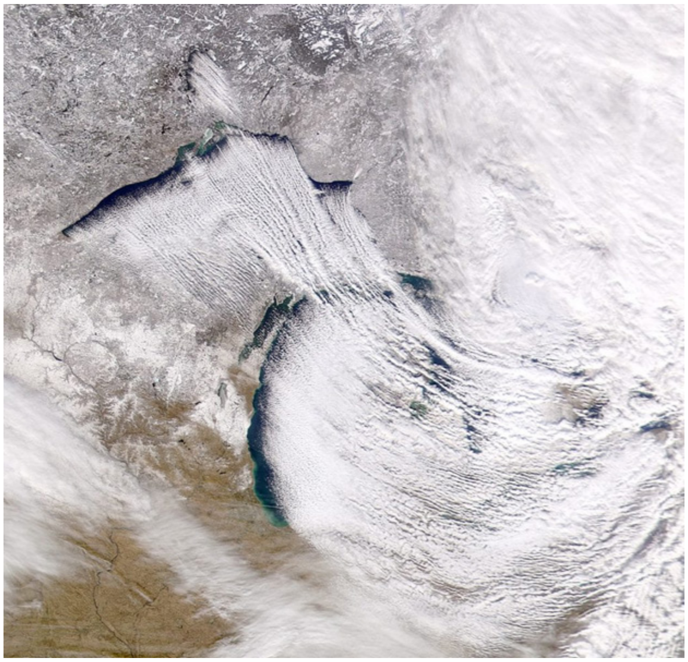
The greenhouse effect.

Source: University of Calgary, Greenhouse effect - Energy Education.

do not mean "good" and "bad." The designation of positive or negative refers to the relationship between the original forcing and the impact of the feedback. If the original forcing and the feedback both push the climate in the same direction, either both warmer or both colder, then the feedback is positive. So, melting polar ice sheets is a positive feedback because warming caused the ice to melt, and losing the ice further warms the climate. Positive feedbacks, therefore, keep pushing climate change in the direction it is headed, either warmer or colder. Positive feedbacks can push climate change to what is called a **tipping point**, at which time the change reaches a point of no return. Negative feedbacks, conversely, serve to moderate climate change.[5]

Examples of Positive Feedbacks

Melting ice sheets around the North Pole offer a good example of a positive feedback. Warmer temperatures in the region impact aspects of the Earth's subsystems. Warmer temperatures affect the cryosphere, the ice portion of the hydrosphere, by melting ice that is on the surface of the ocean. This development alone has harmful consequences within the biosphere by de stroying the habitat of animals such as the polar bear. But the melting ice is classified as a positive feedback because the loss of ice actually warms up the climate in addition to the initial warming.When ice covers the surface of the ocean, it reflects solar energy back away from the surface of the Earth. The ocean water that is exposed after the ice melts, conversely, absorbs more energy from the Sun, and the warming of the ocean is increased.Forest fires in Yunnan are an other example of a positive feedback in climate change. Rising temperatures lead to more droughts in Yunnan, a part of China filled with pine forests. These pines are oily and catch fire easily. When it gets drier, the chance of wildfires spikes. In 2020, Yunnan saw less rain and hotter weather, drying out the forests. That year, 61 wildfires scorched 3,600 hectares, three times more than usual. These fires pump carbon diox ide into the air, a gas that worsens climate change, and a loop is formed: hotter weather sparks more fires, and more fires fuel further warming. Plus, wildfires dam age the for est ecosystem and change how the land re flects sunlight, which can mess with local weather pat the lakes, in turn, becomes cloud cover that cools the surface of the Earth by blocking the sunlight.



Lake-effect snow over the Great Lakes.

Source: University of Michigan, Lake-effect Snow in the Great Lakes Region | GLISA (umich.edu)

Additionally, the clouds formed over the lakes sometimes bring lake effect snowstorms, creating a layer of snow on the ground that, like ice, serves to reflect solar energy away from the surface of the Earth.

The weakening of the polar vortex is another possible example of a negative feedback because it has recently caused colder than typical conditions in North America. Scientists hypothesize that warmer temperatures in the oceans and atmosphere of the Northern Hemisphere have weakened the polar vortex of cold air that circulates around the North Pole. As warmer air moves upward, the cold air in the polar vortex moves out of its usual course and travels farther south into areas of the United States that it had not previously reached. In those places, winter temperatures drop well below regular levels. In this case, the initial warming causes colder conditions in some areas. Since the initial warming and subsequent cooling do not align, it is a negative feedback.

SOURCES FOR RECONSTRUCTING THE HISTORY OF CLIMATE

Even if a scientist or a scholar is equipped with an understanding of how climate operates and how climate change can occur, more information is needed to reconstruct the actual history of climate. Like any other historical investigation, scholars need to have sources that provide evidence of past events. In standard historical practice, historians examine written sources that are collected and stored in an **archive**—a physical repository of documents. Scholars who study climate history have adopted the concept of an archive to describe where they find the sources they use. Sometimes, scholars who study climate also examine written documents for clues about past climatic conditions. Scholars have called the places that hold these types of sources "archives of society." But climate scholars also search nature itself for clues about climate's history. Scholars call these figurative storehouses of sources the "archives of nature."[6]

Sections II and III of this guide will cover the history of human interactions with climate over roughly the past 10,000 years. Rather than merely recount that history, we will now discuss how scholars have figured out what happened.

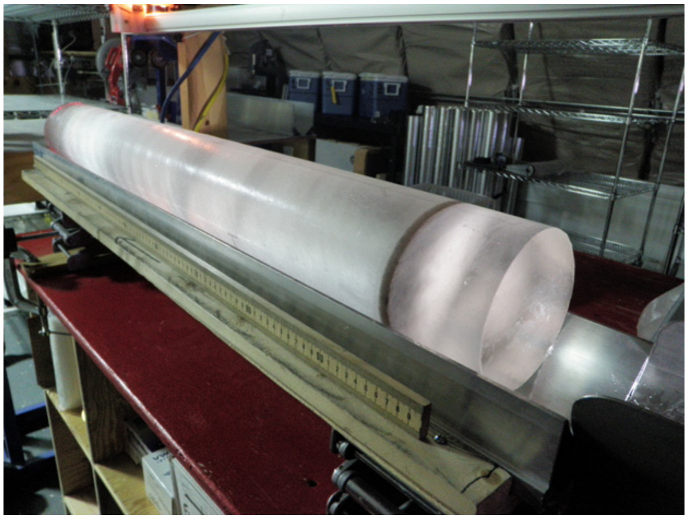
The Archives of Nature

As discussed earlier, climate influences the four subsystems—earth, water, air, and living organisms—in a host of different ways. Certain climatic conditions might correspond with specific gases in the atmosphere, or the growth of certain plants, or certain water levels, and so forth. Traces of those impacts are sometimes discoverable in the natural world. In fact, nature is an archive, or a repository, of clues about how climate and the different parts of our world interacted in the past. Scientists and scholars have creatively developed methods for observing and measuring signs of past climatic conditions that have been preserved in different parts of the world.

Something observable in nature that gives an indication of past climate conditions is called a **proxy**. Proxies are natural features that show evidence of being impacted by specific climate conditions. If scholars can isolate and date these impacts, they will have an idea of what the climate was like at a certain time in the past. Three of the most revealing sources of climate history are ice, trees, and soil.[7]

Ice Cores

Ice core sampling is a technique of drilling long cylinders of ice out of deep glaciers. The ice cores are then analyzed in layers. As the ice was formed, the snowfall from each year that became the new top layer



Ice core sample.

Source: National Science Foundation, [About Ice Cores | NSF Ice Core Facility]

of the glacier would trap particles from the atmosphere and freeze them in the ice. Therefore, an ice core from an old glacier can reveal what the atmospheric conditions were dating as far back as hundreds of thousands of years.[8]

Trees

Trees can also be analyzed for information about past climates. One can identify the age of a tree by counting the rings of a tree that has been cut through. Scientists know how to analyze each of those rings for information about the weather during that year of the tree's life. The tree can reveal if the year was dry or rainy and can even show the weather conditions of individual seasons in a year. The practice of gathering this information from trees is called dendrochronology—a combination of words that refer to trees and time.

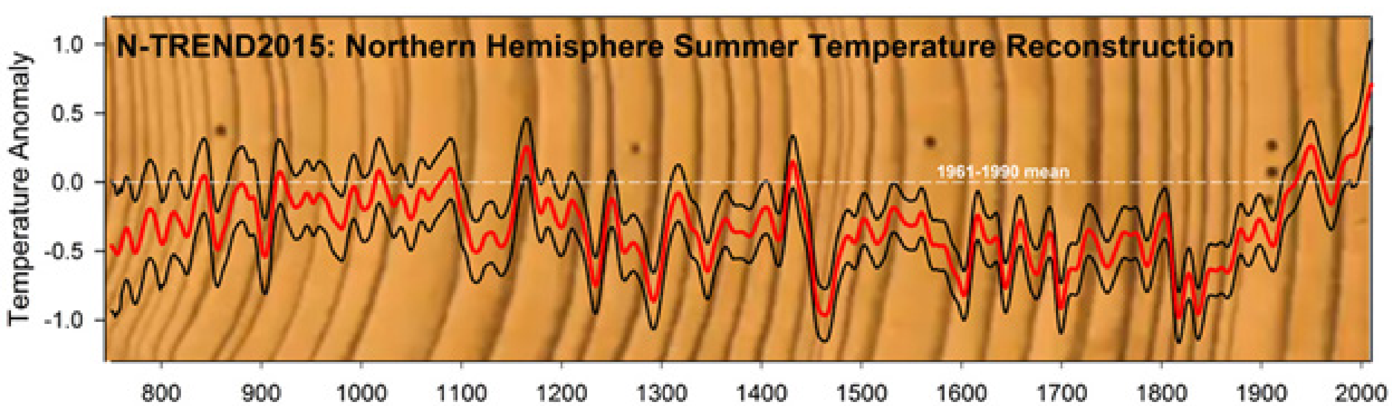
Sedimentation and Other Sources

Scholars also study other parts of the natural world for data that has been collected about past conditions. The layers of sediment or mud on the bottom of lakes and the ocean contain information about the historical composition and content of water such as the amount of pollen. Coral sampling in the ocean reveals similar data as well as past temperatures of the ocean in the location of the coral. Scientists creatively harvest data about the impact of past climate conditions on various parts of the Earth System and its subsystems. Through such discoveries, scholars rediscover the climate of the past.

The Archives of Society

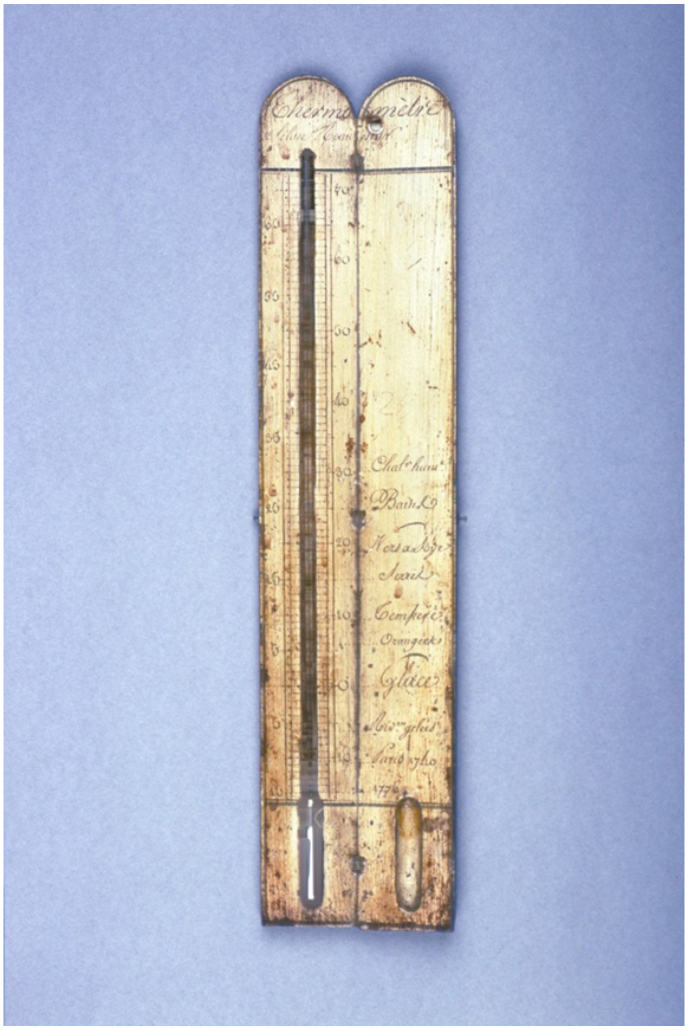
There is also evidence of past climate conditions in the archives of society. The term archives of society refers to sources produced by humans that contain information about the climate. There are some obvious limitations to the information scholars can gain from the archives of society. Human records only describe conditions over the past hundreds or thousands of years, not the past millions of years that some sources from the archives of nature reveal. Additionally, the most precise information from the archives of society, instrumental records such as recordings of temperature, are an even newer development. The oldest instrumental records date back only to the thermometer's invention around 1700.

However, even before humans had modern scientific instruments, they left records that describe the climate. To make use of these records, scholars utilize a system of proxies similar to those used to study the archive of nature. While proxies produce less precise information



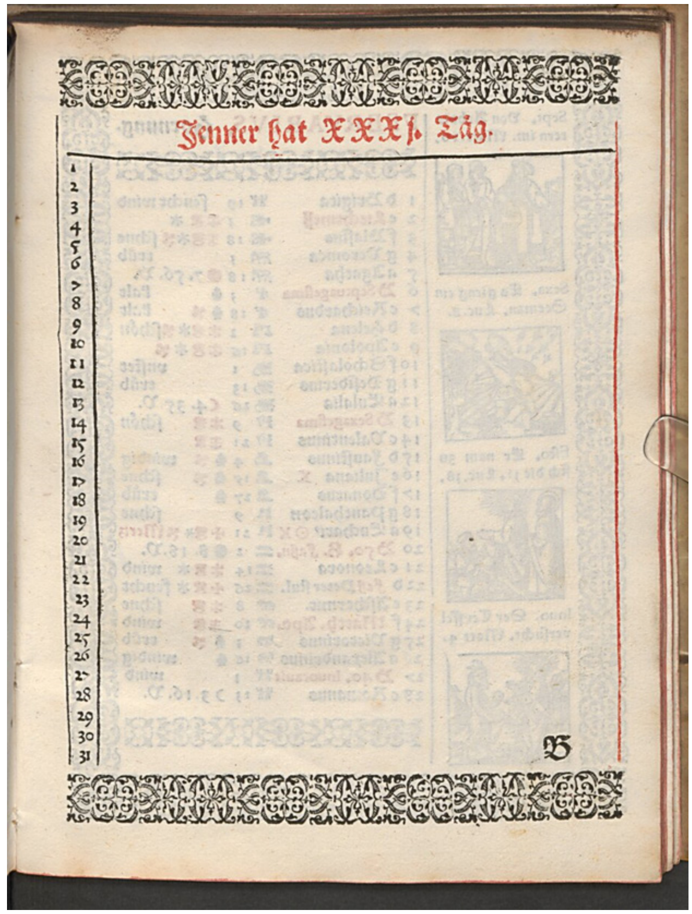
Temperature reconstruction and tree ring sample.

Source: University of Georgia, [New tree ring data set shows NH temperatures | Climate and Agriculture in the Southeast (uga.edu)]



A French thermometer from around 1780.

Source: Whipple Museum of the History of Science, collections.whipplemuseum.cam.ac.uk/objects/11165/



An unused German calendar printed in 1594 with space for users to record weather conditions.

Source: Bayerische StaatsBibliothek

than a reading from a scientific instrument, they can still produce reliable information. For instance, a narrative account of a flood in the 1500s will not contain an exact measurement of how much rain fell, but scholars can reliably know that substantial rainfall did occur. One area in which the archives of society are more specific than the archives of nature is in dating. From tree records we may know that there was a good year of rain for plant growth, but human records can give details down to the day or hour of when it rained.

Instrumental Records

Scholars use precise and accurate information that was recorded by scientific instruments to identify trends in the climate. Today, meteorologists measure daily records of high and low temperatures from many places around the globe. These records are very precise, but they only go back a relatively short period of time. For example, Phoenix, Arizona, received attention in the news for record-breaking heat in the summer of 2023. But temperature records for the city of Phoenix only date back to the year 1896.[9] So, instrumental records can clearly show trends in the climate, but only for a little more than a century.

Additionally, scholars are constantly thinking about where and how measurements should be taken to get the clearest possible picture of what the climate is doing. For instance, scientists record the temperature at different layers of the atmosphere to gain a more precise understanding of climate trends above the Earth.[10] Even though data on things like **precipitation** and temperature taken from reliable scientific instruments seems straightforward, careful analysis is still required to combine points of data from many individual places at specific moments in time into an accurate understanding of climate trends over larger areas and longer timespans.

An eighteenth-century depiction of the Swiss town of Grindelwald and a nearby glacier.

Source: Swiss National Library

Narrative Records

Even before people had the instruments to systematically record temperature and precipitation, they still paid attention to the weather and other climate-related developments. Scholars can examine human-produced records for clues about past climate conditions. These types of sources include weather diaries in which someone might have left short daily entries that document if it was hot, cold, rainy, cloudy, sunny, and so on. Ship logbooks often include such information. Other narrative records might describe whether a glacier is growing or receding, how high the water level is in a river or lake, whether a lake is frozen or not, or detailed descriptions of exceptional occurrences, such as a long drought or a severe storm.

Scholars treat the information contained in narrative records as proxies for estimating the information that modern scientific instruments can record such as temperature, rainfall, or snowfall. Although the narrative records do not contain a temperature recording, if they do contain an account of a cold winter and describe a frozen river, for instance, scholars can devise systems for taking those records and comparing them with other records to gain an idea of climate trends and conditions over time. To return to the glacier example, accounts of glacier expansion are not the same as temperature readings that show an average temperature decline, but the growth of the glacier works as a proxy to suggest that it is growing because temperatures are declining.

Hendrick Avercamp, Winter Landscape with Skaters*, c. 1608.*

Source: Rijksmuseum

Other Types of Records

There are a variety of other types of records that people have left behind that also contain clues about past weather and climate. Sometimes people made a marker of a highwater mark during a flood that one can see on a building. Works of art also show depictions of weather conditions. Scholars need to be careful, however, to consider if the proxy they are using accurately reflects real conditions. A painting is not the same as a photograph with a date. The artist may be depicting a real scene, or the inspiration for the painting may have come from elsewhere.

Grain prices are another proxy that scholars have used as a marker for weather conditions. There is a correlation between certain types of weather and fruitful harvests that would cause prices to decline. Recently, however, some scholars have argued that too many other factors unrelated to weather, like popula tion, farming techniques, policies, social stability, transportation and so on, might also drive grain prices up or down, and that prices do not accurately reflect weather conditions.Debating how to use sources is a healthy part of scientific inquiry and leads to more re-fined and reliable methods for studying past climate conditions.

FIELDS FOR STUDYING THE HISTORY OF CLIMATE

Because there are many different types of sources that contain information about the history of climate, different scholars have focused on becoming experts at analyzing specific types of sources. Professionally trained scientists, social scientists, and historians are all highly specialized. That means a scholar will likely only have expertise in researching climate history in a particular way. Sometimes, when a scholar identifies sources that contain useful information and develops effective methods for analyzing those sources, other scholars begin to join the work, and a new **scholarly field** is born. A scholarly field, in other words, consists of a group of scholars who share common practices for studying the type of evidence they analyze. Usually, a scholar is primarily a member of one field.

Different scholarly fields use different means to study the history of climate. Often, findings about climate history do not perfectly align within a given field, or between fields. Having robust fields with many scholars and lots of research provides many perspectives and the opportunity to reflect on which studies produce the most accurate results. Comparing discoveries that have been made between different fields allows for an outside check on the findings made within a field. Every scholarly field also has its own strengths and limitations. The existence of multiple fields that examine climate history offers us an opportunity to learn from the strengths of different fields and move beyond the limits of each individual field. Overall, the fields of **historical climatology**, **paleoclimatology**, **climate history**, and the **history of climate and society (HCS)** offer complementary views of climate history. Together, they present a reliable picture of past climate and its relationship to human life.

Christian Pfister led the creation of the field of climate history.

Historical Climatology and Paleoclimatology

Historical climatology and paleoclimatology are different names for essentially the same field. Both terms end with the word "climatology" prefaced by either "historical" or "paleo-." Climatology is a study of climate that relies primarily on the archives of nature. That is, climatologists are experts at gathering information about climate from the investigation of the natural world, including the sources described earlier, such as layers of ice or lake sediment or tree rings. The term "paleo-" means ancient or old and conveys a similar idea as the term "historical" in this context. Historical climatology or paleoclimatology investigates climates of the past, particularly before the 1800s when scholars first utilized existing scientific instruments for the purpose of creating widespread and systematic records of climatic conditions.

Historical climatologists must learn and practice many skills, including, but not limited to: collecting samples from nature (such as ice cores or tree samples), operating the machinery or instruments used to analyze the samples, analyzing the data obtained from nature to reconstruct past climate conditions, and communicating findings from nature according to the standard written conventions of the field of climatology.

Climate History

The field of climate history, by comparison, utilizes a different set of practices to analyze different sources. Climate historians collect and study sources from the archives of society, and they analyze those sources according to historians' methods and conventions. The skills that climate historians bring to the study of climate include: the ability to read the language and script of the texts being analyzed; the ability to find the texts, analytical techniques, and the contextual knowledge to interpret the texts accurately; and the ability to formulate and communicate historical narratives based on the textual evidence. The Swiss historian Christian Pfister (b. 1944) was an influential pioneer in the field of climate history and diligently compiled sources and developed methodological approaches to analyze them. He compellingly demonstrated that sources in the archives of society can be used to produce trustworthy climate reconstructions.

A climate historian, therefore, looks in different places

IUGS/ICS time scale (2022)

System / Period	Series / Epoch	Subseries / Subepoch	Stage / Age	GSSP Age	GSSP Location	GSSP Primary guide
				present		
Quaternary	Holocene	Upper / Late	Meghalayan	4250 yr b2k	KM-A speleothem	(4.2 ka climate event)
Quaternary	Holocene	Middle	Northgrippian	8236 yr b2k	NGRIP1	(8.2 ka climate event)
Quaternary	Holocene	Lower / Early	Greenlandian	11,700 yr b2k	NGRIP2	(abrupt warming)
Quaternary	Pleistocene	Upper / Late	Stage 4	~129 ka	In progress	(Termination II?)
Quaternary	Pleistocene	Middle	Chibanian	0.774 Ma	Chiba	(Matuyama–Brunhes reversal)
Quaternary	Pleistocene	Lower / Early	Calabrian	1.80 Ma	Vrica	(top Olduvai Subchron)
Quaternary	Pleistocene	Lower / Early	Gelasian	2.58 Ma	Monte San Nicola	(Gauss–Matuyama reversal)

Time scale ratified by the International Union of Geological Sciences (IUGS). Abbreviations: "b2k" = before the year 2000; ka = thousands of years before present; Ma = millions of years before present.

Source: Subcommission on Quaternary Stratigraphy, [Major divisions | Subcommission on Quaternary Stratigraphy]

than the historical climatologist for information about past climate conditions. A scholar in one field would likely not have the training and expertise to accurately analyze the types of sources that someone in the other field does. The fields complement each other, however, because they broadly explore the same topic from different perspectives.

The History of Climate and Society (HCS)

The history of climate and society (HCS) is a new field that is just beginning to emerge. Environmental historian Dagomar Degroot has led the push to form and name the field of HCS. The need for the field, according to scholars within it, has stemmed from the recent expansion of studies on climate from a variety of fields including climate history and paleoclimatology, as well as other fields such as archeology, economics, geography, linguistics, and genetics. As its name suggests, the field of HCS focuses on the relationship between past climate conditions and human societies.

At that intersection of climate and society, HCS scholars contend that existing research from the various fields just listed often lacks precision. Specifically, HCS scrutinizes scholarly claims about climate's past impacts on human society. HCS seeks to apply rigorous analytical methods from the field of history to make sure that sufficient evidence exists to support claims about climate causing certain social conditions. To put it simply, HCS scrutinizes causal claims.[11]

The field of HCS also pays careful attention to scale. In this context, scale refers to the scope of one's investigation. For instance, different levels of geographical scale could be a city, a large region, a continent, or the entire globe. Different levels of chronological scale could be days, decades, millennia, or millions of years.

HCS emphasizes the significance of precision when it comes to scale. If a scholar only has sources from one town, that evidence may not support a claim about an entire region. Or if the sources are all from a few decades, that data may not support a claim about entire centuries. Geographical scale and chronological scale are very important for the study of climate history and its relationship to human life.

CONCEPTUALIZING CLIMATE CHANGE TODAY

Historians always study the past from the context of their own time. Historians try to be self-aware and consider potential biases from the present that might taint their understanding of the past. The relationship between present climate change and past climate change, however, offers a particularly tricky challenge for scholars. On the one hand, there is a lot of continuity between the past and the present in terms of how the Earth's subsystems interact to create climate conditions. Studying past climate change can help scientists and scholars understand what causes change and how the Earth reacts to forcings that cause climate change.

On the other hand, the primary cause of current global warming—greenhouse gases in the atmosphere re-leased by human—is unprecedented in human and

Quaternary Period with the Anthropocene Epoch

Eonothem/ Eon	Erathem/ Era	System/ Period	Series/ Epoch	Stage/ Age	millions of years ago
Phanerozoic	Cenozoic	Quaternary	Anthropocene[1]		
					1950 CE
			Holocene		
					0.0117
			Pleistocene	Upper	
					0.126
				Middle	
					0.781
				Calabrian	
					1.806
				Gelasian	
					2.588

[1] In August 2016 the Anthropocene Working Group (AWG), a special body created within the International Commission on Stratigraphy (ICS), recommended that the Anthropocene Epoch be made a formal interval within the International Chronostratigraphic Chart. The AWG recommended that the year 1950 be used as the starting point of the Anthropocene Epoch.

Though the Anthropocene Working Group (AWG) has proposed that the Anthropocene should officially be recognized as a new geological time interval, their proposal was not officially accepted.

global history. Starting about two hundred years ago and increasingly since then, people have extracted and released **fossil fuels** at unprecedented levels. So, there are also strong discontinuities between climate change today and climate change in the distant, and not-so-distant past.

What concepts should scholars use to describe the relationship between today's climate change and climate change in a broader chronological context? One answer to this question has come in the form of a term used to describe climate change today: the Anthropocene.[12] The term Anthropocene puts current climate conditions into the broader context of climate history while also placing a strong emphasis on the uniqueness of today's climate.

The concept of the Anthropocene refers to the scientific divisions of geological time that correspond to climatic conditions of the Earth over millions of years through to the present. The current era, or epoch, is the Holocene, which started around 11.7 thousand years ago, at the end of the last global ice age. The epoch prior to the Holocene, the Pleistocene, started around 2.58 million years ago. The use of the term "Anthropocene" to refer to the era we live in today suggests that the climate change we are experiencing at present is so significant that it is comparable to epochs of very substantial climate change in the past. The inclusion of the prefix "Anthro-" at the beginning of the term strongly emphasizes that humans have primarily contributed to the current trajectory of the climate.

Concerns about Using the Term "Anthropocene"

The stakes are high for adopting the term Anthropocene, and some scholars have reservations about doing so. In science, systems of classification are very consequential as these systems organize human knowledge. Accordingly, scholars and scientists take them very seriously. Do scientists really know enough about climate change today to decide that it is on the same level as the transitions into and out of the Pleistocene and into the Holocene? That question is currently being debated by scientists in the geological community. In 2019, the Anthropocene Working Group (AWG), a body of experts on the Earth's geological epochs, proposed that the Anthropocene should officially be recognized as a new geological time interval. They advocated that starting in the mid twentieth century, around 1950, the Holocene ended, and the Anthropocene began. In March of 2024, the International Union of Geological Sciences rejected a proposal to formally name the Anthropocene as a new geological epoch, but noted that the term will "continue to be used not only by Earth and environmental scientists but also by social scientists, politicians and economists as well as by the public at large" and "will remain an invaluable descriptor of human impact on the Earth system."[13] According to the formal geological time scale accepted by the geological community, we still live in the Holocene.[14]

Arguments for Using the Term "Anthropocene"

Is it useful and appropriate to use the term Anthropocene even if scholars in the geological community who could make the term official have not done so? It is, for several reasons. Putting the Anthropocene in the official geological time scale is a very high bar to reach. The fact that geologists are careful before taking such a dramatic step reflects the seriousness of scientists.

Although it has not received that specific recognition, the idea of the Anthropocene is not discredited.An important reason to use the term Anthropocene is that other scholars and the public often refer to the Anthropocene already. The evidence overwhelmingly suggests that 1950 is an appropriate marker for the start of current warming trends in the climate. While scholars are debating the nuances of whether or not the mid-twentieth century qualifies as an epochal change on the scale of the Holocene, something drastic is taking place in today's climate. The term Anthropocene has become a way to name what is happening, making it easier for scholars, journalists, and members of the public to have vitally important conversations about our changing climate.[15]

CLIMATE CHANGE AND NARRATIVES OF GLOBAL HISTORY

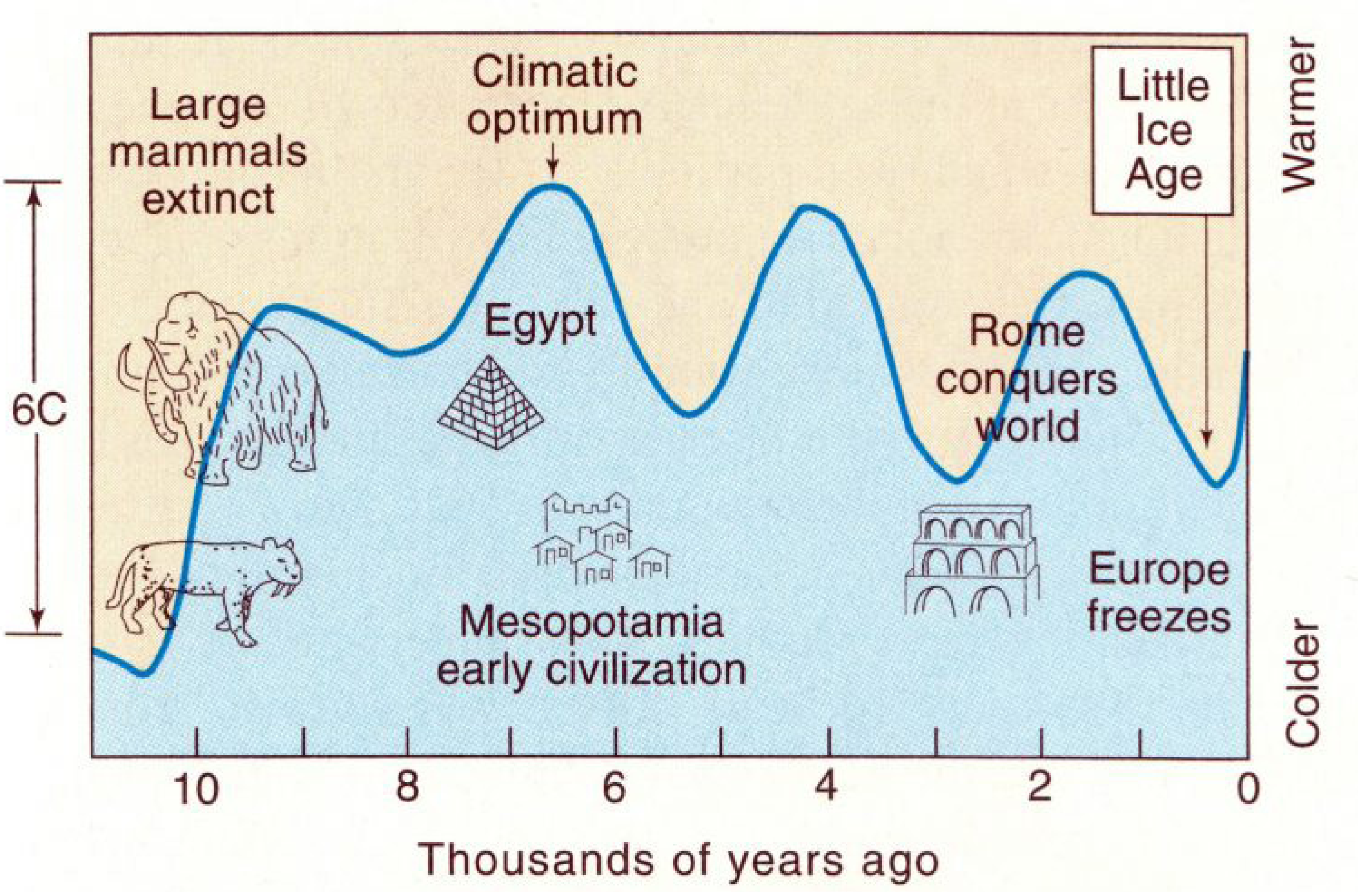
Humans have been telling narratives, or stories, about global history for millennia. Professional historians have been telling stories about global history since the modern historical profession became formalized in Europe in the 1800s. But with a few exceptions, such as the French historians E. Le Roy Ladurie and Fernand Braudel, modern historians did not include climate in their narratives of global history until around the year 2000. Because climate was not included in traditional histories of the world before the past two and a half decades, many things have converged—or crashed together at once—since historians have started trying to incorporate climate into the story of world history. Historians and scientists are still sorting out the combined story of climate history and human history.

Just as scholars in scientific fields constantly revise their conclusions and make them more precise and accurate when new information becomes available, historians also revise the stories they tell when new information becomes available. This process has been very evident as scholars have tried a variety of approaches to integrating climate history and human history. Historians study how climate shaped past civilizations and affects us today, which is fascinating and important.

Mapping Climate onto Existing Narratives

One, perhaps obvious, approach to bringing together existing historical narratives and new information about climate history is to simply place the two histories over one another. Historians have well-estab lished narratives about most parts of the world. Often, these narratives divide history into a series of stages that are marked by political developments. The history of ancient China over the past several thousand years, for instance, can be portrayed neatly as a timeline of successive dynasties: Xia, Shang, Zhou, Qin, Han, and so forth. For the Mediterranean World, a popular narrative moves from the Greek Empire to the Roman Empire to the Middle Ages and so on.

One option for combining climate history with these types of existing historical narratives is to look at notable junctures, such as the transition from one dynasty to another, and then see if any climatic events—such as a period of climate change—occurred around the same period. Often, scholars who employ this technique are able to show striking correlations between large-scale social events, such as mass revolts



Correlations between climatic conditions and human history.

Source: "Climate in Human History" in Donald R. Prothero and Robert H. Dott, *Evolution of the Earth*, 8th ed, McGraw-Hill, 2010.

in a region, and climate-related conditions, like an extended period of drought. The climate events and the social events often line up to such a degree that it seems to be more than a mere coincidence.

A potential weakness of this approach, however, is that it is not the best practice in history or science for scholars to desire to prove too specific of a thesis before they begin to study their sources. This is because such an approach could lead to biases in the results. Sometimes, scholars might find what they are looking for, but miss other important observations. Using existing narratives to guide the study of climate history may put too many constraints on climate history and turn it into little more than the retelling of narratives that are still based on the old frameworks they were always based on, such as dynasties or empires.

Climate Determinism and the Question of Causal Relationships

Attempts to combine climate history and human history have also raised challenging interpretive issues related to causation. In other words, it is hard to determine if one event in history caused a second event, or if they just coincidentally happened. Often, there is not enough evidence to prove the causal relationship definitively or with certainty. So, historians need to use careful reasoning to offer an interpretation of the available evidence that suggests why and how something might have caused something else.[16]

In the context of the history of climate and society, a major question is whether or how climate conditions cause events in human societies. Do favorable climate conditions cause an empire to thrive? Do unfavorable conditions cause revolts and revolutions? Older scholarship—including some of the pioneering works of history that first gave attention to climate history as a factor in human history—made the case that climate determined or dictated the course of human history. More recent studies have cast doubt on such a strong link between climate and the development of human culture and society. Recently, the argument that climate sets the course for human history has been labeled **climate determinism**. Almost all studies of climate and society today say that they are rejecting climate determinism, but some scholars still argue that climate played a major role in shaping human history.

Scholars may be able to show a correlation between a certain climatic condition and a human event, but that does not actually prove that the first one caused the second one. In the field of the history of climate and society, scholars place emphasis on the careful study and analysis of **causal mechanisms**. This approach asks what specifically did the climate impact that then became the exact thing that triggered a human response. Those gears that turn and then make the next gears turn are the causal mechanisms that could offer a more intricate explanation of how climate has affected human behavior and societies.[17]

Multiple Scales

Another major challenge for combining human climate history and human history relates to the matter of scale. Typically, the chronological scale of human histories dates back to around five or six thousand years ago when the first major human societies emerged. These civilizations in places like Ancient Mesopotamia, Egypt, and China have left robust traces of buildings, artifacts, and even written documents that allow historians to reconstruct their histories. Sometimes, scholars use archeological evidence to create historical accounts of what human life was like as far back as around 40,000 years ago. Most often, historical studies focus on human events in the much more recent past, such as the twentieth century. However, a few thousand years of human history amount to little more than a blink of an eye when compared to the millions of years over which the Earth's climate has evolved to become what it is today. In what meaningful way can scholars compare these two vastly different chronological scales?

In the past, scholarly fields that study the science of climate change and scholarly fields that study human histories did not overlap. The current state of climate change caused by human behavior—known as the Anthropocene—has brought these two histories together at the moment in time in which we currently live. Scholars are in the process of figuring out how to tell the story of the Earth's history and the story of human history in a way that illuminates both histories.

Incongruent Chronological Scales

Rachel Carson in 1960.

Source: https://www.rachelcarson.org/

Because humans before the twentieth century did not have the capability to impact climate on the scale that humans do today, humans arguably lacked an awareness that their actions even had the potential to shape the world in such a forceful and lasting way. Although she was not writing about climate, Rachel Carson (1907–64), with her 1962 book *Silent Spring*, is often credited with helping humans develop an awareness of how much potential we now have to alter the Earth. Her book compellingly demonstrated how commercial pesticides destroyed living organisms and radically transformed ecosystems in mere years, whereas natural changes in an ecosystem could only evolve over the course of decades or generations.

Today, scholars such as Dipesh Chakrabarty argue that humans are reshaping the configuration of the Earth's climate systems in a timespan that is exponentially shorter than natural processes.[18] Today's climate change, therefore, is an event that takes place on the small scale of human history because humans are a leading cause of the climate change. But the changes are also so substantial that they have become part of the story of the Earth's climate that has developed over a much greater scale (millions of years). Thus, while these two chronological scales are incongruent, they nevertheless have now intersected in our lifetime.

The Novelty of the Anthropocene

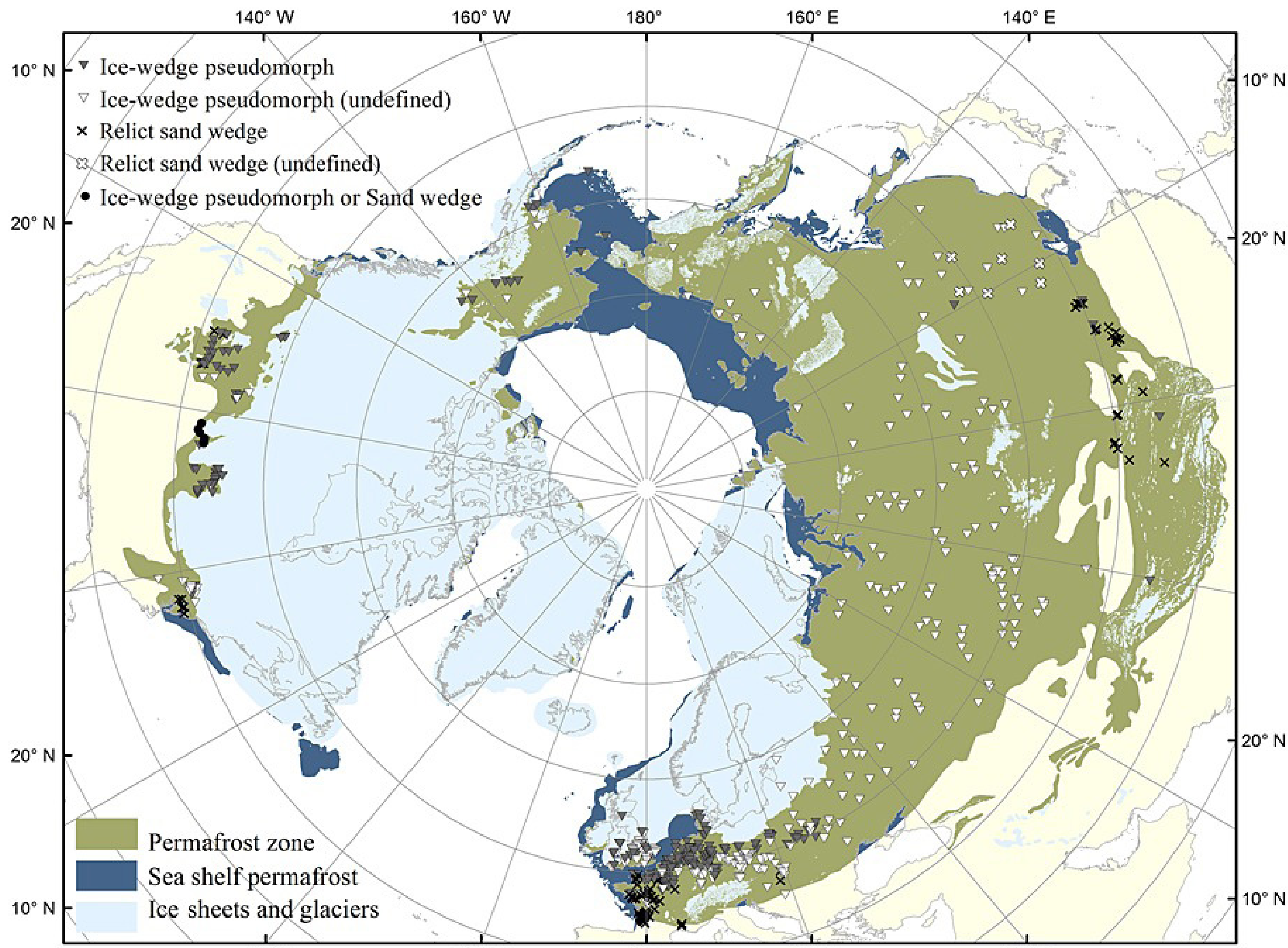
The novelty, or newness, of the Anthropocene presents the social science scholar and the social science student with unprecedented challenges. Old methods and old narratives in the field of history must now account for new discoveries from fields that examine climatology. Scholars and the public wonder if anything from history relates to the Anthropocene and if there are any lessons from the past that can guide us through the challenges of the multiple natural crises that humanity now faces. These are real challenges, but they also offer exciting new opportunities. The project of uncovering and sharing the combined history of climate and humanity is already allowing us to see ourselves and our world in new ways.

SECTION I SUMMARY

- The field of Earth System Science (ESS) provides key concepts for understanding the nature of the Earth's climate and climate change.
- The Earth's four subsystems—earth, water, air, and living organisms—interact on local and global levels to shape climatic conditions.
- Forcings—events such as volcanic eruptions, variations in solar energy reaching the Earth, and rising concentrations of greenhouse gases in the atmosphere—can tip the balance between subsystems and cause climate change.
- Sometimes, climate change causes reactions in nature that drive the change even more (positive feedbacks), while other times reactions in nature act against the initial drivers of climate change (negative feedbacks).
- Scholars and scientists learn about climate history by investigating evidence in the natural world (archives of nature) and in records left by humans (archives of society).
- Historical climatologists and paleoclimatologists study the archives of nature. Climate historians study the archives of society. Scholars in the field of the history of climate and society (HCS) focus on the relationship between climate history and human history.
- The Anthropocene is a term that scholars and the public have begun using to describe the idea that current climate change might mark the beginning of a new climatic and geological epoch in the Earth's history. The term Anthropocene also reflects the reality that current climate change is significantly caused by human actions.
- It is a challenge to combine climate history and human history into a single story, but by attempting to do so, historians help us better understand our changing climate.

Section II

Humans in the Holocene

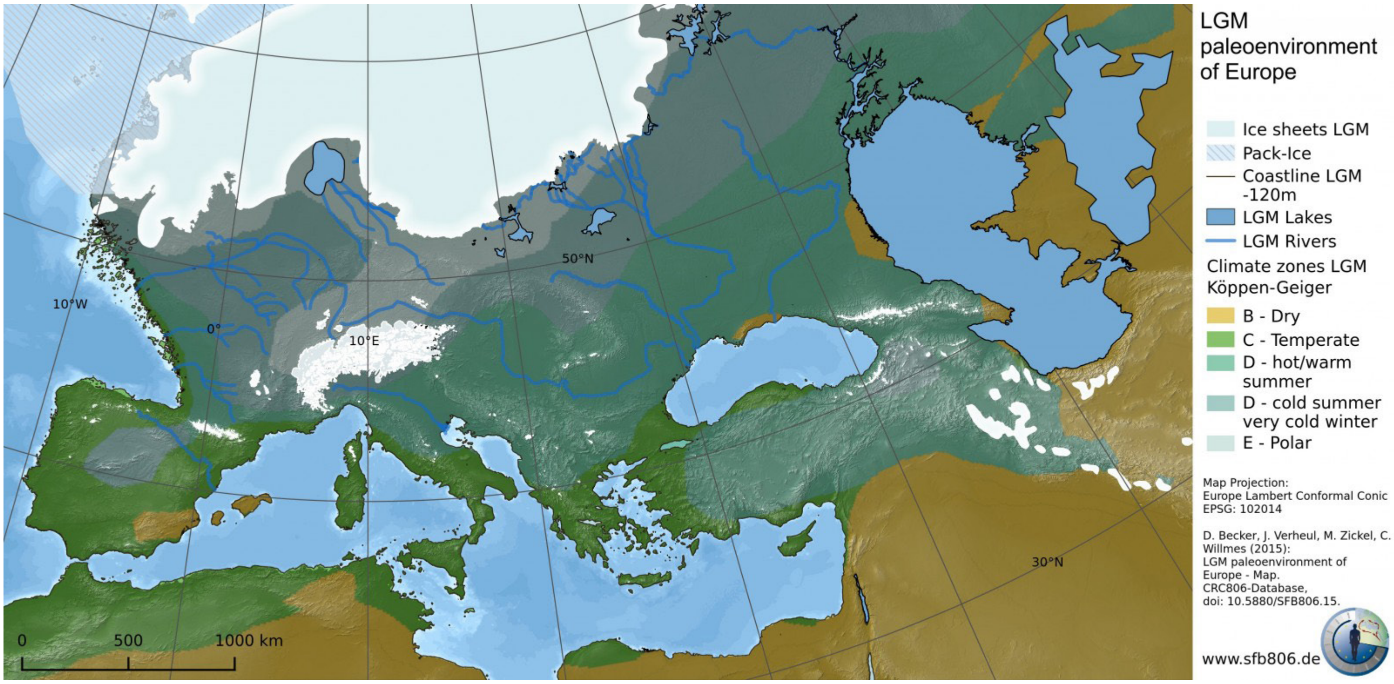


Map of Northern Hemisphere ice sheets, glaciers, and permafrost during the Last Glacial Maximum.

Source: Amelie Lindgren, et al, "GIS-based Maps and Area Estimates of Northern Hemisphere Permafrost Extent during the Last Glacial Maximum," *Permafrost and Periglacial Processes* 27 (2015).

SECTION II INTRODUCTION

According to the International Union of Geological Sciences (IUGS), the geological and climatic epoch known as the Holocene began about 11,700 years ago.[19] During the Holocene, humans have thrived like never before. Whether one dates back to the oldest fossil record of *Homo sapiens* around 300,000 years ago or to around 40,000 years ago when *Homo sapiens* became the last surviving members of the *Homo* genus, humans had a much longer history on Earth before the start of the Holocene than we have had since it began.[20] But our species never came close to building what humans have built during the Holocene. Before



The climate of Europe during the Last Glacial Maximum.

European Geosciences Union. Cryospheric Sciences | Image of the Week — Last Glacial Maximum in Europe (egu.eu)

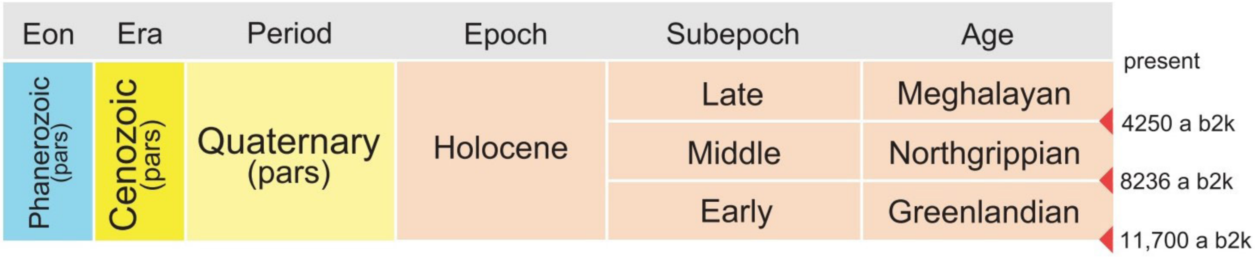
the Holocene, the global human population may not have surpassed ten million.[21] According to the United Nations, the current global human population is over eight billion.[22] Traces of human manufacturing and other activities that impact the environment can be found nearly everywhere on the planet.[23]

Climate and environmental conditions do not solely shape human development or the path of history. However, certain climates are critical for human survival and growth, as many environments are too harsh to support human life. The trajectory of human history over the past 12,000 years during the Holocene epoch—a period of relatively stable and warm climate—was not predetermined by these conditions. Climate did not dictate the specific events, achievements, or challenges that define human history, nor did it guarantee humanity's current global presence. Nevertheless, the Holocene provided a favorable environmental backdrop that enabled humans to thrive and become a dominant global force. Human success has depended on the specific conditions of the Holocene, not in isolation, but as a critical setting. To understand the stage on which human history has unfolded, it is essential to recognize what the Holocene represents and its role in supporting human progress.

The general marker of the start of the Holocene, roughly 11,700 years ago, is the end of the last ice age. The world that humans lived in prior to the Holocene was very different than the environment we have known since. Most of human history took place in the Pleistocene Epoch (2.6 million years ago to 11,700 years ago), a period characterized by cycles of glaciers growing or receding and, at times, covering vast portions of the Earth's surface. The Last Glacial Maximum (LGM) occurred around 20,000 years ago. At that time, much of what is known today as Europe, including Scandinavia, the British Isles, and parts of Central Europe, were covered by ice. Much of North America was as well. Asia experienced permafrost—perennially frozen ground.[24]

So much of the world's water was frozen on the Earth's surface that ocean levels were hundreds of feet lower than current levels. Since the LGM, that ice has receded and melted into the ocean, but not evenly over time. Warming caused by solar cycles contributed to a major glacial meltwater event that started around 14,000 years ago and, in turn, cooled ocean temperatures and reversed warming trends for roughly a millennium. That cold millennium is known as the Younger Dryas period (c. 12,900 to 11,700 years ago). Since its conclusion, the world resumed warming in a new epoch known as the Holocene.

Technically, the Holocene is an **interglacial period**—a warm period between ice ages—that follows a much



The subdivision of the Holocene Series/Epoch.

Mike Walker, Martin J. Head, John Lowe, Max Berkelhammer, Svante Björck, et al, "Subdividing the Holocene Series/Epoch: formalization of stages/ages and subseries/subepochs, and designation of GSSPs and auxiliary stratotypes," *Journal of Quaternary Science* 34 (2019): 174.

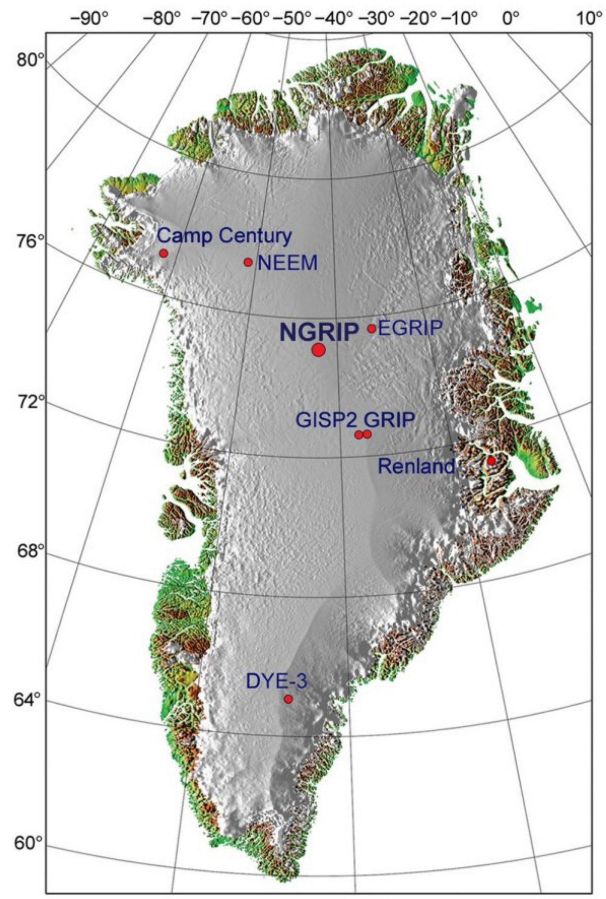
longer pattern throughout the Quaternary Period of ice growing and receding in cycles. The Quaternary Period is a unit of geologic time that began 2.6 million years ago and consists of both the Pleistocene and the Holocene.[25] In the last 800,000 years, there have been eight similar interglacials. Patterns in the Earth's orbit, known as the Milankovitch cycles, affect the amount of solar energy the Earth absorbs at different times. The differences in solar energy over time have been the dominant forcing that moves the climate back and forth between ice ages and interglacial warm periods.[26] According to the factors that had dominated the Earth's climate over the past 2.6 million years, the Earth would likely be cooling right now. Instead, the planet is rapidly warming. That phenomenon will be the subject of section III of this resource guide.

AN OVERVIEW OF THE HOLOCENE

Scholars divide the Holocene into three periods. The three periods are dated according to how many years ago they started and ended, with the year 2000 CE marking the present. The first period lasted from 11,700 to 8,236 years ago and is named the **Early Holocene Subseries/Subepoch** or the **Greenlandian Stage/Age**. The second period dates from 8,236 to **4,250 years ago** and is known as the **Middle Holocene Subseries/Subepoch** or the **Northgrippian Stage/Age**. The third, and final, period runs from 4,250 years ago to the present and is known as the **Late Holocene Subseries/Subepoch** or the **Meghalayan Stage/Age**. Each of these three periods are described in more detail below.[27]

The End of the Last Ice Age and the Early Holocene

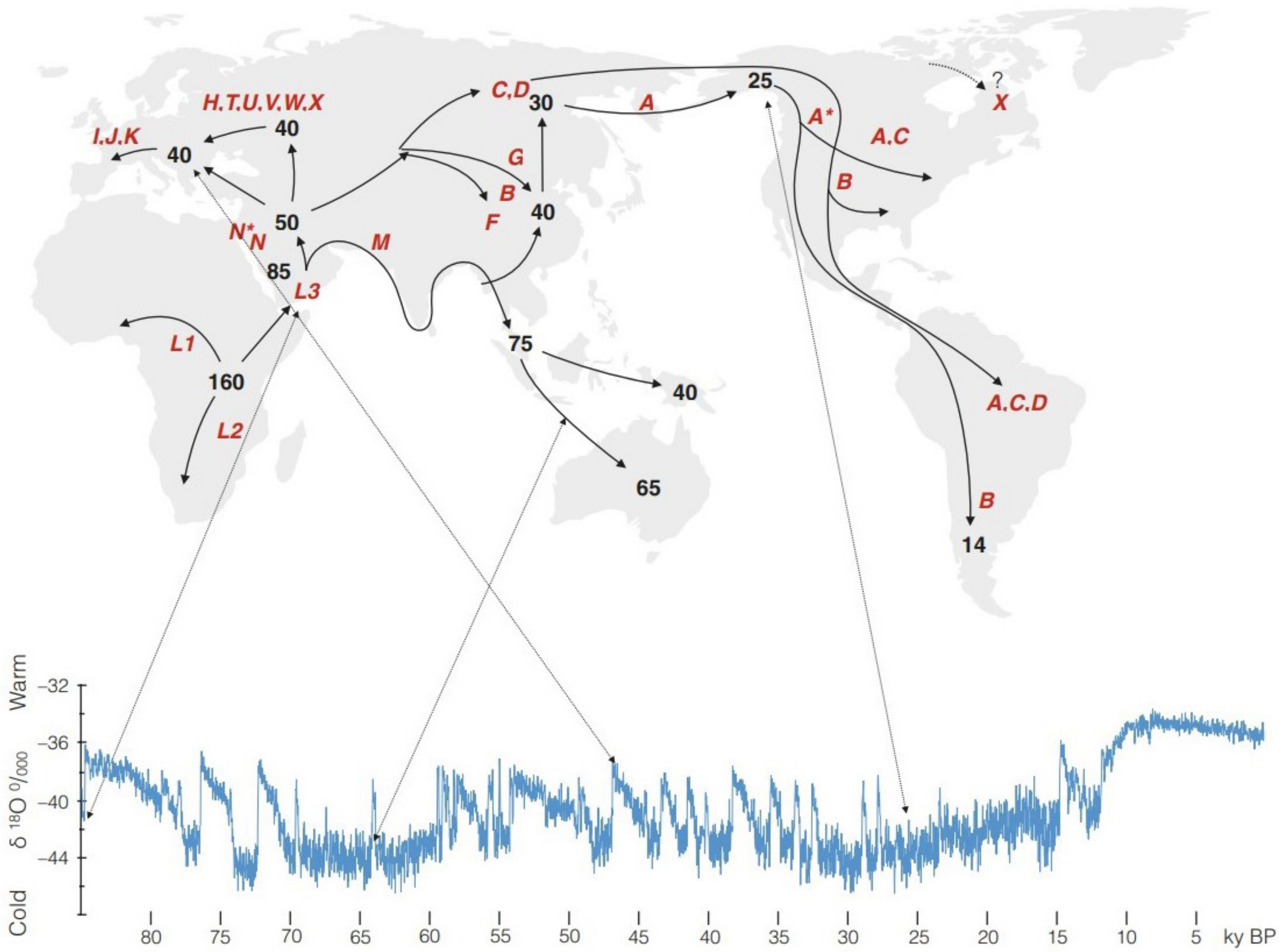
The Early Holocene lasted from around 11,700 to 8,236 years ago. It occurred during part of the warmest period of the Holocene—the Holocene thermal maximum



Location of NGRIP2 ice core sample, indicated in bold font.

Mike Walker, Martin J. Head, John Lowe, Max Berkelhammer, Svante Björck, et al, "Subdividing the Holocene Series/Epoch: formalization of stages/ages and subseries/subepochs, and designation of GSSPs and auxiliary stratotypes," *Journal of Quaternary Science* 34 (2019): 176.

from around 11,000 to 7,000 years ago. The ending date of the Early Holocene coincides with a meltwater event in Canada when a glacial ice sheet collapsed. The classification of the Early Holocene as the Greenlandian Stage/Age refers to the ice core in Greenland, NGRIP2, which bears evidence of the onset of the Holocene at a depth of 1,492.45 meters into the ice.

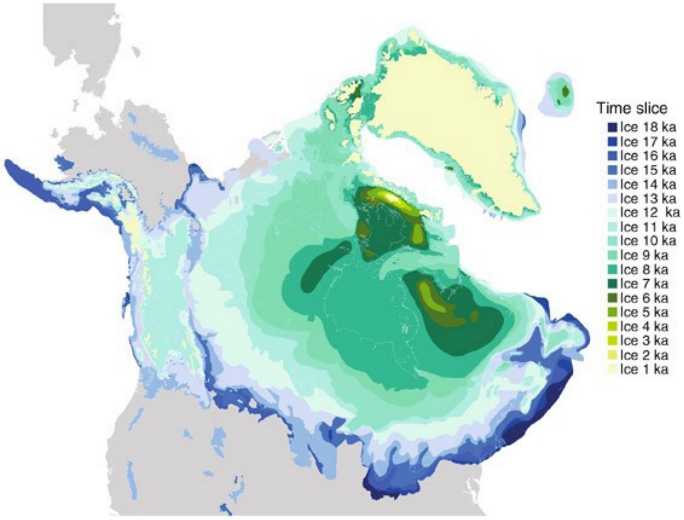


Peopling of global regions in the last 100,000 years. Numbers indicate thousands of years (ky) before present (BP). Graph on bottom shows proxy indicator of warm and cold periods.

Franz Mauelshagen, "Migration and Climate in World History," 415.

By the start of the Holocene, humans had already migrated out of Africa across Eurasia to the east and west, to Australia, and to the Americas from the north to the south.[28] In other words, humans were already all over the globe. Therefore, the question of how the transition from the Pleistocene to the Holocene intersected with human life must be answered at the local level because no two places in the world are identical. New research has investigated human practices during the Early Holocene in the Iberian Peninsula at the western edge of Europe and in Madagascar off the east coast of Africa, to take two examples.[29] In North America, the beginning of the Holocene appears to coincide with the extinction of dozens of large mammals. One popular explanation for their disappearance is that humans adapting to the new conditions of the Early Holocene hunted them to extinction. Other scholars point to a lack of evidence for that hypothesis and suggest that the animals' decline was the result of their own inability to adapt to a changing climate.[30]

These few examples highlight that climate change was not the same everywhere on the globe, and even in a specific location, the interactions between climate, nature, and humans were complex. Local environmental conditions and factors that are unique to individual human communities make it necessary for scholars and scientists to study specific geographical regions in depth. Generalizations have limited accuracy. Broadly speaking, though, during the Early Holocene humans still largely hunted and maintained mobile communities. Agricultural practices—including staple crop production in the Americas—increased, but the large agricultural societies typically



The retreat of the Laurentide Ice Sheet over the past 18,000 Years.

Dyke, A.S. (2009). Laurentide Ice Sheet. In: Gornitz, V. (eds) *Encyclopedia of Paleoclimatology and Ancient Environments*. Encyclopedia of Earth Sciences Series. Springer, Dordrecht.

Bronze axe heads from third millenium BCE Mesopotamia.

Source: The Sulaimaniya Museum, Iraq

counted as the first of their kind in human history did not emerge until the Middle Holocene period.[31]

The Middle Holocene

The Middle Holocene, also known as the Northgrippian Stage/Age, began around 8,236 years ago and ended 4,250 years ago. Its name also refers to an ice core from the same location in Greenland bearing evidence of cooling temperatures compared to the Early Holocene. The beginning date of the Middle Holocene, 6200 BCE, coincides with the collapse of the glacial **Laurentide Ice Sheet** in Canada, which accelerated the flow of icy water into the ocean in what is known as a "meltwater event." Despite the flow of cold water into the ocean, temperatures in the Northern Hemisphere remained high for another thousand years. Around 7,300 years ago, between 5300 and 3700 BCE, temperatures began to decline due to weakened solar energy, among other factors. Known as the Mid-Holocene Transition, this period marks a turn toward the cooler temperatures that would continue in the Late Holocene.[32]

During the years of the Middle Holocene, humans lived—as before—in diverse environmental and social conditions around the globe. In Eurasia, the Middle Holocene saw the development of some practices that have traditionally been regarded as key historical markers for humanity. In the later centuries of the Middle Holocene, which historians consider as the transition from the Stone Age to the Bronze Age, humans began utilizing metal instruments in new ways. Larger and more socially stratified agricultural societies also emerged, including, most notably, the Mesopotamian civilization around 5,500 years ago, or 3500 BCE. Scholars continue to investigate the role of climate in the development of Eurasian settlements during the Middle Holocene, but it appears clear that local social dynamics—and not only climate—influenced the fates of burgeoning human communities.[33]

The Late Holocene

The Late Holocene, also known as the Meghalayan Stage or Age, started approximately 4,250 years ago and extends to the present day. The term"Meghalayan" originates from a cave in northeastern India, where mineral deposits provide evidence of a significant climatic event that marks the onset of this period. At a time known as the Holocene Turnover, roughly 4,250 years ago, abrupt changes in atmospheric and oceanic conditions led to sudden alterations in weather condi tions around the world.[34] The question of whether or not the Late Holocene has ended and given way to the Anthropocene is a matter of debate. Some scholars favor an official adoption of the Anthropocene and a starting date of 1950, which would then mark the end of the Late Holocene and the Holocene altogether.[35] However, in 2024 the International Union of Geological Sciences rejected a proposal to formally name the Anthropocene as a new geological epoch.

With the Late Holocene's starting date of roughly **2250 BCE**, it spans a period of human history that is well documented. Moreover, historical narratives of various parts of the world date back to the start of the Late Holocene or earlier. The challenge of evaluating these historical records and narratives in relation to the climate history of the Late Holocene will be discussed in the next part of Section II.

Sediment sample from Mawmluh Cave, Meghalaya, northeast India, showing the position of the 4.2-ka event.

Source: Mike Walker, et al, "Subdividing the Holocene Series/Epoch," 178.

CLIMATE AND THE DEVELOPMENT OF HUMAN CIVILIZATIONS

A good start for beginning to understand the relationship between climate history and human history is to try to synchronize the timelines of our narratives of the two histories. Yet, even this seemingly simple task has pitfalls. One pitfall is that scholarly fields in climatology typically date past events—as we have been doing so far in this section—according to how long ago they happened, using 1950 CE or 2000 CE as the present. So, the Holocene began 11,700 years ago, using the year 2000 CE as the marker for the present.

In our narratives of human history, on the other hand, Western scholars, along with global scholars engaging with Western traditions, use a calendar system that designates years based on a pivotal historical event: the estimated birth of Jesus Christ, marked as year 1. This system counts years back ward from this point for events before it and for ward for events after, leading to significant his torical milestones such as the founding of the Roman Empire around 27 BCE, the American Revolutionary War's conclusion in 1783 CE, and China's hosting of the Olympic Games in 2008 CE. So, according to this system, the Holocene began in 9,700 BCE. As confusing as it may be, scholars and students engaged in the interdisciplinary task of studying climate history and human history to gether need to hold two dating systems in their heads simultaneously and be ready to make quick calcula- tions. For instance, the Late Holocene began 4,250 years ago which is the same as 2250 BCE.

Another pitfall when trying to synchronize climate history and human history is that it is hard to conceptually grasp the different scales of the historical timelines for each. The Pleistocene is only one of the most recent geological epochs, and it lasted a miniscule 2.6 million years out of the Earth's four-billion-year his tory.[36] Yet, from the perspective of human life, 2.6 million years is a staggering amount of time that dwarfs the en tire history of humanity. (Because climatologists deal with such large numbers, it is conventional to round to a number such as 2.6 million.)

One may also notice that sometimes scholars in climatology use 1950 CE and sometimes they use 2000 CE as the date they count back from when dating an event that happened many thousands of years ago. While fifty years feels like a lifetime in our daily

The spread of agriculture in the Near East.

Source: *Teaching the Middle East*, The University of Chicago Library

experience, it is largely within the margin of error when estimating the dates of past climatic events throughout the Holocene. So, the student and the scholar must keep in mind that when dealing with human history since the so-called start of civilization in **Mesopotamia** around 3500 BCE, the whole of human history from then until now falls within a tiny sliver of climate history. Conversely, climate typically changes at such a slow rate that some scholars have proposed that people throughout history were largely unaware of any changes that might have occurred during the span of a single human life.[37]

This next part of our discussion will examine human history from the rise of the Mesopotamian civilizations around 3500 BCE to the era known as the late **Middle Ages** in Europe around 1350 CE. In terms of climate history, this timespan runs from the last centuries of the Middle Holocene into the Late Holocene, just before the onset of the **Little Ice Age (LIA)**.

Early Agrarian Societies

Traditionally, scholars have counted the development of large agrarian societies to be a milestone in human history. Although humans had cultivated some crops throughout the Holocene, starting around 3500 BCE and in the centuries following, new agriculturally based societies sprang up with unprecedented features, and these societies were larger than previous human settlements. They remained stationary over long periods of time and made striking modifications to their local environments, including the building of impressive structures. People in these societies had different roles to play and formed classes. These societies developed forms of writing, initially for recordkeeping, that were often employed for other purposes as well, including storytelling. In the past, scholars have made a value judgement that the emergence of features such as these represents the dawn of human civilization. However, more recently, some scholars have critiqued the practice of ranking civilizations based on such markers.[38]

Because the early agrarian societies had an obvious connection to nature through agriculture, drawing connections between the societies and climate can appear deceptively simple. According to this line of thought, if a society relied on agriculture, any climatic impact on crops should mark a direct tie between climate and human society. In reality, the relationship between humans and the broader natural world

Bronze head depicting an Akkadian ruler.

Egyptian artifact from around 3100 BCE.

Source: Ashmolean Museum, University of Oxford

has never been this simple.[39] The complexity of the interactions between humans and climate can be seen in even a brief glimpse at some of the more famous early agrarian societies that arose in Mesopotamia, Egypt, India, and the Americas.

Mesopotamia

Mesopotamia, located in and around the current national boundaries of Iraq, was home to several of the most influential empires in Eurasian history. The region received more rainfall during the establishment of its early societies than it does today. The region's water supply came from the Tigris and Euphrates Rivers and was augmented by human-built irrigation systems. The Sumerians in Mesopotamia developed writing, a priestly class, and planned cities between around 3600 and 3000 BCE.

A notable climate-related event occurred in Mesopotamia around 2250 BCE, the aforementioned boundary between the Middle and Late Holocene. In Mesopotamia and nearby regions, the abrupt climate changes of that moment may have resulted in cooler and dryer conditions.[40] The ruling empire in Mesopotamia at the time was the **Akkadian Empire**, which had been established roughly a century before, around 2334 BCE. However, the empire collapsed by around 2218 BCE. Was climate change and drought the cause of its demise? Some scholars argue that it was, and they point to societal collapses at the same time in Egypt and other regions as well.[41] Others, however, take a more geographically focused approach and have argued that even within Mesopotamia individual cities reacted differently to the changing climate, and that not all cities in the area even declined.[42] This scholarly debate shows the challenges of harmonizing data from the archives of nature and the archives of society. It also shows that scholars may get different results depending on the scale or scope of their inquiry. The picture changes depending on whether one looks at empires from a broad perspective or examines individual cities up close.

Remains of Mohenjo-Daro, a city in the Indus Valley built around 2500 BCE.

Source: Woods Hole Oceanographic Institution. Ancient Indus Valley Civilization & Climate Change's Impact (whoi.edu)

Egypt

Not far from Mesopotamia, another powerful society arose on the **Nile Delta** in Egypt. The Egyptians utilized their own form of writing by around 3200 BCE and built colossal monuments, including iconic pyramids by 2500 BCE. The climatic changes around 2250 BCE have also attracted the attention of scholars of Egypt where the onset of drought is often linked to the collapse of the Old Kingdom (c.2700–2200 BCE) in Egypt.[43] Due to well-preserved records of ancient Egyptians' irrigation practices and manipulations of the Nile River, scholars also study Egypt for insights into the history of human efforts to protect sustainable farming conditions.

India

The Indus Civilization in the **Indus Valley**, located in modern Pakistan, was one of the largest and most developed societies in the world. Scholars have divided the civilization into three periods. The Early Harappan phase lasted from around 3200 to 2600 BCE and was characterized by population growth, new settlements, and urbanization. The Mature Harappan phase from 2600 to 1900 BCE saw the most advanced features of the society, including highly stratified social groups, complex city planning with gridded streets and drainage, and trade between multiple villages and urban centers. During the Late Harappan phase from 1900 to 1000 BCE, populations moved out of urban centers in favor of smaller village settlements. Scholars do not know the causes of the decline of the once powerful society. A lack of evidence of military conflict leaves open the possibility that climatic factors played a role. Once again, some scholars point to 2250 BCE as a key turning point in the history of the Indus Civilization.[44]

Olmec Figure from the ninth to twelfth century BCE.

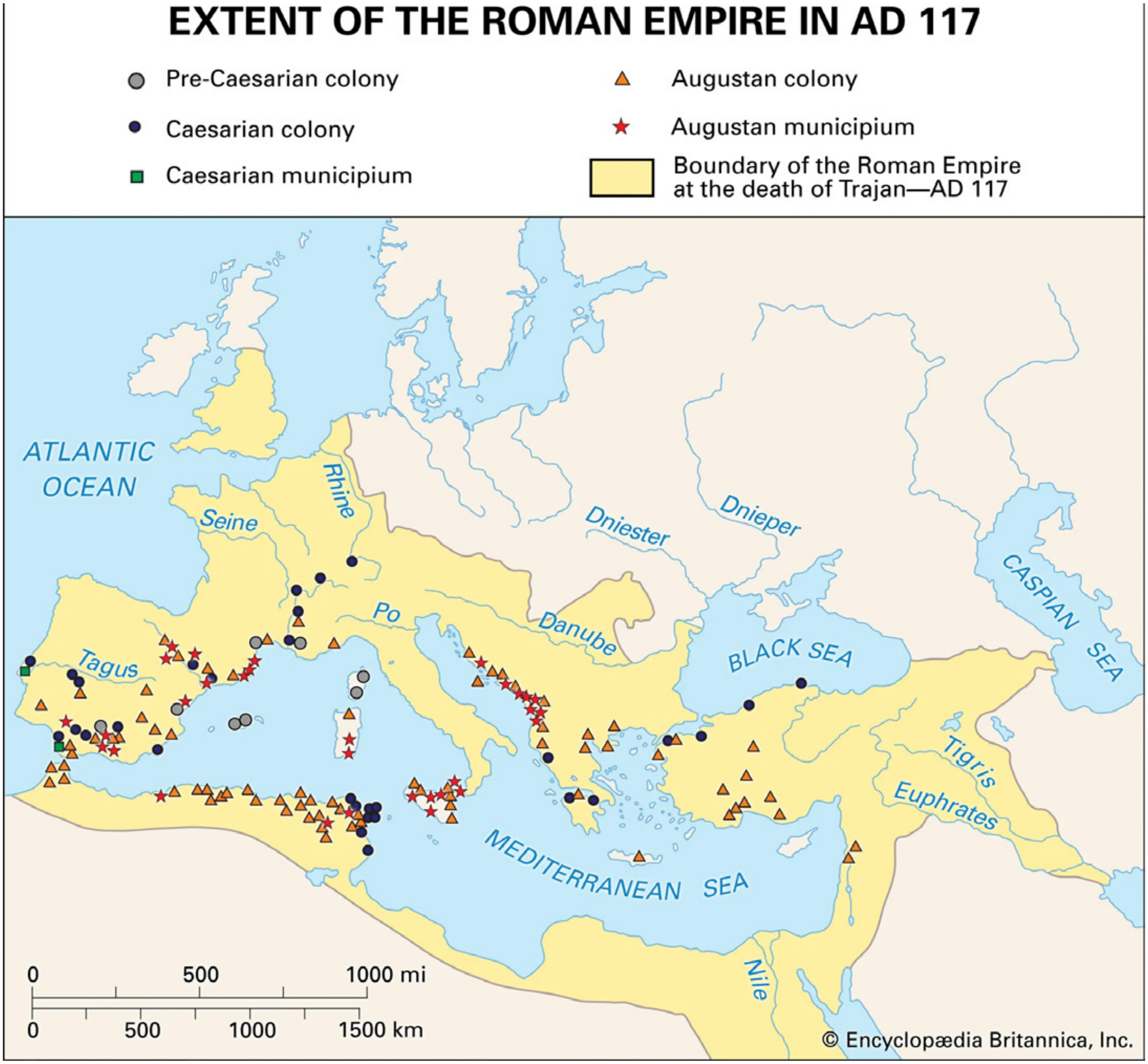
Source: The Metropolitan Museum of Art.

The Americas

In the Western Hemisphere of the globe, humans formed societies with many of the same features as the early agrarian societies of Mesopotamia, Egypt, and India. These societies emerged in the region of the **Andes Mountains** along the west coast of South America in modern-day Peru. They also emerged in **Mesoamerica**, a large region spanning parts of Mexico and Central America. The Andean civilizations included the Chavín culture from around 900 to 200 BCE, which was followed by a series of societies until the Inca Empire consolidated control over the entire region in the 1400s CE. In Mesoamerica, the Olmec ruled from around 1200 to 400 BCE, followed by various societies, including the Maya and the Aztec. Climatologists are reconstructing the climate history of the Americas, an environmentally diverse part of the world with high mountains and extensive vegetation. As in the Eastern Hemisphere, the Holocene in the Americas was marked by fluctuating climatic conditions. Scholars continue to investigate the relationship between the climate and human life in the Americas.[45]

The Mediterranean World During Antiquity

Alexander the Great (356–323 BCE), a Greek king from Macedonia, conquered Egypt and Persia and then traveled as far as India with his army. His conquests established a shared common language, Greek, and shared cultural points of reference throughout all the



Map of the Roman Empire in 117 CE.

regions he visited. The Roman Empire inherited the region that Alexander had united, and at its greatest extent around 117 CE, the Romans ruled much of Alexander's realm to the east and even more land to the west on both the African and European boundaries of the Mediterranean Sea. Greek and Rome have had an enduring legacy in the history of the Western world, including in the Islamic empires that governed much of the former Roman Empire.

Scholars and public audiences have been interested in discovering if climate trends played a role in the rise and fall of the Greek and Roman Empires. There is not a conclusive answer to this question. It is possible to make compelling links between important eras in imperial history and the broadest of climatic reconstructions. From the eighth to fifth centuries BCE, the Mediterranean region experienced cooler and wetter conditions compared to earlier times. These conditions supported productive agriculture and population growth, which likely contributed to the rising power of Greek city-states. Similarly, during the height of the Roman Empire, from approximately 150 BCE to 250 CE, stable climate and consistent rainfall in key regions of the empire fostered agricultural prosperity and sustained Rome's dominance.[46]

Mosaic depicting Alexander the Great, c. 100 BCE.

Ring from France, c.450–525.

Source: The Metropolitan Museum of Art

It is fair to question, however, how meaningful it is to sketch correlations between well-established historical narratives based on the history of empires and such geographically and chronologically broad climate reconstructions. Does adding climate to an essentially unchanged narrative of the rise and fall of the Greek and Roman Empires explain anything new about human history or the history of nature? Even more problematic for the approach of linking large-scale social histories with large-scale climate history, is the question of whether or not there are actually any causal connections between the two. An event as massive as the rise of the Roman Empire has an almost incalculable number of causal factors that brought it into being. Did climate truly impact any of those factors? A promising direction for climate research is to investigate if, on a smaller scale, climate conditions impacted people's daily lives and in what ways it did so. That type of research might allow scholars to connect the dots between local climate conditions and large-scale social events.

536 CE: The Worst Year to Be Alive?

The collapse of the Western Roman Empire, a gradual process that is often tied to the date of 476 CE, marks a defining moment in the traditional narrative of European or Western history. The fall of Rome represents the transition from antiquity to the Middle Ages or the medieval period. Traditionally, the Middle Ages are viewed as lasting until 1450 or 1500 when the so-called early modern period began. In this telling, the Renaissance, running from around 1300 to 1600, is the cultural event that bridges the transition from the medieval to the early modern period. The Middle Ages—especially between the fall of Rome and the Renaissance—have long been described as the Dark Ages because of a supposed decline in culture and technology. That entire narrative is increasingly recognized as being unique to European history and as only telling part of the story, even for Europe.[47]

Incorporating climate history into the history of medieval Europe presents an opportunity for re-shaping how we tell the story of Europe's history. Or, climate could serve merely to offer a slightly different vantagepoint for viewing the same story. A climate event that nicely fits the narrative of the fall of Rome ushering in a dark era of human history is the climate anomaly of **536 CE**. Volcanic eruptions served as the forcing factor that ushered in a cold "year without a summer" in Europe in 536 CE, and several of the coldest summers in the Northern Hemisphere over the next decade and a half. On the basis of sources from the archives of nature and archives of society, scholars have debated whether 536 CE was a short-term blip on the radar or the start of a longer period of significant cooling. Compelling arguments based on available evidence can be made that 536 CE was very historically

Xia (c. 2070–1600 BCE)
Shang (c. 1600–1050 BCE)
Zhou (1050–256 BCE)
"Spring and Autumn" period (770–476 BCE)
Warring States period (475–221 BCE)
Qin (221–207 BCE)
Han (207 BCE–220 CE)
Three Kingdoms (220–280 CE)
Jin (265–420 CE)
Northern and Southern Dynasties (420–589 CE)
Sui (581–618 CE)
Tang (618–907 CE)
"Five Dynasties and Ten States" period (907–960 CE)
Song (960–1279 CE)
Yuan (1271–1368 CE)
Ming (1368–1644 CE)
Qing (1644–1911 CE)

Dynasties of Imperial China.

Source: Quansheng Ge, Zhixin Hao, Jingyun Zheng, and Yang Liu, "China," 190.

significant, or that it was not very significant.[48] When historian Michael McCormick boldly asserted that 536 CE was the worst year to be alive, the claim caught fire with media outlets.[49] It appears to have resonated with a public that still views the early Middle Ages as a rough time to be alive.

The Climate in China and the Mandate of Heaven

Telling the history of climate in China presents a similar challenge to that of incorporating climate into the history of Europe. A narrative of Chinese history organized largely around political history and dynasties already exists. Will incorporating climate history offer new perspectives on the history of Chinese dynasties? Will it offer an alternative way of framing or organizing the story of China's history altogether? Ultimately, time will tell, but scholars of climate and society in China are already making nuanced and compelling discoveries.[50]

China is a uniquely interesting and significant subject for the combined study of climate and human history. China has a remarkable collection of human records that date back to the Shang dynasty (c.1600–1050 BCE). Scholars have analyzed human records and concluded

A bell from the Zhou Dynasty, c. 500 BCE.

Source: National Museum of Asian Art, Smithsonian

that over China's history, warming periods coincided with population growth.[51] Other scholars have studied plant growth throughout the Holocene and determined that from around 2000 to 1600 BCE agricultural practices and a complex society allowed people to withstand the stress of a dry spell in China.[52] The analysis in these studies moves beyond a simple correlation between bad climatic conditions and social turmoil.

Even when addressing the matter of whether climate impacted the collapse of dynasties, another recent study offers a complex explanation based on robust methodologies. It incorporates a cultural idea known as the **mandate of heaven** that became established during the Zhou dynasty (1046–256 BCE). The mandate of heaven was the concept that external conditions such as prosperity in the land and perhaps even favorable weather validated the authority of the current ruler. The study accounts for this unique cultural feature of Chinese history in assessing how climate may have influenced some of the many factors that created stability for a ruler.[53]

Portuguese map of the world from 1573.

THE LITTLE ICE AGE (LIA) AND THE COLONIAL WORLD

The Little Ice Age (LIA) is one of the most well-known and well-researched climatic events of the Late Holocene. Typically dated around 1300–1850 CE, the LIA was a period of cooler global temperatures on average. It is important to note that the LIA did not impact the entire globe evenly—an individual location may not have been as cold as somewhere else. The cooling of the LIA was also not consistent over the centuries it spanned. Yet, during those years, temperatures were cooler often enough and cooler in enough places that they brought down global temperatures on average.

The years of the LIA spanned some very important developments in human history. The 1300s, at the beginning of the LIA, witnessed the so-called Black Death in Europe, a devastating plague that decimated the European population. Climatic conditions may have played a role in the timing of the outbreak of the plague, and subsequent outbreaks, which spread via trade networks spanning Eurasia.[54] The European population rebounded, however, and was still growing when European states, starting with the expedition of Christopher Columbus in 1492, made contact with the Western Hemisphere and immediately began a program of conquering land and people under the system of **colonialism**. The LIA, therefore, provided the climatic backdrop to foundational events in the shaping of the modern world. And how the LIA impacted the formation of the modern world is a matter of historical interpretation.

Phases of the LIA

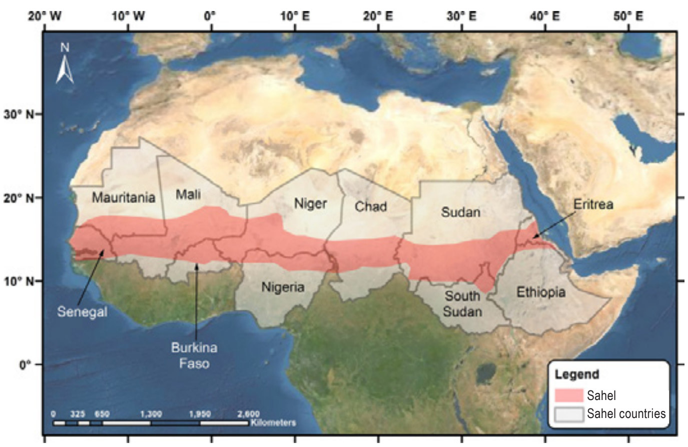
The LIA does not fit neatly into distinct time periods: instead of being a single, unified event, it was a series of climatic shifts that occurred over several centuries.The leading causes of the LIA were volcanic forcings and periods of lower solar activity identified by a lack of sunspots. Those occurrences are known as **solar minima**, and they happened in 1280–1350 (Wolf minimum), 1645–1715 (Maunder minimum), and 1790–1820 (Dalton minimum). The minima roughly corresponded to three periods of glacial advances in the European Alps that took place from the late 1200s to around 1380, from the 1580s to around 1660, and from around 1810 to 1860.[55] In other words, cooling during the LIA occurred in three distinct periods. The effects of this cooling were particularly noticeable in the Northern Hemisphere.

The LIA around the Globe

The LIA occurred at a time when humans around the globe became connected to an unprecedented degree. European contact with the Americas made the Atlantic Ocean a route heavily traveled by humans. Europeans were also trying to discover new routes to reach parts

Drawing by Tang Yin (1470–1524) from Ming China.

Source: National Museum of Asian Art, Smithsonian



The Sahel region in Africa.

Source: Mohammad Al-Saidi, S.A. Saad, Nadir Ahmed Elagib, *Environment Development and Sustainability.*

of the globe that had trading partners or resources that Europeans coveted. Eventually, new routes opened around the southern boundaries of Africa and South America, and regular travel spread to the Pacific Ocean. Other attempted routes, especially near the North Pole, never materialized.

Europeans were relatively ill-equipped for making sense of the geography that they encountered. Deeply entrenched erroneous ideas about climate and geography that they had inherited from antiquity were only slowly discarded. Europeans expected that the whole globe shared the same climatic conditions they had experienced around the Mediterranean Sea. As Europeans encountered more of the world, they struggled to incorporate new regions into a revised conception of the Earth's climate.[56] The world that Europeans first discovered was the world during the LIA. Colder than usual conditions also impacted the sites of interaction between Europeans and Indigenous populations during the first centuries of contact.

European Empires

Throughout the 1500s, several European states built overseas empires via various measures of conquest.[57] These empires included the Portuguese, the Spanish, the Dutch, the British, and the French. Impacts of the LIA influenced the trajectories of these overseas empires in many ways at home and abroad. A cold spell in the 1590s interacted with cultural and economic conditions in Spain and England in ways that encouraged impoverished Spanish residents to stay in Spain and move to urban centers, while English residents were incentivized to move overseas and participate in settler colonialism.[58] Meanwhile, frozen seas to the north blocked the Dutch Empire in its attempt to find a maritime route to China.[59] Climate alone did not determine the fate of European empires, but it did influence the course of their development in various ways.

Africa

Recent scholarship confirms that the LIA also impacted Africa, though its effects varied across the continent. One outcome of the LIA for parts of Africa was a decrease in rainfall and a rise in drought conditions. In particular, the Sahel region experienced drying events around 1600 as well as from 1800 to 1850.

Asia

Now we turn our attention to Asia, where climatic shifts mirrored some of the challenges and adaptations seen in Europe and Africa. Similar to Europe on the other side of the Eurasian landmass, China experienced cooler temperatures during the LIA as well. Like other parts of Eurasia, China also faced social unrest in the seventeenth century, but eventually managed to diversify its crops and stabilize food prices during this period. In South Asia, the LIA disrupted weather patterns, primarily through weakened monsoon rains, leading to significant climatic changes.[60]

North America

Indigenous survival practices in North America equipped Native Americans for the LIA much better than the settlers from Europe who first arrived between the fifteenth and seventeenth centuries. As the Spanish, English, and French fought each other to gain a foothold in Florida, their navies were repeatedly stymied by the weather itself because of their inability to adapt to and anticipate local conditions. Cold weather in Florida—yes, Florida—during the LIA slowed the process of colonialism there, but also exacerbated violence against local populations. When harsh weather limited available resources, colonists sought to forcibly take food supplies that Indigenous populations had shared during times of plenty.[61] On the western side of North America where Spain claimed vast territories, European newcomers often became reliant on local populations for survival after venturing out to new regions.

From the "Seventeenth-Century Crisis" to the Nineteenth Century

The LIA has been incorporated into a traditional narrative of the **modernization** process that supposedly took place in Europe.[62] The narrative will likely be familiar to many. Its key markers are the Middle Ages as a period of superstition and blind religious fanaticism, the Renaissance as the moment when humans centered themselves in the world, the Scientific Revolution when a modern scientific outlook replaced superstition, with the culminating event being the Enlightenment, when all the pieces were put together and modern society with a democratic state arrived in the world.

This narrative has been disputed by scholars on a variety of fronts, but popular books by some scholars have kept the narrative intact and weaved the LIA into its fabric. In this telling, the middle phase of the LIA with colder average temperatures from around 1560 to 1630 put a strain on superstitious beliefs about the weather and religion. The seventeenth century, which was indeed a violent century throughout Europe, further strained premodern sensibilities about the weather and helped usher in the Enlightenment, when people finally rejected outdated ideas about the weather based on knowledge from antiquity in favor of knowledge based on empirical observation. Additionally, more sophisticated infrastructure and stable governments were able to mitigate the dangers posed by climate change during the last decades of the LIA in the 1800s.

St. Augustine in Florida was settled by the Spanish in 1565. Castillo de San Marco, seen here, was constructed from 1672 to 1695.

This narrative of the LIA appears to resonate with the public based on its popularity, but it does have flaws. As some recent studies have shown, common ideas about the weather cannot be so clearly divided between pre- and post-Enlightenment mentalities.[63] Additionally, from the perspective of the Anthropocene to day, trust that modern statecraft and infrastructure can protect humans from the dangers of climate change appears misguided. Perhaps a more important story about the developments from the seventeenth century to the nineteenth century is the story of the path to the globe's current predicament in which humanity appears bound to exacerbate an ongoing climate crisis. That is the story of the transition from the Holocene to the Anthropocene, which will be taken up in the next section of this resource guide.

SECTION II SUMMARY

- The Holocene began around 11,700 years ago at the end of the last ice age.
- Humans had already spread around the world by the start of the Holocene.
- The Holocene is divided into three stages: the

Early Holocene (11,700 to 8,236 years ago), the Middle Holocene (8,236 to 4,250 years ago), and the Late Holocene (4,250 years ago to the present).

- Humans began forming large agrarian societies around the end of the Middle Holocene.
- It is not clear if climate played a major role in the rise or fall of the Greek and Roman Empires or in most large-scale societal events.
- Integrating climate history into human history can be an opportunity for fresh historical perspectives, but sometimes climate is merely added to existing historical narratives.
- The Little Ice Age (LIA) was a period of cooler global temperatures, on average, from around 1300 to 1850 CE.
- A popular historical narrative describes how humans overcame the LIA by building modern states, but that narrative is complicated by the current conditions of the Anthropocene.

Section III

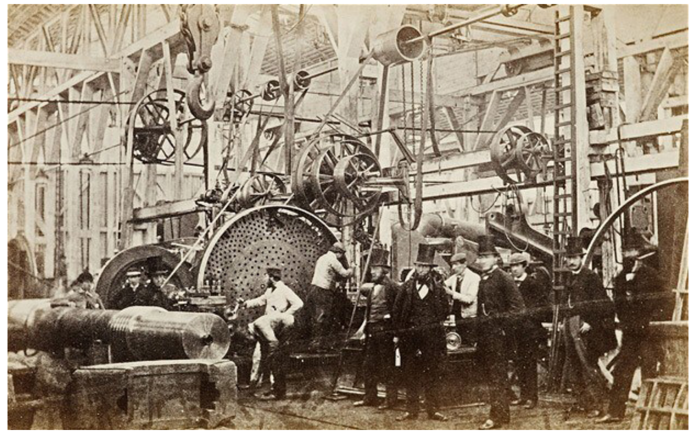
The Anthropocene

SECTION III INTRODUCTION

In one context, the Anthropocene is a technical term denoting that since around 1950, the human impact on the Earth has been substantial enough to leave a mark in the geologic record. The word derives from the Greek *anthropos*, meaning "human," and *kainós*, meaning "new."[64] As a result of the nuclear weapons testing in the 1950s, for example, traces of plutonium-239—an artificial element—can be found in sediment core samples.[65] Other environmental changes linked to human activity, like higher concentrations of CO_2 in the atmosphere, also create unique signatures in the rock strata, or layers of rock, that geologists examine in order to decide the boundaries of various geological epochs.[66]

In a broader context, the "Anthropocene" is linked to contemporary climate change, or global warming, in both scientific and popular discourse. Additionally, news reports and educational content raising awareness of the realities of climate change often refer to the Anthropocene. In popular usage, the "Anthropocene" is a term that evokes the human behaviors that are causing climate change and the impacts of climate change on our world and our lives.

Section III of this guide will explain what the Anthropocene is according to scholars and then will examine the causes and consequences of the Anthropocene. **Paul J. Crutzen** and **Eugene F. Stoermer** first used the term "Anthropocene" in the year 2000. For them, the term designated a new geological epoch, following the Holocene, that recognizes the "central role of mankind in geology and ecology."[67] They proposed the late eighteenth century as a starting date of the Anthropocene to coincide with the popularization of the steam engine and early records of growing greenhouse gas concentrations in the atmosphere.[68] Historians commonly label the same period as the beginning of the **Industrial Revolution**.



A London factory that built ships' boilers and engines, 1863.

Source: Dickens's London: in pictures (telegraph.co.uk).

Since Crutzen and Stoermer introduced it, the Anthropocene has not only become a widely used term, but scholars in geological fields have also moved toward formally recognizing the Anthropocene as a geological epoch. In 2019, an authoritative body of scholars—the Anthropocene Working Group (AWG) acting under the guidance of the Subcommission on Quaternary Stratigraphy (SQS) and the International Commission on Stratigraphy (ICS)—voted in favor of treating the Anthropocene as a formal geological epoch beginning in the mid-twentieth century.[69] Their proposed starting date of around 1950 corresponds to the so-called "**Great Acceleration**" when many of the developments that originated around the start of the Industrial Revolution amplified. While their proposal was not accepted by the International Union of Geological Sciences, the term nonetheless has currency and relevance as a means of describing the era in which humans have significantly impacted the global environment.

The Anthropocene and the climate are interrelated. The **Intergovernmental Panel on Climate Change (IPCC)** defines the Anthropocene as "a proposed new geological epoch resulting from significant human-driven changes to the structure and functioning of the

Earth System, including the climate system."[70] The AWG offers a robust list of phenomena associated with the Anthropocene, many of which are climate-related:

> *an order-of-magnitude increase in erosion and sediment transport associated with urbanization and agriculture;*
> *marked and abrupt anthropogenic perturbations of the cycles of elements such as carbon, nitrogen, phosphorus and various metals together with new chemical compounds;*
> *environmental changes generated by these perturbations, including global warming, sea-level rise, ocean acidification and spreading oceanic 'dead zones';*
> *rapid changes in the biosphere both on land and in the sea, as a result of habitat loss, predation, explosion of domestic animal populations and species invasions;*
> *and the proliferation and global dispersion of many new 'minerals' and 'rocks' including concrete, fly ash and plastics, and the myriad 'technofossils' produced from these and other materials.*[71]

The AWG further notes that many of these changes will last for at least thousands of years and might permanently alter the Earth System.[72] The idea of the Anthropocene allows scholars, educators, and journalists to gather the changes that humans are making to the Earth and the climate and speak of those changes together in a comprehensible way. The Anthropocene consists of many significant changes to the Earth System that are happening simultaneously due to unprecedented human activities.

THE ORIGINS OF THE ANTHROPOCENE

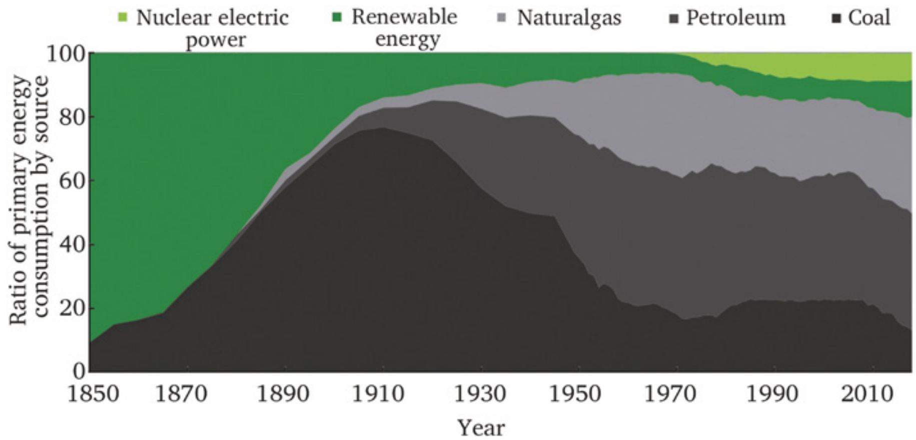
In terms of climatic conditions, the Anthropocene marks a stark contrast to the earlier years of the Holocene. The Holocene began around 11,700 years ago and is best described as an interglacial period. It is merely the latest in a series of many warm periods nestled between two ice ages, or glacials, characterized by significant ice cover across the globe. Scholars cannot precisely predict when the Holocene interglacial would have ended and the next glacial inception would begin.[73] The orbit of the Earth, which influences the interaction between solar energy and the Earth System, has led to previous cycles of ice ages and interglacials. Those cycles correspond to the Milankovitch cycles—patterns of the Earth's movement in relation to the Sun. While scholars do not know exactly when the Holocene would have given way to another ice age, the Anthropocene has disrupted that trajectory.[74] According to existing natural factors, the Earth would currently be headed toward a cooling period, but anthropogenic—or human-related—causes have set the Earth's climate in the direction of rapid warming instead.

The characteristics of the Anthropocene's climate change are also unique compared to climate history during the prior years of the Holocene. A study of temperature variability over the past two thousand years finds that the largest warming trends during that period have occurred since the middle of the twentieth century.[75] In addition to unprecedented levels of warming, another unique feature of global warming during the Anthropocene is its global synchrony.[76] In other words, global warming today is happening more evenly around the globe than past climate change. During the Little Ice Age (LIA), global temperatures were cooler from around 1300 to 1850 on average. One must note, however, that cooler temperatures *on average* does not mean that temperatures were cooler everywhere at once.

During the LIA, global cooling did not have the same geographic consistency that global warming does during the Anthropocene. The reason for this difference is the causes of the climate change. The LIA was caused by solar forcings and volcanic forcings, which had more local impacts than current climate change caused by greenhouse gases spread throughout the atmosphere.[77] The sources of those greenhouse gases are linked to well-known developments in human history.

The Industrial Revolution and the Burning of Fossil Fuels

Discussion of the Industrial Revolution, beginning in the late eighteenth century and flourishing in the nineteenth century, brings to mind large coal-powered factories and soot-covered workers who were sometimes still children. The British writer Charles Dickens (1812–70) popularized this image of the Industrial Revolution through his vivid descriptions of such scenes in his literary works, including *A Christmas Carol* (1843) and *Oliver Twist* (1837–39). The term "Industrial Revolution" was already in use in the 1800s.[78] Even people alive at that time recognized that machine manufacturing and accompanying changes in the



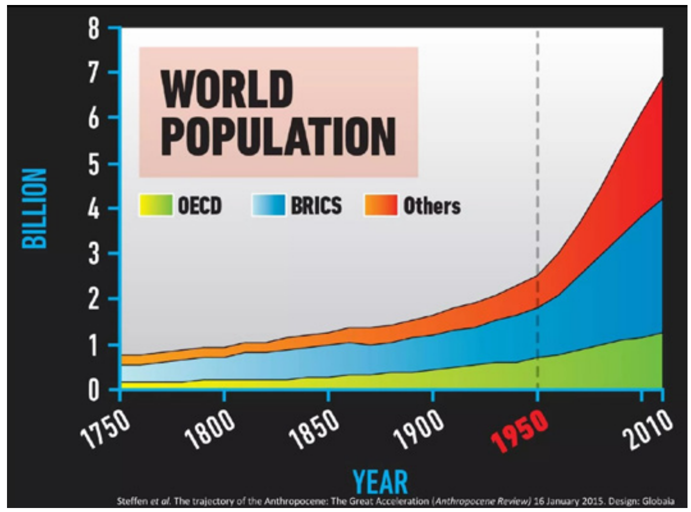
Changes in U.S. energy consumption, 1850–2018.

Source: Yang, et al., "Evolution of energy and metal demand."

Charles Dickens.

economy signaled a major transformation in society.

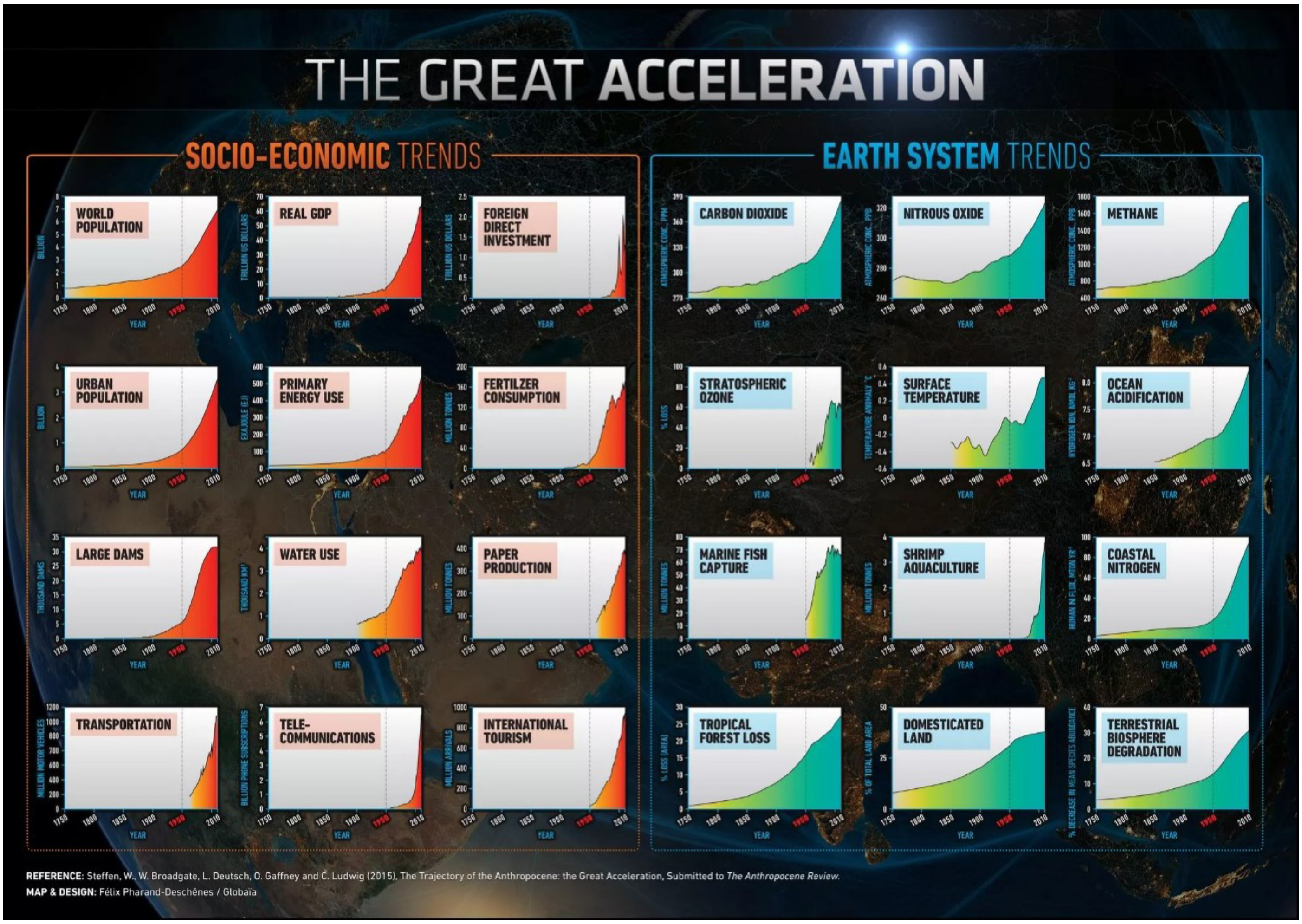
The Industrial Revolution has taken on a new layer of significance in light of the Anthropocene. As mentioned previously, Crutzen and Stoermer argued for making the popularization of the steam engine the starting date of the Anthropocene. Indeed, the introduction of the coal-powered steam engine was a landmark moment in the history of humans burning fossil fuels to produce the energy that powers our machines. Some scholars still argue for the importance of recognizing the era around 1800 to 1850 as being foundational to the Anthropocene.[79] Nevertheless, a scholarly consensus has formed around making the mid-twentieth century, roughly 1950, the starting date of the Anthropocene. This decision was influenced by a more detailed understanding that the Industrial Revolution itself had a historical development. While 1800 might mark the start of the technology that has resulted in large greenhouse gas emissions, the bulk of those emissions came only later.[80]



Global population of humans, 1750–2010. OECD refers to member countries of the [Organisation for Economic Cooperation and Development](). BRICS refers to Brazil, Russia, India, China, and South Africa.

Source: IGBP, "Great Acceleration"

Scholars today are grappling with the challenges posed by climate change during the Anthropocene, and that experience may provide fresh perspectives for reconsidering how historians can tell the history of the Industrial Revolution.[81] In traditional narratives, the Industrial Revolution was a story of the human conquest of nature and of the human capacity for perpetual growth. The conditions of the Anthropocene call both of those narratives into question as it appears that humanity may now be stretching the limits of the Earth's natural resources and resilience. It is possible that in the future, the history of the Industrial Revolution will be less of a success story and more of a cautionary tale.

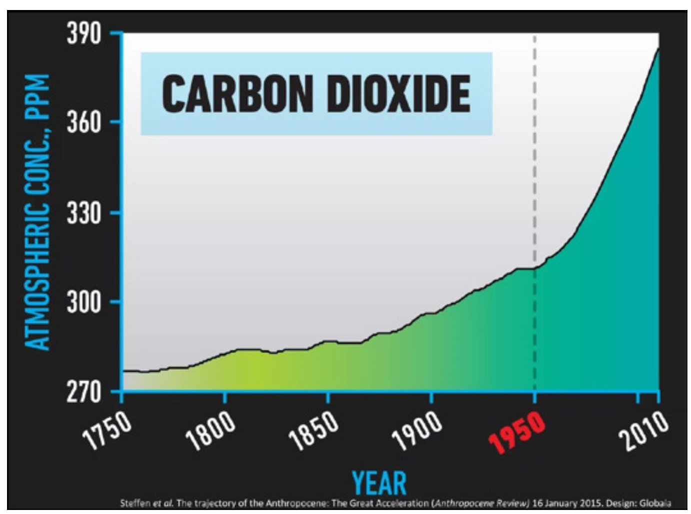


Correlation of socio-economic trends and Earth System trends, 1750–2010.

Source: IGBP "Great Acceleration."

The "Great Acceleration" and the Beginning of the Anthropocene

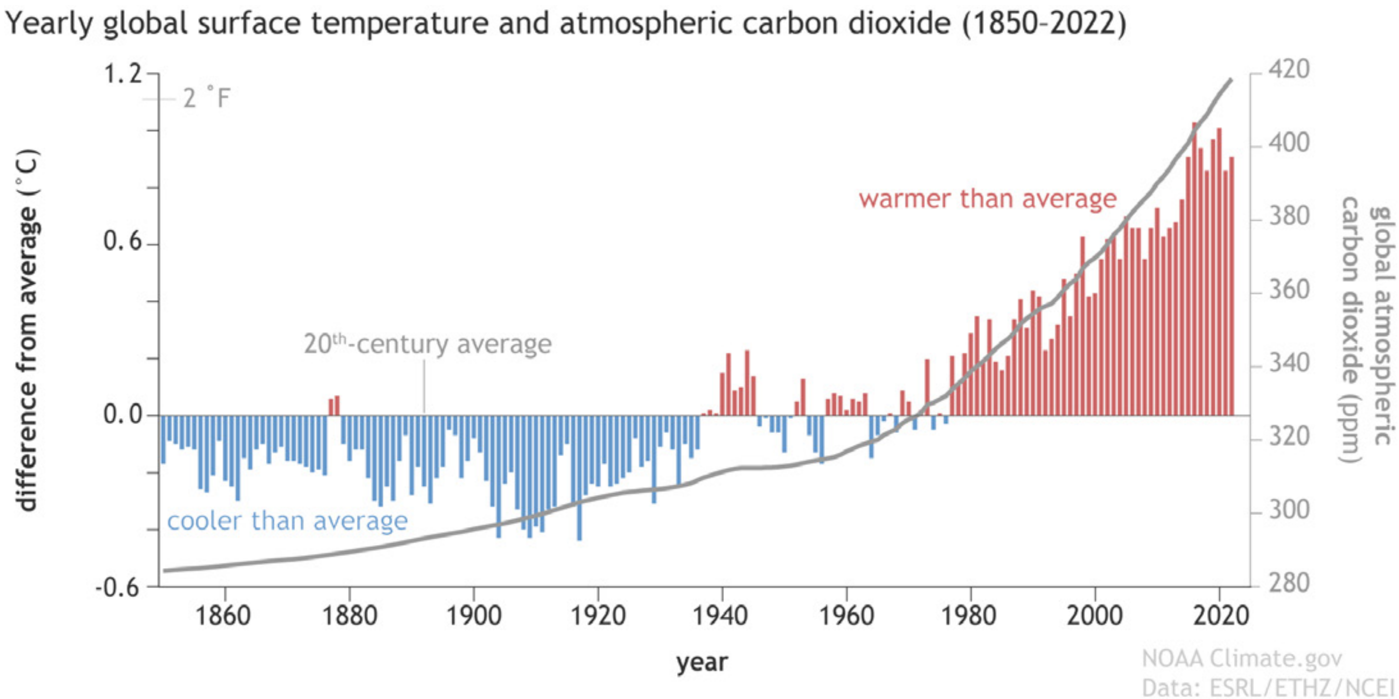
Scholars have recognized that the features of the industrial world that characterize the Anthropocene have not been stable since the start of the Industrial Revolution around 1800. In fact, around 1950 the scale of human activities that altered the Earth System began to greatly increase. In other words, more people began doing more of the things that are changing the planet. Environmental historian **J. R. McNeill** labeled the increase that occurred starting around 1950 as the "Great Acceleration."[82] A key factor in this acceleration, McNeill notes, is the rise in the human population. The human population in 1780 around the start of the Industrial Revolution was less than a billion. By 1930 it had more than doubled to around 2 billion, and between then and now it has nearly quadrupled to more than 8 billion in less time.[83]



Atmospheric concentration of carbon dioxide, 1750–2010.

Source: IGBP "Great Acceleration."

The decades since 1950 are unique in human history in terms of how quickly and how greatly people have transformed the natural world.[84] Humans left little



Correlation between yearly temperatures and atmospheric carbon dioxide levels.

Source: NOAA.

trace of their existence in the world for hundreds of thousands of years before the Holocene began around 12,000 years ago. Since that time, humans have altered landscapes and built monuments that are still visible today. Yet, since the mid-twentieth century, humans have interacted with the planet on a scale that has never been seen before. Although it is difficult to conceptualize, the transformations that humans are currently driving in the natural world match or surpass the natural causes of climate change in the past. Humans are pushing the planet out of the Holocene and into a new geological and climatic epoch.

Eunice Newton Foote.

THE CAUSES OF THE ANTHROPOCENE

New research continues to support the scientific finding that higher concentrations of carbon dioxide and other greenhouse gases in the atmosphere are causing global warming. Scholars are refining their understanding of how this process works and adding precision to their knowledge of how human-caused changes are interacting with the Earth System. For instance, scientists have determined that rising carbon dioxide levels throughout the atmosphere cause different types of changes at different layers of the atmosphere.[85] Temperatures are rising in the troposphere, the lowest layer of the atmosphere that is about seven miles thick, which is where the Earth's weather occurs. Conversely, the higher layers of the atmosphere, where the air is thinner, are cooling. The diverging temperature reactions in different layers of the atmosphere could create instability. New findings such as this confirm that human greenhouse gas emissions are altering the Earth System even as more detailed information complicates the picture of the relationship between rising global temperatures and human behavior.

A graph showing the correlation between higher greenhouse gas concentrations in the atmosphere and rising global surface temperatures only shows part

The 10 largest oil[1] producers and share of total world oil production in 2022

Country	Million barrels per day	Share of world total
United States	20.21	20%
Saudi Arabia	12.14	12%
Russia	10.94	11%
Canada	5.70	6%
China	5.12	5%
Iraq	4.55	5%
United Arab Emirates	4.24	4%
Brazil	3.77	4%
Iran	3.66	4%
Kuwait	3.02	3%
Total top 10	73.36	73%
World total	99.89	

[1] *Oil* includes crude oil, all other petroleum liquids, and biofuels.

Ten largest oil producers in 2022 by nation.

Source: U.S. Energy Information Administration

of the story of the Anthropocene. It does not explain why or how humans are releasing those gases into the atmosphere. To understand that part of the story, one must look at past human behavior in the context of human history. It was only after mechanisms for extracting and burning fossil fuels were developed that people become aware of the challenges of the Anthropocene. It is true that as early as the mid-nineteenth century, the American scientist **Eunice Newton Foote** (1819–88) had argued that atmospheric water vapor and carbon dioxide could interact with solar rays to raise the Earth's temperature.[86] Nevertheless, on a scale that dwarfed the insights of any one individual or group of researchers, humans at that same time were building a global infrastructure that ran, and continues to run, on fossil fuels.

Global Production of Greenhouse Gases

There is a complex process involved in taking carbon stored inside the Earth and releasing it as a gas into the atmosphere. So-called fossil fuels are formed from the remains of living organisms from past geological epochs. These remains are stored within the Earth, so the carbon contained in these fossil fuels does not interact with the Earth's subsystems on the surface of the Earth and influence the climate. Since the Industrial Revolution, however, humans have utilized fossil fuels as a means of producing energy by burning them. In the nineteenth century, coal was burned to power steam machines. In the twentieth and twenty-first centuries, coal has continued to be used to power electric plants, and petroleum, or oil, has been used to power machines, including automobiles. As fossil fuels are burned for these various purposes, carbon and other chemicals within the fossil fuels that had been sitting within the Earth are then released as gases into the atmosphere. Once in the atmosphere, they serve as greenhouse gases and warm the surface temperature of the Earth.

Geography is an important factor in the extraction of fossil fuels and their emission as greenhouse gases. Fossil fuels can only be extracted from locations where they are found. Where they are used has historically depended on which regions of the world have had the wealth and political power to build the infrastructure for them. In the nineteenth and twentieth centuries, the West—mostly regions in the Global North, including Western Europe and North America—industrialized and created cities and transportation systems that required great quantities of fossil fuels to maintain. Corporations and governments from those regions also developed the transnational infrastructure that was required to extract and transport fossil fuels from wherever they were found around the globe to where the fossil fuels were consumed. Today, the countries that sit on deposits of fossil fuels still extract them, while the countries that consume fossil fuels in large quantities have multiplied. If one takes oil as an

The 10 largest oil[1] consumers and share of total world oil consumption in 2021

Country	Million barrels per day	Share of world total
United States	19.89	21%
China	14.76	15%
India	4.79	5%
Russia	3.67	4%
Japan	3.41	4%
Saudi Arabia	3.35	3%
Brazil	2.96	3%
South Korea	2.58	3%
Canada	2.26	2%
Germany	2.13	2%
Total top 10	59.80	62%
World total	96.66	

[1] *Oil* includes crude oil, all other petroleum liquids, and biofuels.

Ten largest oil consumers in 2021 by nation.

Source: U.S. Energy Information Administration

Freeways and smog in twentieth-century Los Angeles.

Source: *Los Angeles Times*

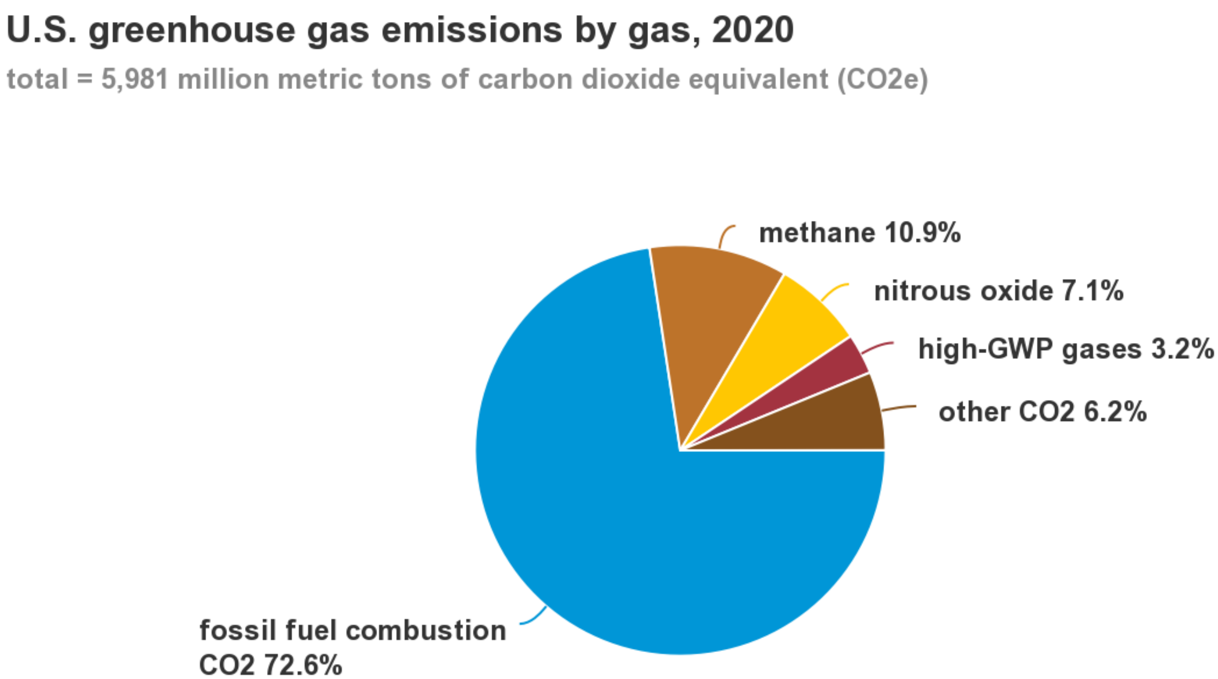
example, the United States was both its top producer and top consumer in the early 2020s.[87]

The History of Oil Extraction

The extraction and possession of fossils fuels by those who want to use them has often been a contested and fraught process.[88] The history of oil extraction in the Middle East, particularly Iraq, in the early twentieth century offers an example of this. When oil was discovered by Europeans in the region shortly before World War I, Western corporations and governments quickly acted to establish a means of controlling the oil. The history of oil in the twentieth century is intertwined with the history of European colonialism

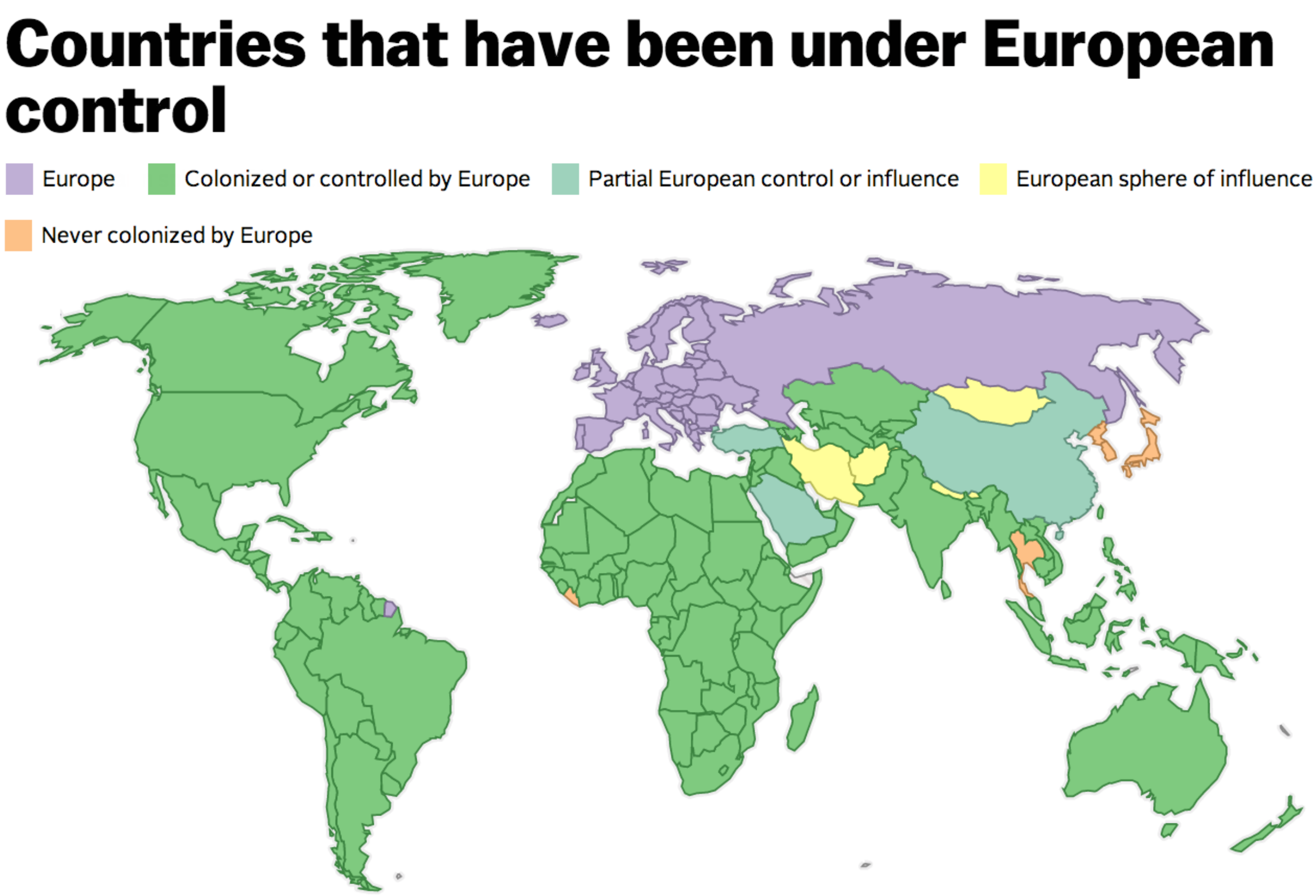
An oil field in Kirkuk, Iraq.

and imperialism. In their dealings with local leaders, Western powers prioritized gaining access to oil over supporting democracy in oil-rich regions.[89] In the Iraqi city of Kirkuk, where oil was discovered in 1927, global geopolitics and British colonial influence shaped the contours of local urban life, straining relationships between different groups within the diverse city's population.[90]



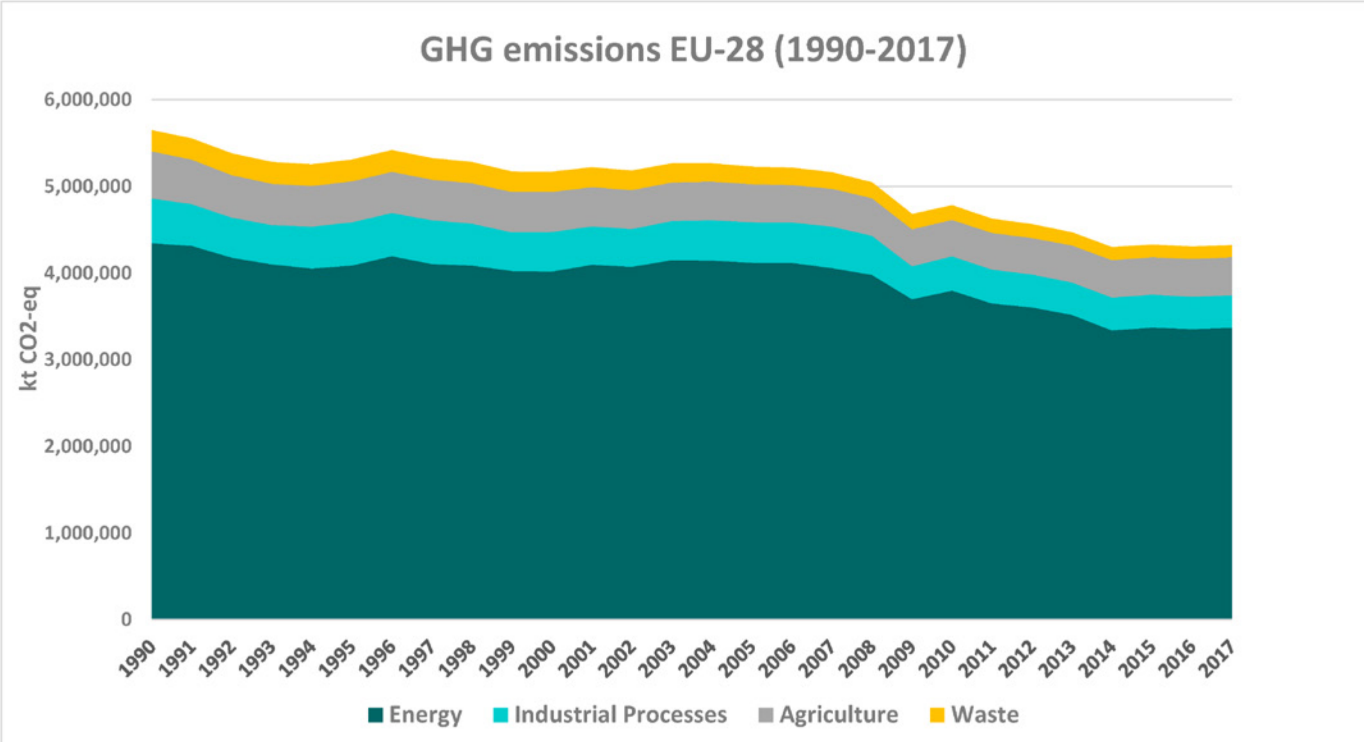
U.S. sources of greenhouse gas emissions.

Source: [U.S. Energy Information Administration]



Map of the globe showing extent of European colonialism.

Source: [Vox]



Greenhouse gas emissions in Europe, 1990–2017.

Source: Mariano González-Sánchez and Juan Luis Martín-Ortega, "Greenhouse Gas Emissions Growth in Europe: A Comparative Analysis of Determinants," *Sustainability* 12 (2020).

The ability and willingness of imperial powers to exert force in order to secure access to oil in the early twentieth century illustrates the complexity of the human side of the Anthropocene. Even before humanity had a widespread awareness of the adverse environmental consequences caused by the burning of fossil fuels, the value of **black gold** spurred some people to build wealth at the expense of other people's freedom and economic viability.[91] Seen in this light, the economic obstacles to addressing climate change today have continuity with the earlier centuries of the Industrial Revolution. The oil industry has a long history of successfully overcoming challenges that threaten its profitability.

Current Sources of Fossil Fuels

The burning of fossil fuels makes up the majority of global greenhouse gas emissions. The **United Nations (UN)** reports that fossil fuels—coal, oil, and gas—comprise over 75 percent of global greenhouse gas emissions and almost 90 percent of carbon dioxide emissions.[92] It should be noted that coal, oil, and gas are sources of fossil fuels that humans still widely extract and utilize. All three sources are still very relevant to the current discussion of greenhouse gases.

Some gases besides those produced by fossil fuels are greenhouse gases as well. These include methane(CH_4) and nitrous oxide(N_2O), which are released in a variety of human activities, including agriculture, landfill use, coal mining, and oil and natural gas production.[93] Yet, in the United States in 2020, methane and nitrous oxide combined for less than 20 percent of greenhouse gas emissions while the burning of fossil fuels contributed over 72 percent of greenhouse gas emissions.[94]

Historic Greenhouse Gas Emissions in the West

We can better understand the historical context of greenhouse gas emissions in the West by considering some important features of human history over the past several centuries. Starting around 1500, European empires spread around the world and created colonies that extracted valuable natural resources and human labor that was used to create great wealth, which was concentrated in Europe itself. When industrialization started around 1800, European nations—and nations such as the United States—used the wealth that they had accumulated to build factories and industrial infrastructure. Western powers maintained relationships with colonial subjects to produce natural resources such

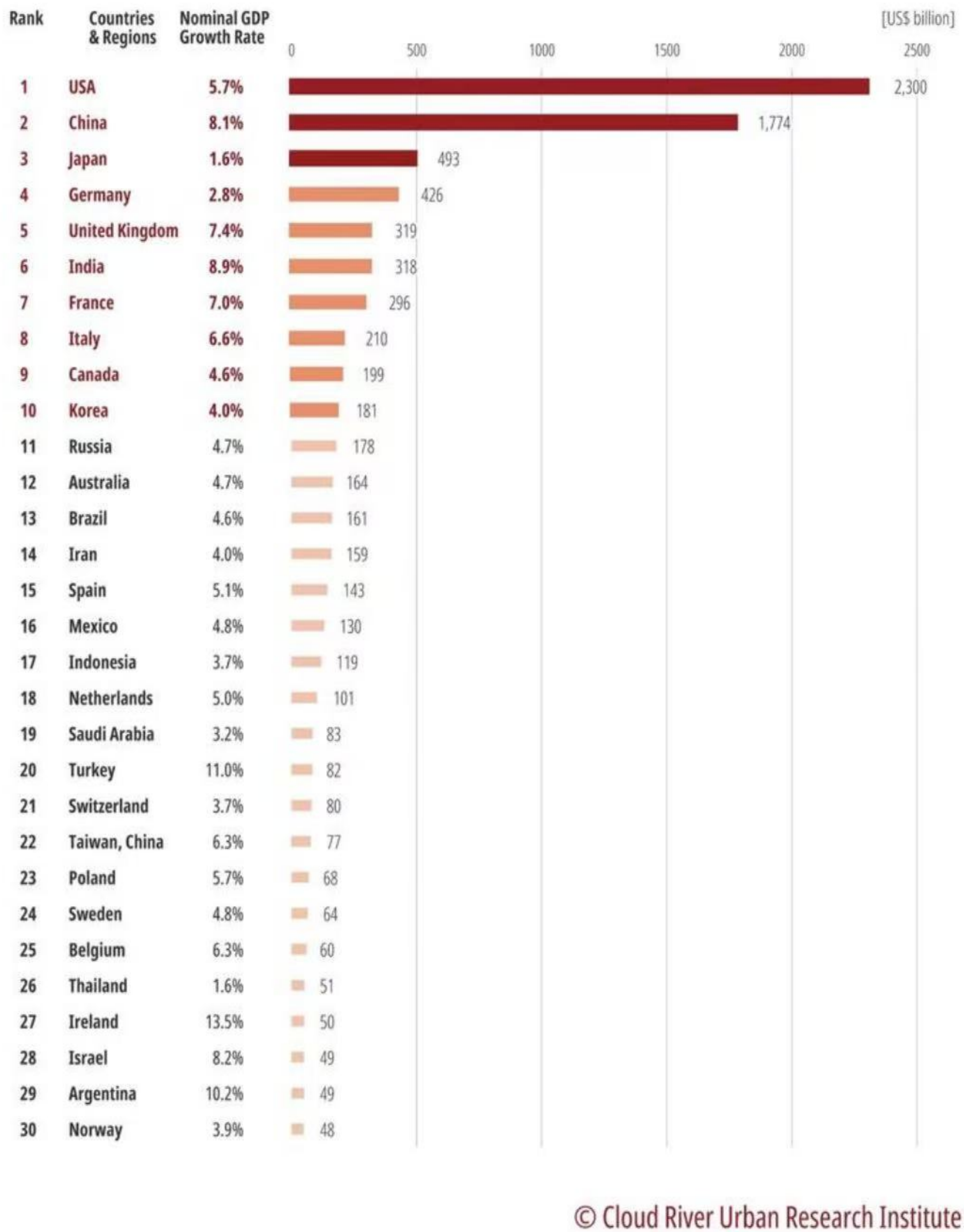
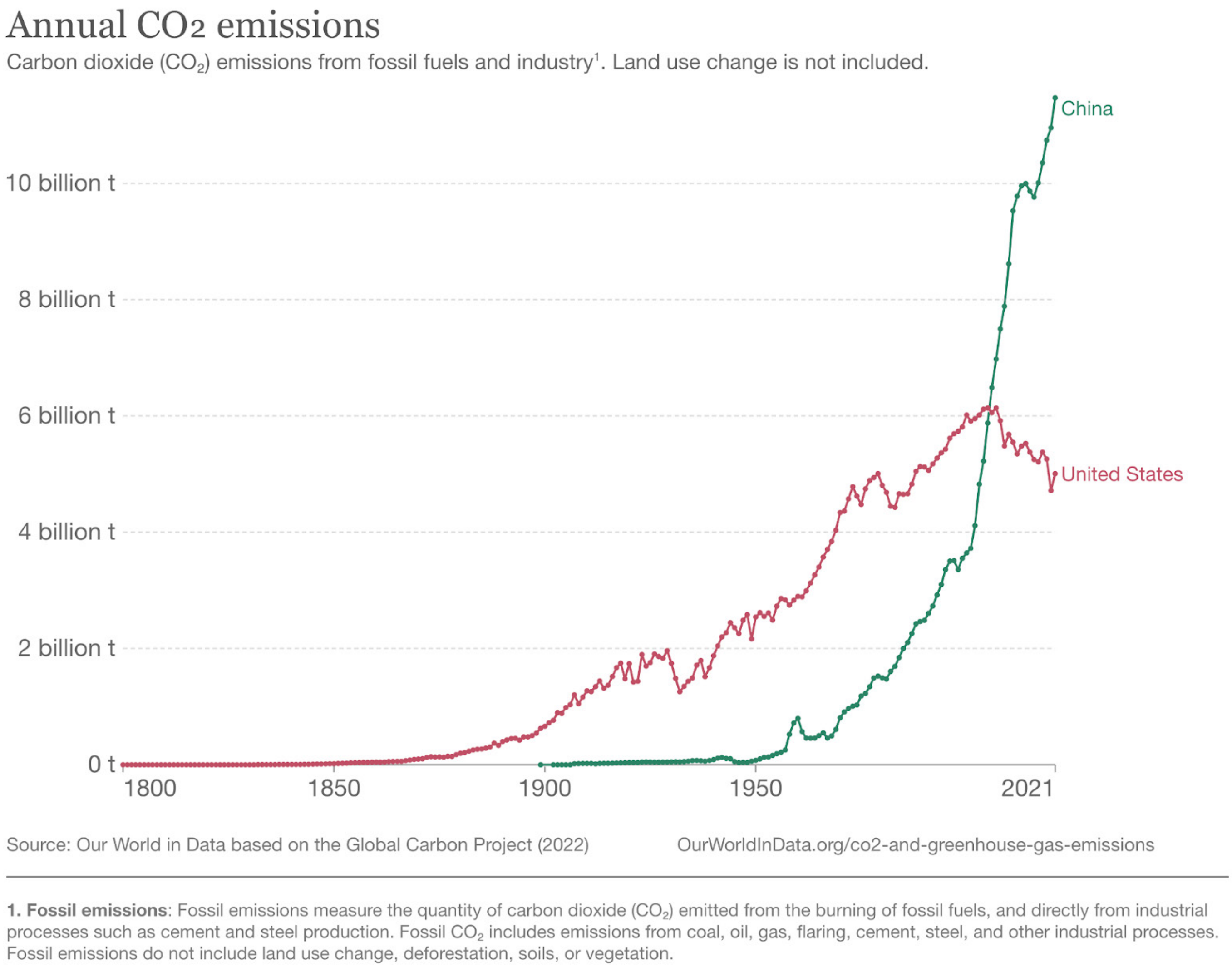


Chart of global GDP by region in 2021.

Source: http://english.scio.gov.cn/m/indepth/2022-12/05/content_78551552.htm



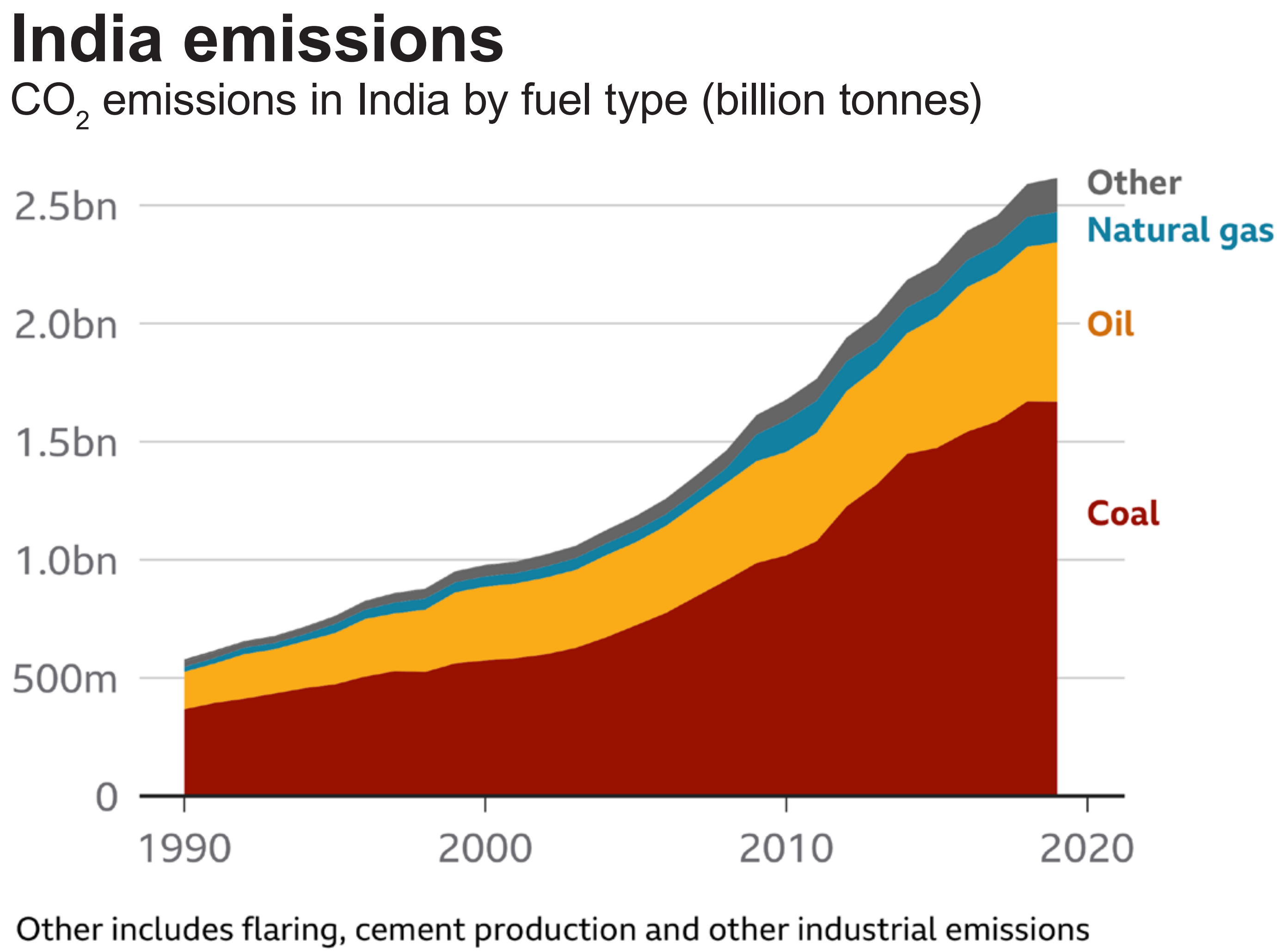
Annual carbon dioxide emissions in China and the United States since 1800.

Source: PolitiFact

as food, cotton, or oil in colonies and transport them to the West for consumption or manufacturing.[95] The consumption of fossil fuels and emissions of greenhouse gases were concentrated in the Western world, which housed the urban centers of industrialization.

Although the process of Western industrialization was a global process that relied on resources and human labor from around the world, traditional narratives of the modernization of the world focused on developments within the West in isolation. Europe and the West came to see itself as modern or "developed," while other parts of the world that did not industrialize were considered to be "developing." The use of fossil fuels and emissions of greenhouse gases as a byproduct, therefore, have become intrinsically linked to perceptions of what a modern nation or society looks like. The issue, however, is not merely one of symbolic value—a value judgement of who is "modern" and who is not; housing industrial centers within the global economic system produces wealth locally. Using gross domestic product (GDP) as a marker of wealth, one can see that global wealth is still concentrated in Western countries, which benefited from industrialization for centuries.[96]

The history of global European colonialism from around 1500 and industrialization from around 1800 has had lasting impacts on the situation in the twenty-first century as humans endeavor to muster a response to global warming. In Western Europe, wealthy countries have invested in technology to reduce greenhouse gas emissions and have cut emissions.[97] The United States has also cut its greenhouse gas emissions from higher levels in the early 2000s.[98]



Carbon dioxide emissions in India.

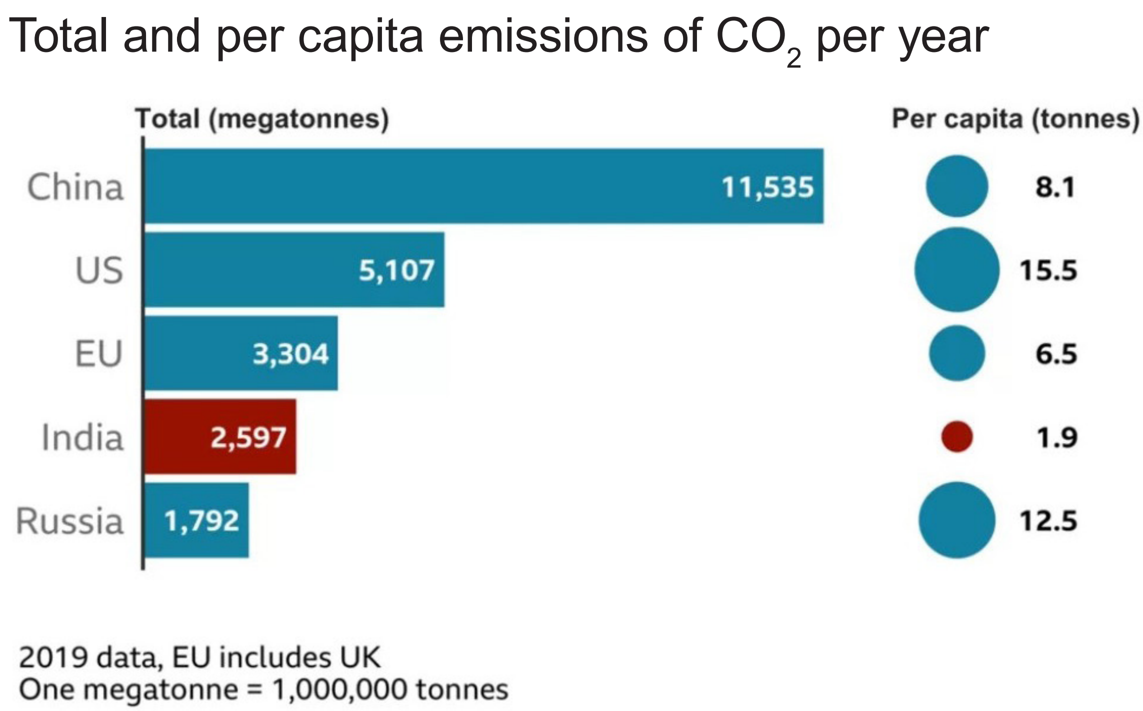
Source: BBC

However, even with the economic resources that European and North American countries possess, spending money to significantly address climate change is politically difficult. Globally, the picture becomes even more dynamic and complex.

Global Greenhouse Gas Emissions

While the United States and the European Union have cut greenhouse gas emissions, China and India have increased theirs. Taking carbon dioxide emissions as a marker, China has become the world's leading emitter, and India has moved up to the fourth place behind the U.S. and the EU, respectively. On the surface, this situation may appear to be a simple case in which China and India need to take the same path as the U.S. and Europe in cutting emissions. But it is not that simple. The United States and Europe used fossil fuels as a means of achieving prosperity for their inhabitants and as a means of building societies that are still held up as models of modern life. Now, as China and India have the opportunity to achieve those same ambitions, the rules are suddenly being changed, and those nations are sometimes blamed as the new culprits responsible for climate change.[99] Such a verdict seems unfair to some. Yet, at the same time, climate change is a serious matter, and the United States and Europe cannot sufficiently cut global emissions alone.

The global situation regarding fossil fuel emissions is complex, and even the narrative that the United States and Western Europe are seemingly headed in the right direction while China and India are going the other way is too simplistic. When one considers population, the argument that India and China are now chiefly responsible for climate change falters. On a per capita basis, China's carbon dioxide emissions are nearly half those of the United States,



Comparison of the five leading carbon dioxide emitters.

Source: BBC

while India's emissions are less than one-seventh of the United States' per person.[100]

The centuries-long legacy of the Industrial Revolution and deeply entrenched models of what the features of a modern nation should be are influencing people's reactions to climate change today. The Anthropocene has put humans in an unprecedented situation, but on a cultural level, global societies remain deeply connected to the geopolitical legacies of the past few hundred years. A sizable portion of China's emissions are linked to its exports, which are still largely consumed by people in the United States and Europe.[101] Moreover, China has only produced about a quarter of the cumulative amount of CO_2 emissions of Western nations since 1750.[102]

THE CONSEQUENCES OF THE ANTHROPOCENE

The name "Anthropocene" deliberately evokes the role of humanity in creating a new epoch in climate history. Human-caused changes to the chemical configuration of the Earth System have led to the global warming that characterizes the Anthropocene. The relationship between humans and nature in the Anthropocene, however, is not unidirectional. A robust understanding of the Anthropocene involves an awareness that while humans have impacted nature, the relationship between humanity and nature is reciprocal. Nature also impacts the human experience.

It is difficult to precisely identify any direct consequences of the Anthropocene or current climate change, but evidence suggests that both the natural world and human societies are being influenced by climate change. The Earth System is inherently complex with many different subsystems interacting with each other, creating feedbacks that are difficult to anticipate or even recognize. Drawing a direct connection between climate conditions and individual events such as a flood or a fire can be tenuous. As those types of events compound over a relatively short amount of time, however, scientists can begin to observe potential causes in the climate that contribute to their occurrence.

The impacts of climate change on humans and the societies we have organized ourselves into can be equally fraught. We are billions of people, but each one of us acts individually with immense complexity. Individual weather-related events can drastically change, or even end, a person's life. Yet, gradual changes in temperature over decades are difficult for anyone to detect within the comparatively short time span of a human life. Cultural factors such as media and social beliefs can powerfully color people's perception of the natural world, even during times of striking climatic change.[103] Even when weather does strike in the most powerful ways, how people respond is heavily influenced by things such as communal ties and deeply held beliefs.[104] Nevertheless, the evidence that the Anthropocene and changes in the natural world are impacting human life has become nearly impossible to ignore. As scholars and the public continue to observe natural anomalies, scientists will continue to refine our understanding of the connection between those events and climate change.

Climate Change as Part of Compounding Ecological Crises

While climate change, or global warming, may be the characteristic of the Anthropocene that garners the most public attention, the impacts of climate change are best understood within a broader context of changes that are happening in the natural world. Scientists and other concerned individuals have grouped three troubling developments together in the shared framework of the **triple planetary crisis**.[105] The three components of the triple planetary crisis are: 1) climate change, 2) air pollution, and 3) biodiversity loss.

The causes of these three events and the locations where they are felt range in scale. For instance, the loss of certain animals or plants is often related to disruptions in a local habitat. Similarly, air pollution varies from location to location, and even within a single city. Scientists actively study how the three components are linked, but regardless of the causal factors that unite them, human behavior provides a common connection. It can be helpful to keep these three crises distinct in our minds and use the categories to create some order amid sometimes daunting "bad news" about the environment.

Stress on Human Habitats

If one pays attention to local and global news headlines, it seems apparent that human habitats around the world are currently experiencing stress from natural forces. Places where humans have lived for generations are facing conditions that test the limits of human resilience. In the summer of 2023, the heat in Iran pushed the boundaries of human capacities with the heat index reaching 158 degrees Fahrenheit.[106] While a packed news cycle of striking stories does not prove that such events are happening more frequently or that those events are caused by climate change, scientists are diligently collecting robust records from around the world that do provide systematic data about the frequency and severity of abnormal weather-related conditions.[107] While conclusions are tentative and more evidence will bring greater clarity, scientists agree that the world is already experiencing the consequences of climate change.[108] Three notable areas that are particularly impactful for human societies are fires, floods, and droughts.

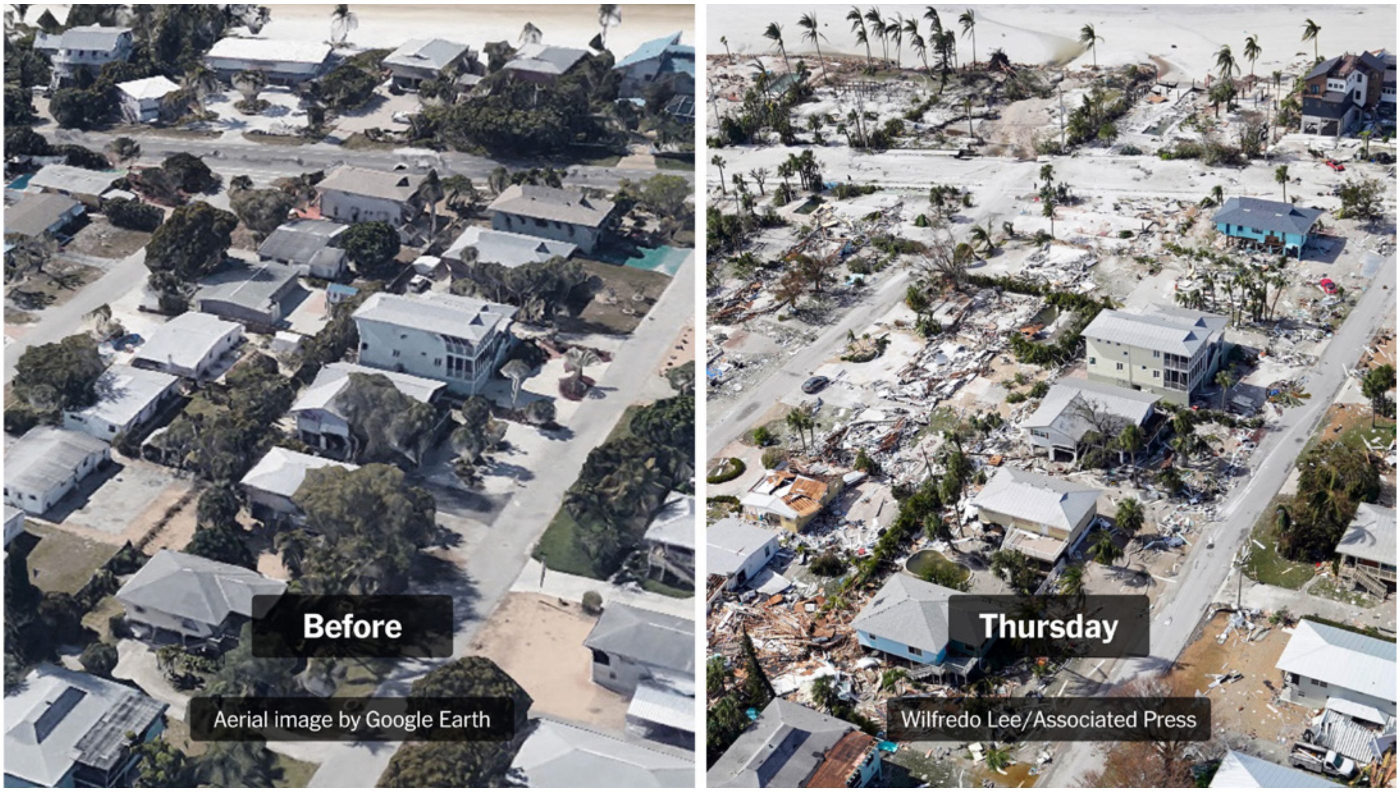
Fires

In just the past few years, abnormally severe fires have ravaged Greece, Australia, California, Canada, and Hawaii. The frequency and massive scale of fires today are staggering.[109] Global warming contributes to the conditions that have led to these and other fires.[110] Warmer temperatures amplify the dry and windy conditions that fuel and spread fires. Other factors also play a role in fires, including practices of forest management. Additionally, the human toll that the fires exact depends on many aspects of human interactions with nature. Building housing communities deeper into forests leaves those new developments vulnerable to local fires.

Floods

A warming climate exacerbates many of the variables, like rain and snowmelt, that contribute to flood risk. Rising sea levels—the result of melting glaciers and warmer ocean temperatures—threaten densely populated coastal communities in the U.S. and elsewhere in the world. According to the National Oceanic and Atmospheric Administration (NOAA), "high-tide flooding is now 300% to more than 900% more frequent than it was 50 years ago."[111] Sea level rise also accelerates coastal erosion, increasing the vulnerability of communities further inland to storm surge flooding.

Moving forward, people will face difficult decisions regarding land use. Will people invest in safeguarding their communities from new ecological dangers or retreat to new locations? Either option promises



Damage caused by Hurricane Ian in 2022.

Source: The New York Times

Satellite view of smoke from fires in Greece, 2007.

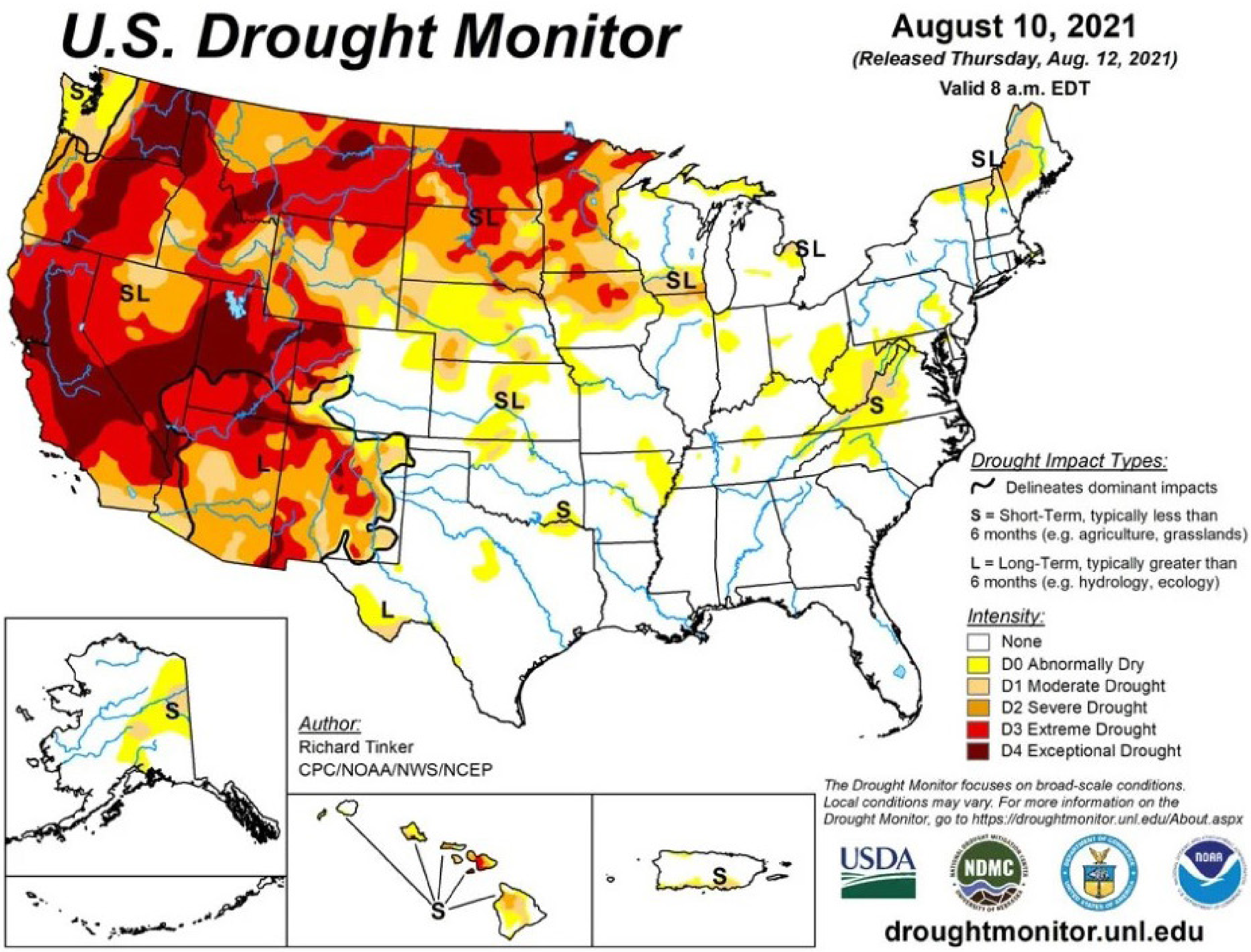
Source: [NASA]

disruptions and challenges. To take one example, in 2022, Hurricane Ian struck Florida, causing 156 fatalities and resulting in $112 billion in damages.[112]

Now, as in other coastal U.S. cities before, local inhabitants must decide whether to rebuild in areas that have been destroyed by flooding. Increased flooding amplified by climate change is expected to continue in India as well.[113] In both of these regions, land use will be a factor in the damage caused by future flooding.

Droughts

While warmer temperatures over large bodies of water can result in more water in the atmosphere and increased precipitation, warmer temperatures over dry land can wick away what little water there is. In North America, climate change exacerbates existing susceptibility to drought in the dry Western United States. By 2022, a megadrought in the region was the worst drought in 1,200 years.[114] Studies have also found that the Horn of Africa is a hundred times more likely to experience drought and the Western Mediterranean is more likely to experience heat as a result of climate change.[115] In both of these regions, as in other vulnerable places around the world, drought can drastically impact the viability of a region for human life. Water is an absolute necessity and essential to food production.



In the summer of 2021, 40 percent of the United States experienced drought.

Source: [Yale Climate Connections]

Stress on Human Life

When the places that people live are threatened or destroyed, humans pay a price in multiple ways. Even the experience of living in an environment that is less hospitable to humans can take a toll on the people in impacted regions. Climate change is already adversely influencing people's health and financial wellbeing around the globe. In some cases, people are moving out of their homes and home communities because of factors including climate impacts.

Health

Many of the negative effects of climate change on people's health are obvious. The conditions mentioned above—fires, floods, and droughts—pose a tremendous threat to people who live in the vicinity of those phenomena. Increased heat and colder winter temperatures in areas that experience disrupted weather conditions, such as disruptions to the polar jet stream, also pose a risk to many people, especially in populated areas that do not have the infrastructure, such as adequate heating or cooling, to mitigate abnormally severe temperatures. Some of the health threats posed by climate change may be less obvious. For instance, a 2017 report from the United Nations identified five important insights related to health risks associated with climate change: 1) Certain groups have higher susceptibility to climate-sensitive health impacts; 2) Many infectious diseases, including water-borne ones, are highly sensitive to climate conditions; 3) Climate change lengthens the transmission season and expands the geographical range of many diseases; 4) Climate change will bring new and emerging health issues, including heatwaves and other extreme events;

and 5) Malnutrition and undernutrition influenced by floods and drought.[116]

Financial Impacts

The financial costs of climate change are diverse and widespread. In regions that depend on specific ecological conditions for farming or fishing, climate change can threaten the livelihood of entire communities. The destruction wrought by climate-related disasters obviously impacts those hit by such disasters. Additionally, the threat of future damage is disrupting existing systems for mitigating and overcoming disasters when they do occur. In the United States, climatic conditions have been one facet of the decision to stop insuring property in regions susceptible to fire or flood.[117] When people are unable to secure insurance for their property, they are more vulnerable to financial ruin as a result of a natural disaster. Even if such a disaster does not materialize, disruptions to insurance markets can cause damage on their own to the financial stability of a region.

Migration

While migration is a multifaceted phenomenon that involves many different factors, climate change is already driving many people from their homes. A United States government report estimates that almost 30 million people move their homes due to extreme weather events every year.[118] The number of people who choose to move or are forced to move due to climate conditions is bound to rise as long as current conditions persist or worsen.

Impending Risks

It is easy to study the realities of climate change today, to see the impacts it is already having on the Earth System and to become alarmed about the possibilities of where the world could go from here. Scientists are constantly revising models of what future climate change will look like depending on different levels of greenhouse gas emissions. While many people are sounding the alarm and calling for decisive action now, others favor maintaining the status quo. Life in the Anthropocene has sparked a variety of human responses to climate change, from dread to apathy. The impending risks of humanity's current course are a defining feature of the conversation about what we should do in response to our changing climate.

SECTION III SUMMARY

- The "Anthropocene" is a term that describes a new era in climate history that recognizes the central role of humankind in geology and ecology today.
- The Anthropocene has been linked to the beginning of the Industrial Revolution around 1800.
- Most scholars today agree that the Anthropocene began around 1950 with the "Great Acceleration" and the start of a rapid growth in the human population and the rapid rise in greenhouse gas emissions.
- The burning of fossil fuels is the leading cause of the Anthropocene and the global warming that characterizes it.
- Through European colonialism and industrialization, Western countries in Europe and North America accumulated great wealth. Their success also made the robust use of fossil fuels a marker of the modern nation.
- Due to their historical backgrounds and current resources, countries in different parts of the world today face unique challenges in the effort to reduce greenhouse gas emissions.
- Climate change is already impacting the natural world and the human population.
- Climate-related migration is already underway and will likely continue to be a major factor in the human response to climate change moving forward.

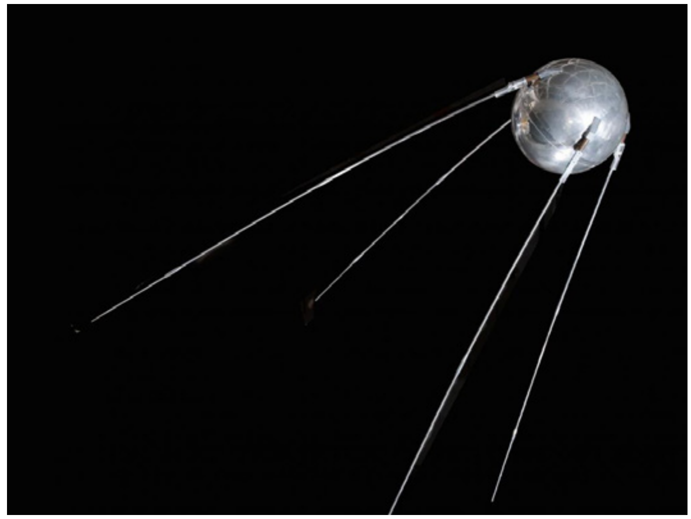
Section IV

Responding to the Climate Crisis

SECTION IV INTRODUCTION

Although the idea is not universally accepted, climate scientists and much of the public around the United States and the globe agree that the world is currently experiencing a climate crisis. Climate experts are appropriately cautious about linking individual weather-related events to climate because many different factors can influence a single flood, heatwave, or other event. However, evidence continues to accumulate that historically rare and/or extreme weather phenomena are occurring more frequently. Additionally, scientists are observing changes in the Earth System—including ice sheet loss, warmer ocean and atmospheric temperatures, sea level rise, and local ecological disruptions—that pose a threat to both human and nonhuman life and to the habitats of an increasing number of species of animals and plants. Taken together, these future climate-related threats and compounding weather-related disasters around the globe can reasonably be seen as a single climate crisis. This final section of the resource guide explores when and how people became aware of the climate crisis and some of the most influential responses to the climate crisis through the present day.

One could craft a historical narrative about the origins of the current climate crisis that dates back to the beginning of the Industrial Revolution or even farther back to the development of large-scale human societies. Yet, no one living six thousand years ago could have anticipated the social and technological changes that have led to the present climate crisis. Not even insightful nineteenth-century scientists such as Eunice Newton Foote, who recognized that carbon dioxide in the atmosphere could raise temperatures on Earth, could anticipate the exact form the climate crisis would take. Instead, this section focuses on those people who since the late 1950s recognized that a climate crisis was unfolding. Humanity's response to having an unprecedented awareness of our changing climate is the final, dramatic, chapter of this resource guide. It is a drama that continues to unfold and that has an uncertain ending.

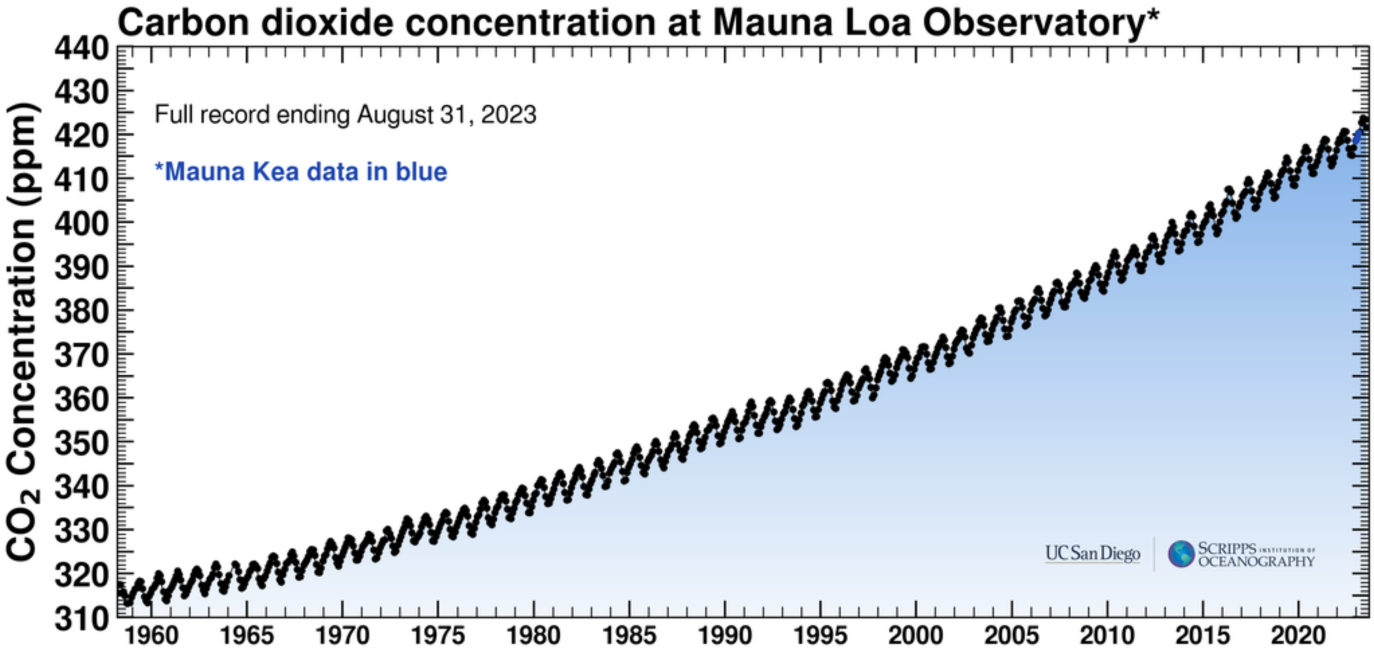


Sputnik was the first human-made satellite to orbit Earth in 1957.

Source: National Air and Space Museum

RECOGNIZING THE CLIMATE CRISIS

In the late 1950s, growing numbers of scientists began to systematically measure carbon dioxide levels in the Earth's atmosphere and realized that its rise could cause global warming. Throughout the 1960s, 1970s, and 1980s, scientists steadily expanded their knowledge that human behavior, particularly the burning of fossil fuels, was changing the Earth's climate in ways that are dangerous for humans. A fascinating process allowed humans to achieve an unprecedented understanding of the natural world. Scientific knowledge does not develop in a vacuum; rather, cultural factors specific to the global conditions of the second half of the twentieth century influenced how humans learned about climate change and how humans responded to that knowledge.



The Keeling curve, showing CO_2 concentrations in the atmosphere since systematic records began.

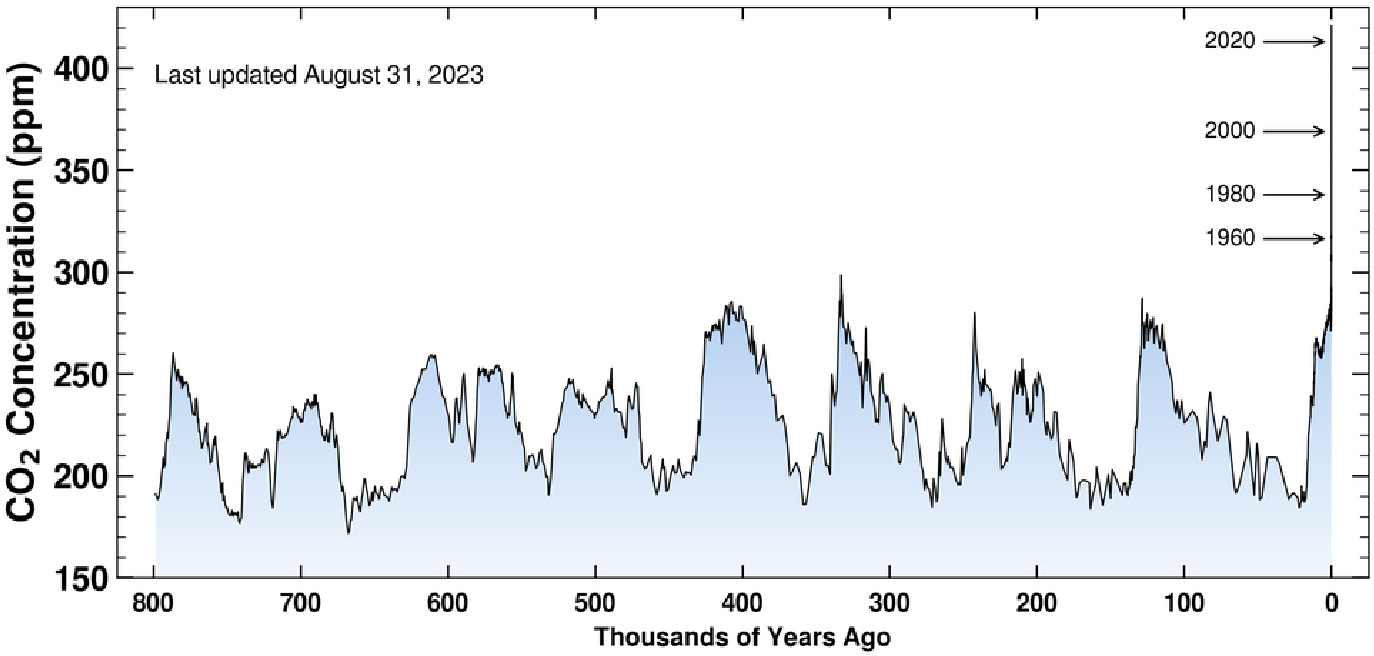
Source: Scripps Institute of Oceanography

Scientists work in communities.[119] As in the study of any community, tracing the causes of human behavior is complicated and fascinating. The story of how communities of scientists recognized that fossil fuel emissions might cause global warming and then organized their activities to determine that those emissions truly were warming the globe is an interesting story. To understand the story, one must realize that around 1950 scientists did not really know the appropriate questions to ask about climate change, nor did they have the tools and teams in place to answer those questions. The path to understanding contemporary climate change was one of steady discovery and constant building on recent gains in knowledge and understanding. It was as if scientists were building the ladder as they were climbing it.

Before diving into more specific details of the progress of climate science from the mid-twentieth century on, we will first consider more broadly how scientific communities organize themselves. On a broad level, scientists specialize their training and research according to fields such as biology, astronomy, or physics. These fields, however, are themselves created by scientists as the result of efforts to organize and systematize knowledge of nature. Additionally, organizations such as universities or research institutes specialize in various fields and bring together groups of scientists from a common field and provide those scientists with the resources to study their respective research interests. The fields of study and the organizations that house and support the members of those fields are not static. Rather, they continually evolve as new research interests raise new questions, which, in turn, necessitate new approaches to answering those questions.

The story of how scientists, from the late 1950s to around 1990, came to understand that the world might face a climate crisis involved lots of dynamic interaction within and between different scientific communities, and it was also influenced by major world events of the twentieth century. In the immediate aftermath of World War II (1939–45), the international community established the United Nations (UN) in 1946. The Charter of the UN is considered an international treaty, and preventing the use of force in international relations is a chief aim of the organization. The UN can also act on an array of issues and has been a facilitator of international cooperation for studying and responding to climate change.[120]

In the 1950s, **the Cold War** (1946–91)[121] became entrenched as a defining feature of geopolitics. The United States and the Soviet Union (USSR) emerged



The Keeling Curve with a reconstruction of CO_2 concentrations over the past 800,000 years.

Scripps Institute of Oceanography

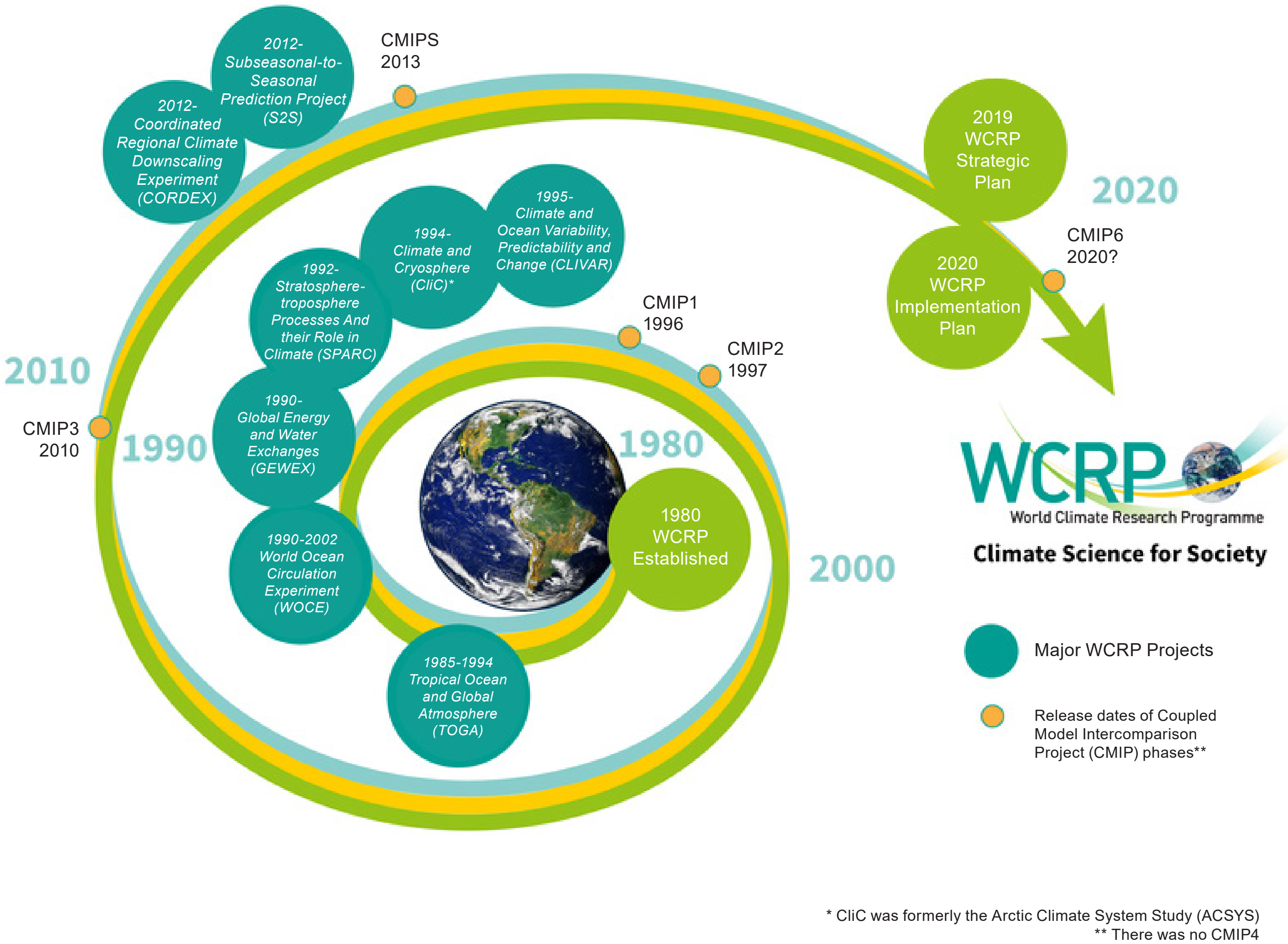
NASA, founded in 1958, became an important hub for scientific research.

from World War II as competing global powers. One of the fronts on which the two sides competed was science. The **Space Race** was a driving force behind the scientific battle. In 1957, the Soviet Union launched the first artificial Earth satellite, known as Sputnik, into orbit. The United States countered with its own space program that successfully landed humans on the moon in 1969.

Space exploration propelled the reorganization of scientific communities in the Soviet Union and the United States, as well as many other nations. New institutions, including the **National Aeronautics and Space Administration (NASA)**, founded in 1958, became important hubs for scientific research. Also, scientists across disciplinary fields collaborated in new ways, giving rise to new fields, including Earth System Science (ESS). A defining characteristic of ESS is that it brings the subjects of different fields—geology, biology, and meteorology, for instance—into a single view of Earth and Earth's relationship to other celestial bodies, including the Sun. It was in these scientific, geopolitical, and cultural contexts that scientists seriously began to study the effect of greenhouse gas emissions on the Earth's climate.

Research Programs

Besides the local institutions, such as a university or NASA, where a scientist might work on a daily basis, scientists working on similar subjects voluntarily organized themselves into more nebulous groups comprised of scientists from all over the world. Typically, these groups meet from time to time at conferences, where scientists, academics, and members of other interested groups, such as government officials, discuss recent advances in research and plan future



Graphic depicting the major projects of the World Climate Research Programme.

research goals.

Occasionally, when scientists and other interested parties identify a topic that they think will be important to study, they formalize a plan for carrying out that study and create a research program. Participants in a research program agree to contribute their time and resources from various institutions to accomplish a collaborative research agenda. Two very consequential research programs in the study of contemporary climate change are the **World Climate Research Programme (WCRP)**, established in 1979, and the **International Geosphere-Biosphere Programme (IGBP)**, established in 1987. The story of how these two research programs came into existence and what they discovered about the global climate offers a good overview of how humans came to recognize the climate crisis.[122]

The World Climate Research Programme (WCRP)

The 1950s was a foundational decade for human understanding of the Earth's climate and anthropogenic climate change. The International Council for Science (ICSU, after its former name: International Council of Scientific Unions), founded in 1931, was one garden from which knowledge of the climate blossomed. In 1952, the American physicist and engineer Lloyd Berkner (1905–67) suggested that ICSU embark on a systematic and comprehensive study of global geophysical activities over a set period.[123]

The ICSU took up Berkner's suggestion in what became the International GeoPhysical Year (IGY). The "year" ran from July 1957 to December 1958, a period of time that was selected to correspond with a high point in the eleven-year cycle of sunspot activity.

The IGY combined scientists from a dizzying array of fields and nations to study a vast number of subjects, including a variety of forms of solar energy, the upper atmosphere, meteorology, oceanography, gravity, and geomagnetism, among many others. The IGY was foundational for the launch of NASA and the United States space program. It also provided the support for what would become a pivotal observation in the development of our understanding of climate change.

As part of the research conducted for the IGY, **Charles David Keeling** (1928–2005) initiated a systematic measurement of atmospheric carbon dioxide (CO_2). He conducted his research from Hawaii at a base on Mauna Loa.[124] At this time, scholars were beginning to understand that rising levels of carbon dioxide in the atmosphere could cause climate change in the form of global warming. In fact, in 1959 Swedish meteorologist **Bert Bolin** (1925–2007) traveled to Washington, D.C., to alert the National Academy of Sciences that increasing CO_2 levels throughout the rest of the century could have serious consequences.[125] In 1961, Keeling released the data of his CO_2 measurements, documenting that carbon dioxide levels were indeed rising. A graph showing the ever-increasing levels of CO_2 in the atmosphere since Keeling's initial records has become known as the "**Keeling Curve**" and is an iconic image that distils the causes of the current climate crisis.[126]

The emerging data about CO_2 concentrations in the atmosphere combined with an awareness of carbon dioxide's potential for warming global temperatures sparked concern among scientists and, naturally, generated a research interest in the subject. Bert Bolin was a leader in consolidating researchers who were interested in this topic. As a result, new communities of scientists and formal organizations dedicated to the study of climate began to emerge. In 1964, Bolin became the first chair of the ICSU's new committee on atmospheric sciences.[127] Three years later, in 1967, another collaboration and consolidation occurred. The ICSU and the World Meteorological Association (WMO), an agency of the United Nations (UN) created in 1950, jointly founded the Global Atmospheric Research Program (GARP). GARP was at the center of advances in knowledge of the weather and climate in the 1970s and 1980s.

In 1978, at an International Workshop on Climate Issues hosted in Austria by the ICSU, the WMO, and the United Nations Environment Programme (UNEP), scientists recognized a need for an organization that could do even more than GARP. In 1979, the World Climate Research Programme (WCRP) succeeded GARP as the new center for the study of climate change.[128] Throughout the 1980s, the WCRP led research that greatly increased knowledge of how the oceans and atmosphere interact to create weather.[129] WCRP also significantly raised awareness of climate change outside the scientific community.

The International Geosphere-Biosphere Programme (IGBP)

In the 1980s, as the WCRP led the scholarly community in studying meteorology and climate change, scientists once again evaluated whether or not their current organizations and methods of study could adequately answer emerging research questions. Bert Bolin and other scientists argued for the necessity of a new scientific program that studied global change more broadly in order to contextualize climate change and its causes. This push led to the creation of the International Geosphere-Biosphere Programme (IGBP) in 1987.[130] As its name suggests, the IGBP studied climate change within an Earth System Science framework and added breadth and detail to our scientific knowledge about the complexities of the interactions between the Earth's subsystems and how they are impacted by global changes. The IGBP did not supplant the WCRP; rather, the two programs complemented each other and coordinated their scientific investigations of climate from the broader perspective of the Earth System.

Raising Awareness of Climate Change

While it may be a cliché to say that the ivory towers of academia are inaccessible to the general public, the results of scientific research can often seem esoteric. The disconnect—between the knowledge of experts and the knowledge of the average person who has not devoted their lives to studying a certain scientific topic—means that even if a select group of scientists conclusively discovers something about the world, the general public will likely be unaware of what they have discovered. In order for the public or general population to learn about what a community of scientists has learned, that knowledge needs to be transferred to wider audiences in a way that they can comprehend it.

In the 1980s, scientists became more convinced that the climate was warming, that the warming was caused by human practices, and that it could have significant impacts on the planet. Logically, they concluded that this information was important for the public and government leaders to know. As a result, scientists set about raising awareness of climate change.

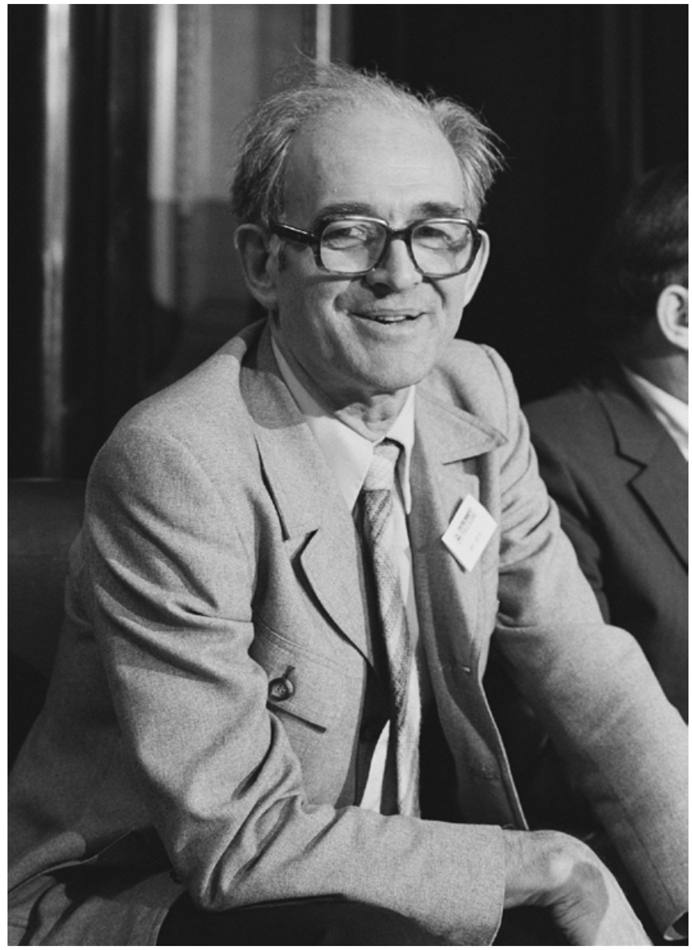
Early Public Warnings

In 1985, the ICSU, WMO, and UNEP organized a major conference in Villach, Austria, around the topic "Assessment of the Role of Carbon Dioxide and of Other Greenhouse Gases in Climate Variations and Associated Impacts."[131] The participants in the conference agreed that greenhouse gas emissions could raise global temperatures and that the consequences could be serious. An ICSU committee, the Scientific Committee on Problems of the Environment (SCOPE), issued a report following the conference titled "The greenhouse effect, climatic change and ecosystems." The report warned of the possibility of "substantial warming" as a result of CO_2 emissions that "were attributable to human activities."[132] It also recommended policy actions that could stem climate change caused by human actions. In order to create an avenue for sustained communication of scientific knowledge to policy makers and the public, scientists formed the Advisory Group on Greenhouse Gases (AGGG).

In 1988, the WMO contributed to another major conference, this time hosted in Toronto, Canada, entitled, "The Changing Atmosphere: Implications for Global Security."[133] The conference marked another very public moment when scientists made the claim that climate change should be taken seriously by policymakers across the globe. In the same year, NASA scientist James E. Hansen testified before the United States Senate that the human-induced greenhouse effect, not natural fluctuations, was the cause of climate change.[134] By the late 1980s, knowledge of the risks of climate change was not only shared among a diverse body of scientists from numerous different fields, but word about climate change was also getting out to the public and to government officials.

The Intergovernmental Panel on Climate Change (IPCC)

One of the most consequential developments from the late 1980s regarding the spread of climate change science was the creation of the Intergovernmental Panel on Climate Change (IPCC) in 1988. A group of scientists, including Paul J. Crutzen and Bert Bolin—who served as the panel's first chair from 1988 to 1998—were instrumental in the establishment of the organization. As envisioned by Bolin, the IPCC was established as a formal institution dedicated to coordinating between scientists who recognized the risk of climate change and international policymakers who would be vital to a global, intergovernmental response that could mitigate the dangers posed by climate change. Bolin also anticipated the need to provide a bulwark for scientific knowledge against those who would attack it. Accordingly, the IPCC was also created as a separate entity from the WCRP and the IGBP and was designed to serve as a body that independently assesses the science produced out of those programs.[135] To this day, the IPCC remains the most authoritative source of information about the science of climate change.



Bert Bolin, the first chair of the Intergovernmental Panel on Climate Change (IPCC).

By around 1990, scientists had already known for decades that rising levels of carbon dioxide and other greenhouse gases in the atmosphere were heating the Earth, and they had begun to raise awareness among the public. But in the 1990s, some powerful people reacted to the growing knowledge of global warming in a way that has significantly complicated humanity's response to the climate crisis ever since.

OPPOSITION TO CLIMATE ACTION

The scientific evidence for anthropogenic climate change is overwhelming. However, many individuals and even many politicians, particularly in the United States, maintain that climate change is not real. How did the trajectory of increasingly detailed scientific knowledge about climate change from the 1950s through the 1980s result in widespread public skepticism of climate science by the turn of the twenty-first century? In 2010, historians of science Naomi Oreskes and Erik M. Conway published a bestselling book titled *Merchants of Doubt: How a Handful of Scientists Obscured the Truth on Issues from Tobacco Smoke to Global Warming* that helps explain why so many Americans do not believe in climate science.[136] In the book, Oreskes and Conway carefully retrace, and amply document, an episode in the early 1990s in which a small group of strategically placed individuals were able to halt action combating climate change. The book helps illustrate the mechanisms through which people have effectively contested credible scientific knowledge and presented alternative information to the public instead.

Oreskes and Conway show that in addition to scientific knowledge being produced within the types of organizations and working groups outlined in this section of the guide, the scientific community as a whole exists within a larger cultural setting. Some of the most influential people that make up this larger cultural setting include those advocating for the interests of businesses or industries, government or political actors, and members of the news media. All three sectors played a role in creating a disconnect between the reality of climate science and what many Americans choose to believe about climate change.

Nierenberg and the Marshall Institute

In 1984, William Nierenberg retired as the Director of the Scripps Institute of Oceanography—a very well-respected scientific institution at the forefront of the study of climate. Upon retirement, Nierenberg joined the Board of Directors of a recently founded think tank in Washington, D.C., called the George C. Marshall Institute. The Marshall Institute was a very different type of institution from the Scripps Institute of Oceanography. Although scientists worked at both places, the Marshall Institute had a political agenda to achieve: namely, to defend certain Cold War policies—such as the Strategic Defense Initiative (SDI)—of then United States President Ronald Reagan against critiques from the scientific community. When the Berlin Wall came down in 1989, signaling the end of the Cold War, the Marshall Institute did not stop its practice of using scientists to attack other scientists; it turned its attention to a new target: climate science.[137]

In 1989, President Reagan's former Vice President and successor, President George H. W. Bush, was considering taking action to mitigate climate change. Nierenberg and his colleagues at the Marshall Institute used their ties to the president and the Bush administration to present an unpublished paper about climate change that did not have the support of the broader scientific community. The paper argued that variability in solar energy, not greenhouse gases, was causing warming. The argument was not scientifically sound, and the IPCC published a report the next year saying as much.

In retrospect, it may seem that in a dispute about a scientific finding between a retired scientist who now works for a politically motivated think-tank (Nierenberg and the Marshall Institute) and the most authoritative body of experts on that topic from around the world (the IPCC), the body of scientists should win the argument. And the IPCC and its climate scientists did win the dispute—at least among scientists themselves. But within the Bush administration and among the broader United States public, the argument was a draw, at best.

Prominent members of the Bush administration enthusiastically adopted the scientifically dubious argument of the Marshall Institute paper, and the Marshall Institute was able to leverage its support from the U.S. government and manipulate news coverage of the dispute in newspapers, including the *Wall Street Journal*. As a result of such efforts, in the 1990s the inaccurate idea that a significant number of scientists disagreed about the fundamental facts of climate change became firmly entrenched in the public's perception.[138] With the public doubting that scientists agreed on the climate, politicians in the United States retreated from taking action to stop climate change. In the 1990s, influential narratives that were not produced out of the scientific community came to dominate United States politicians' rationale for not acting on climate.

U.S. Opposition to the Kyoto Protocol

A revealing marker of the attitude among those politicians at that time was the U.S. government's response to the Kyoto Protocol, an international treaty adopted in 1997 in which nations committed to reducing greenhouse gas emissions. The Marshall Institute had close relations with the presidential administrations of two Republican Presidents, Ronald Reagan and George H. W. Bush, but opposition to formally agreeing to the Kyoto Protocol was bipartisan—that is, both Republicans and Democrats opposed it. On July 25, 1997, by a vote of 95–0, the United States Senate agreed to the following resolution stating:

> *[The U.S. Senate] declares that the United States should not be a signatory to any protocol to, or other agreement regarding, the United Nations Framework Convention on Climate Change of 1992, at negotiations in Kyoto in December 1997 or thereafter which would: (1) mandate new commitments to limit or reduce greenhouse gas emissions for the Annex 1 Parties, unless the protocol or other agreement also mandates new specific scheduled commitments to limit or reduce greenhouse gas emissions for Developing Country Parties within the same compliance period; or (2) result in serious harm to the U.S. economy.*[139]

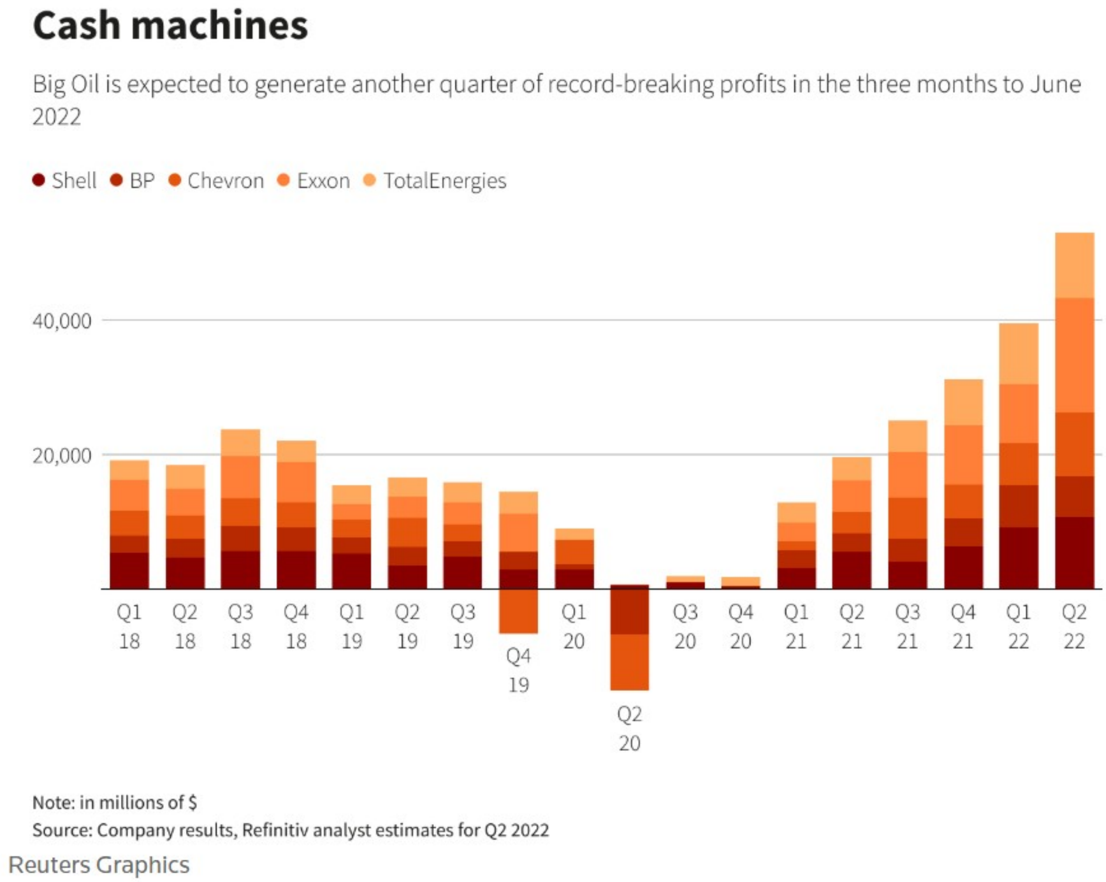
The language of the resolution contains two ideas that influenced United States public policy regarding climate change. The first is that other countries—including those with less wealth than the United States—need to commit to action before the United States will commit. And the second is that taking action to slow climate change will hurt the U.S. economy. Then President Bill Clinton directed the United States to sign the treaty in 1998 as a gesture of support, but with the understanding that it would not be ratified by the U.S. Senate, and therefore would not be binding for the nation.[140]

In the year 2000, President Clinton's vice president, Al Gore, ran for president against George W. Bush, former President George H. W. Bush's son. Vice President Gore ran on a campaign that pledged to be more proactive about combating climate change. A *New York Times* article from November 3, 2000, published four days before the election, counted Gore's support for the Kyoto Protocol and Bush's opposition to it as the leading issue that distinguished the candidates on the environment.[141] In a very close race, Gore received more popular votes than Bush, but lost the election after a prolonged legal battle regarding vote counts in the state of Florida that was settled in the United States Supreme Court. The new Bush administration reaffirmed the United States' rejection of the Kyoto Protocol and pursued a policy of expanding global oil production in cooperation with the oil industry.[142] In many ways, the developments of the 1990s still reverberate today in terms of how humans are responding to the climate crisis.

Chinese students promote the Kyoto Protocol in Beijing on Feb. 16, 2005, the day the agreement went into effect.
Source: China Newsphoto / via REUTERS

Business and Industry

While proponents of climate action argue that mitigating greenhouse gas emissions can produce new jobs and industries, others are concerned that climate action is bad for the economy. Whether or not the broader economy would suffer, the oil industry has viewed a move away from fossil fuels as a threat for decades. In the 1970s and 1980s, Exxon Mobil participated in the scientific study of climate change, but when governmental regulations of fossil fuels appeared on the horizon, the company began funding organizations that opposed business regulations and promoted skepticism of climate science.[143]



The yearly combined profit of largest Western oil companies from 2007 to 2015.

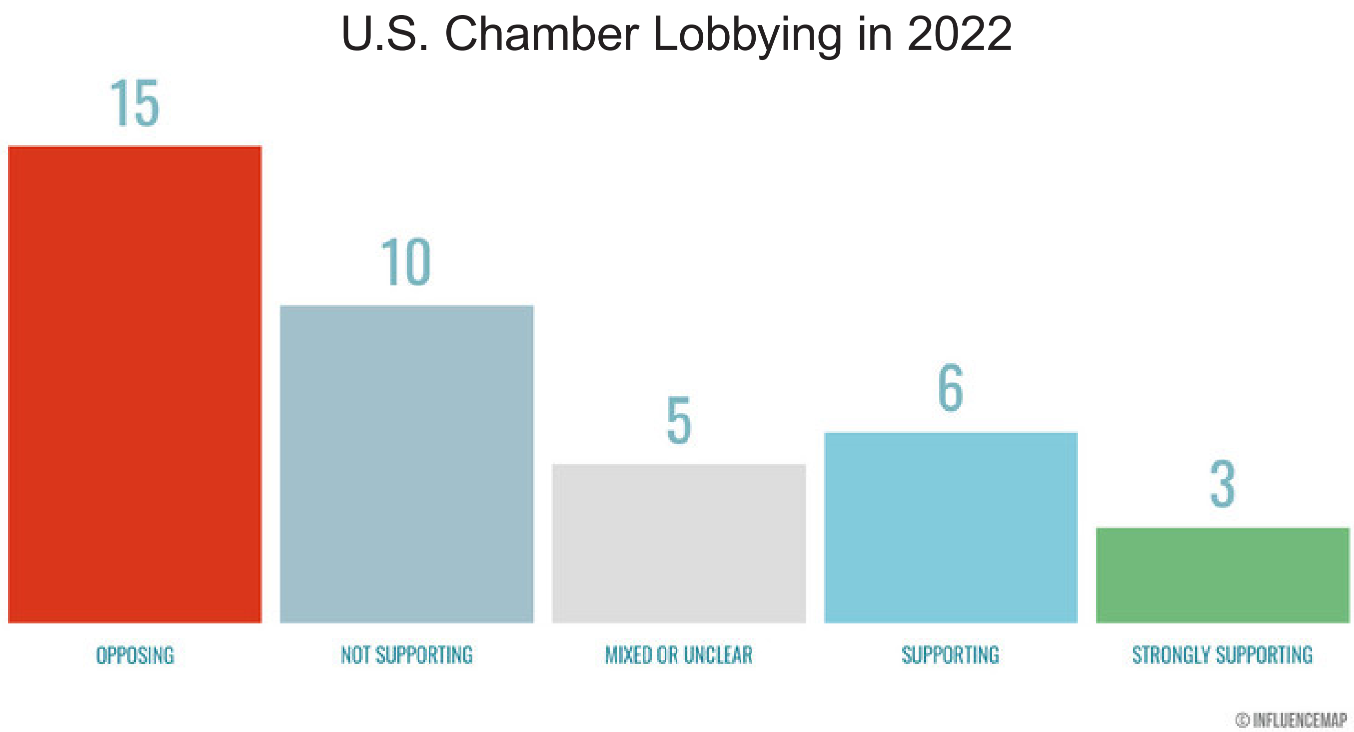
Reuters

The reality of the situation is that just as scientific knowledge of climate change advanced from the 1950s to the 1990s, it has continued to advance from the 1990s to the 2020s. Climate skeptics and those who spread disinformation about climate change do not statically argue that scientists still have gaps in their knowledge. Promotors of climate skepticism actively keep up with new scientific discoveries and the consequences of climate change and cast doubt on the latest developments.

The oil industry has deep pockets for funding such efforts. The year 2022 was the most profitable year ever for the biggest Western oil companies. A combination of factors, including the Russian-Ukrainian war, added to the record year.Remarkably, six com panies earned a combined $219 billion, from which they paid out $110 billion in dividends and shares to in vestors.[144] Such enormous amounts of money give the oil industry, and those who profit from it, the resources and motivation to promote climate skepticism.

The tactics of the oil companies can be very sophisticated and are not limited to convincing people that they should not be concerned about climate change. One effective approach by British Petroleum and a hired marketing company produced the well-known concept of the "carbon footprint."[145] The idea of individual people reducing their own fossil fuel emissions, or "**carbon footprint**," sounds appealing to those who care about the climate crisis. The problem, however, is that sufficient global reductions in greenhouse gas emissions will not be accomplished by concerned individuals cutting back here or there. The effort requires cooperation at the governmental level and substantial investment in new alternatives to global societies that currently run on fossil fuels. Those are the changes that the oil companies oppose for the purpose of self-preservation. By leading concerned people to focus on themselves and reducing their own "carbon footprint," the oil companies took attention away from collective action that would pressure governments and companies to make substantial changes that could make a meaningful global impact.[146]

The oil industry is not the only sector of the business community that has raised obstacles to action against



The U.S. Chamber of Commerce's stance on policies to combat climate change in 2022.

Source: www.influencemap.org

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climate change. The United States Chamber of Commerce is not part of the United States government but is a powerful lobbying group that broadly advocates on behalf of the interests of business. The United States Chamber of Commerce finally acknowledged that humans contribute to climate change in 2019, the same year that it formed a climate change task force to explore the relationship between business and climate change.[147] Those steps did not, however, lead to a complete change in direction on the issue for the U.S. Chamber of Commerce. An independent analysis found that in 2022, out of thirty-nine climate policies that the Chamber took a stance on, it advocated against climate action twenty-five times.[148]

The UNFCCC was founded by the United Nations at the 1992 Earth Summit in Rio de Janeiro, Brazil.

Source: The United Nations

News Media

News organizations can help spread scientific knowledge about climate change, but they can also spread misinformation, either intentionally or unwittingly. For decades, mainstream news outlets, without intentionally promoting climate skepticism, unintentionally suggested that scientists disagreed on the causes of climate change. Although since at least the 1980s an overwhelming majority of scientists with relevant expertise have shared the opinion that human actions are contributing to climate change, as dissenting voices arose in the 1990s, news outlets reported both sides of the debate almost equally—even though that equal representation did not accurately reflect the scientific community.[156] The reason for this coverage was well intentioned. Reputable journalists aim for unbiased coverage. Nev ertheless, the repeated message that came across was that some scientists think one thing, and others think the opposite, and since both sides received roughly equal airtime, it gave the impression that an equal number of scientists were on both sides even though that was never the case.

Today, many news organizations have become aware that those journalistic practices were problematic and have attempted to change course, but there are still challenges. A recent study of news broadcasts in the United Kingdom found that in 2022 nearly a third of British broadcasters cast doubt on climate science.[157] One spokesman for a news organization defended the decision to air such views by continuing to suggest that it was simply good journalism to report both sides of controversial issues. Another reality of the news landscape today, however, is that many outlets openly endorse one political viewpoint or another. The cable news channel, Fox News, for instance, reports from a conservative or Republican-leaning perspective and reaches a wide audience. In the summer of 2023, as smoke from massive Canadian forest fires poured into the United States, Fox News host Laura Ingraham gave airtime to known supporters of the oil industry who denied scientific consensus on the dangers of smoke.[158] This is one of myriad examples of how business, politics, and news media are still interacting in many of the same ways they have since the 1990s to offer the public a view of the world that denies climate science.

MITIGATING THE CLIMATE CRISIS

While some people have denied the science of climate change, many others have recognized that the environment is changing and that humans need to adapt in order to protect ourselves and our planet. People have worked toward climate solutions from almost every part of human societies. Every contribution makes a meaningful difference, and everyone's participation is important. When it comes to slowing and stopping global warming caused by greenhouse gas concentrations in the atmosphere, however, there are certain actions that need to happen on a global scale that will likely require substantial involvement from governments around the world and powerful industries and businesses.

Political (In)Action

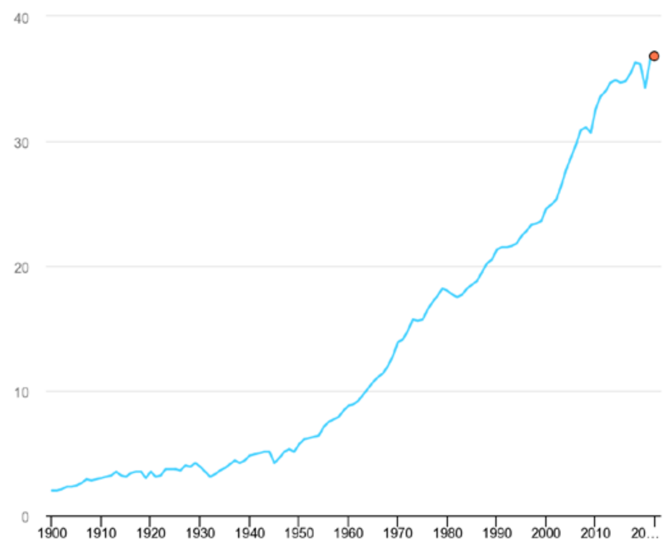
Depending on one's perspective, the work of governments on climate change can be credibly described as action or inaction. On the one hand, many nations have worked collaboratively and within their own countries to curb greenhouse gas emissions. This section will briefly describe the most significant multinational climate agreements enacted so far. On the other hand, if the standard for action is taking the necessary steps to limit the most dangerous possibilities for future global warming, then one must admit that global governments are not meeting that target.

The United Nations Framework Convention on Climate Change (UNFCCC)

In 1992, at the United Nations Conference on Environment and Development hosted in Rio de Janeiro, Brazil, the UN founded the **United Nations Framework Convention on Climate Change (UNFCCC)**, also known as UN Climate Change. The founding of the UNFCCC marked a major milestone in the coordination between scientific organizations and governmental organizations on the issue of climate change. While scientists had been organizing themselves for many years in order to create institutions to study climate change, the UNFCCC became an organizational center of government action to address climate change. The UNFCCC was a catalyst for the Kyoto Protocol.[159]

The Kyoto Protocol

The **Kyoto Protocol** is a binding agreement among thirty-seven nations that was adopted in 1997 and put into effect starting in 2005. The participating nations had individual targets for emission reductions that taken together as a whole came out to a 5 percent emission reduction over the years 2008 to 2012, compared to 1990 levels. The agreement included



Global CO_2 emissions rose to new levels in 2022.

Source: International Energy Agency

provisions for countries that did not meet the targeted emission reductions within their own borders to help reduce emissions internationally. Importantly, the Kyoto Protocol included procedures for monitoring greenhouse gas emissions, including registry systems to track and record emissions, reporting of annual emission inventories, and a compliance system to help parties meet their commitments.[160]

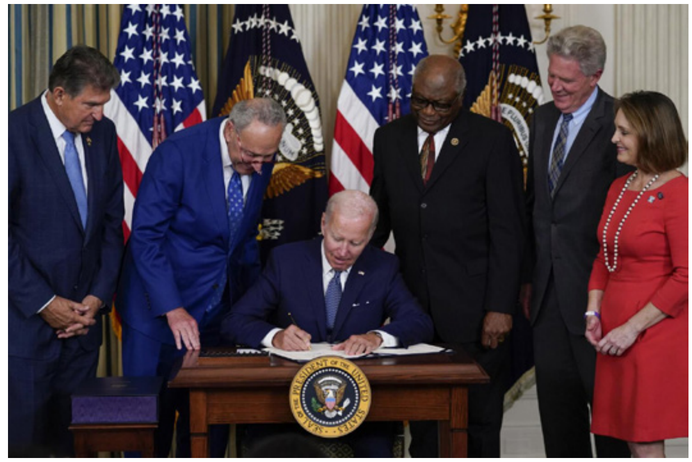
The Paris Agreement

In 2015, the Paris Agreement (also known as the Paris Accords or Paris Climate Accords) was adopted by 196 parties. The Paris Agreement, following scientific guidelines from the IPCC, sets the goal of limiting the increase in global average temperatures to well below 2°C above pre-industrial levels. More recently, global leaders have emphasized the importance of limiting the increase to **1.5°C** by the end of the century. In order to accomplish that outcome, the following emission goals are included: a peak in greenhouse gas emissions before 2025 and a 43 percent decline by 2030.[161]

As of 2022, however, global carbon dioxide emissions were still rising, indicating that the goal of hitting a peak in greenhouse gas emissions has yet to be achieved as the window before 2025 grows very small. As far as the longer-term goals are concerned, a United Nations report from late 2022 indicated that the current pace of emission reductions is not even close to putting the world on track to meet the 1.5°C target by the end of the century.[162]

U.S. Congressional Representative Alexandria Ocasio-Cortez and other lawmakers campaign for the Green New Deal Legislation on February 7, 2019.

Source: *New York Times*



U.S. President Joseph R. Biden signs the Inflation Reduction Act in 2022.

Source: AP News

The Green New Deal and the Inflation Reduction Act (IRA)

Within the United States during the Trump administration, Democratic lawmakers advanced discussions on how the U.S. government might take substantial action on climate change. A Congressional Representative from New York, Alexandria Ocasio-Cortez, and Senator Ed Markey of Massachusetts, championed legislation known as the **Green New Deal**. The Green New Deal could not become law without the support of the president at that time, but it did cast a vision for robust investment in clean energy and other forms of new infrastructure that burn fewer fossil fuels. The name "The Green New Deal" references the major public works investment in the 1930s known as the New Deal, which is credited with revitalizing the U.S. economy during the Great Depression. A key idea of the Green New Deal is that financial investment in mitigating climate change could actually boost new and existing industries and the economy as a whole.[163]

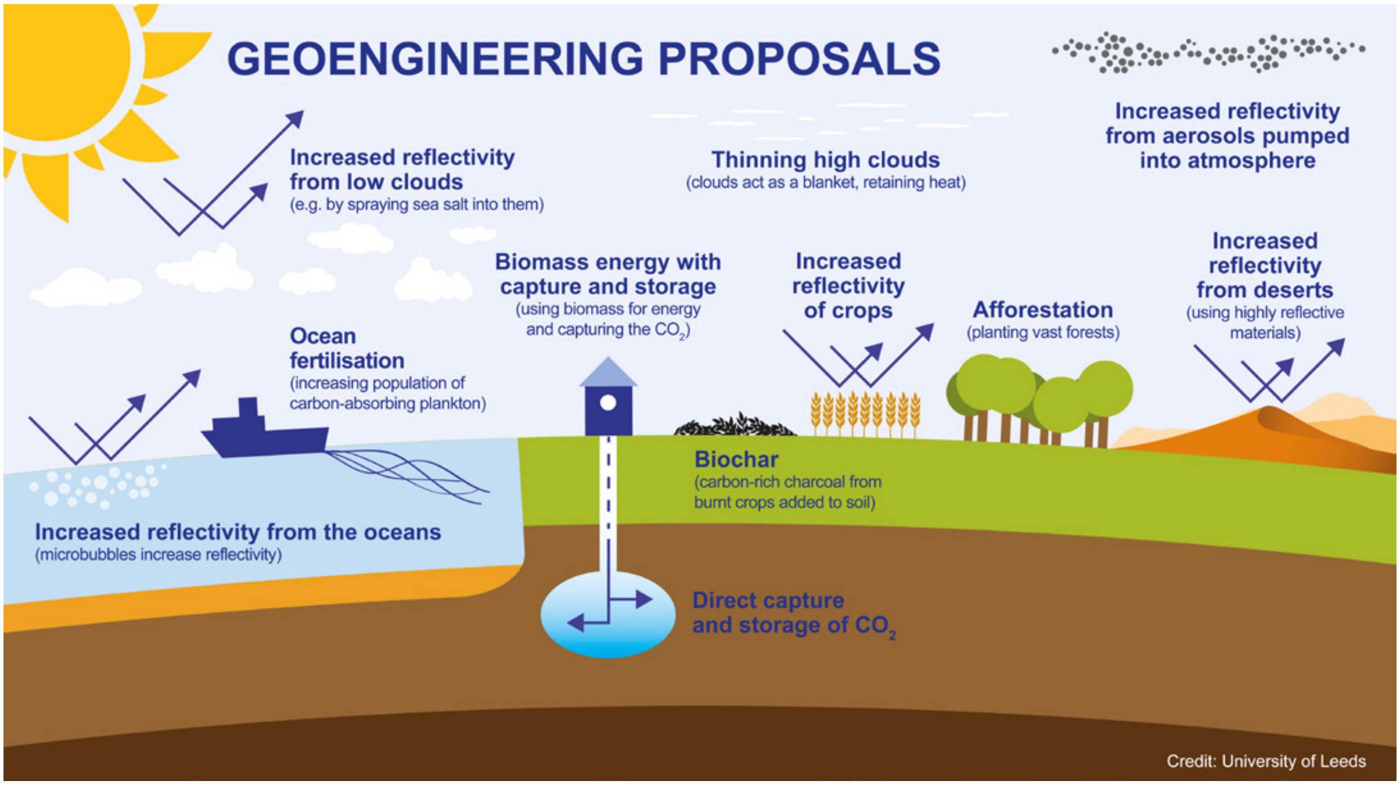
While the branding of the Green New Deal was not adopted by President Joe Biden, with Democratic majorities in Congress he signed the **Inflation Reduction Act (IRA)** into law on August 16, 2022. Although its name does not signal it, the IRA provided the largest source of government funding for climate-related issues in U.S. history. The IRA provides around $369 billion in climate funding that is being directed to energy, transportation, and agriculture, among other areas. This spending aims to help develop new technologies and support investments in green industries.[164]

New Technology and Industry

While experts warn that misplaced trust in technological solutions that have yet to be invented should not be an excuse for continued fossil fuel use with the hopes that the damage will be undone later, developments in new technology and industry are already slowing greenhouse gas emissions and are fittingly generating a lot of excitement.

Geoengineering

Scientists argue for the necessity of cutting carbon emissions, but other actions could also help diminish the causes of global warming. Geoengineering involves manipulating the Earth's environment in ways that counteract the current trends of climate change. One of the leading directions in geoengineering is **carbon sequestration** or carbon capture, which involves capturing carbon dioxide in the atmosphere and storing it.[165] Capturing the carbon and locating safe places to store it are two challenges involved in this effort. Another form of geoengineering includes solar radiation management. This approach would involve producing sources that act similarly to clouds and that could reflect solar energy away from the surface of the Earth or ocean surfaces. Scientists warn that people should use caution when endeavoring to manipulate the environment on a large scale, but continued investment in geoengineering could produce new ways to counteract climate change.[166]



Geoengineering involves manipulating the Earth's environment in ways that counteract the current trends of climate change.

Source: University of Leeds

New Sources of Energy

If humanity is going to reduce its greenhouse gas emissions and burning of fossil fuels, we will need to either stop those practices that use fossil fuels or find ways to continue them albeit without fossil fuels. Since the second of these options is more appealing to most individuals and governments, alternative sources of energy that reduce carbon emissions offer a promising path forward. Hydrogen, solar power, wind power, and biofuels are directions that people are actively exploring or already using as sources of power. Wind and solar generated a record amount of clean energy in 2022, producing 12 percent of global energy.[167] Here, let's take China's developments and achievements in wind power as an example. By 2024, China led with over 40% of global wind capacity, surpassing 520 gigawatts, powering millions of homes. It is the result of investments in technology and policy which have slashed costs and boosted efficiency. Additionally, offshore wind farms in Jiangsu and Guangdong showcase advanced engineering, harnessing coastal winds. China's efforts in developing wind power cut millions of tons of carbon emissions annually, aiding climate change mitigation.Additionally, the use of electric energy and developments in batteries for storing electric energy may be on the cusp of transforming humans' use of fossil fuels for powering the machines that we operate.

As scientists and business industries work together to create new sources of energy, it is important to stall efforts at change and protect the status quo. Such arguments might emphasize things like the coal energy plants that are used to power an electric vehicle, the environmental impacts of mining the resources used to create batteries, or the risks wind turbines pose to wildlife. When one comes across an argument of this kind, it is important to carefully scrutinize the source of the argument and consider what motivations might be behind it. At the same time, it is true that addressing climate change is complex and will inevitably involve choosing between competing interests. Focusing exclusively on keeping global warming to a certain level is only part of the broader picture. The next section examines the people who remind us that it does not just matter if we address the climate crisis. How we address it matters, too.

CLIMATE ACTIVISM

There has been a third response to the ongoing climate crisis that deserves separate consideration. Some people have denied the crisis, or aspects of our scientific knowledge about it. Some have tried to take action to mitigate the crisis. Others have thought about what the climate crisis might mean for humanity and nature—what the climate crisis' moral significance is—and they have spoken out about these issues. Their perspective on climate change and willingness to raise their voices on behalf of themselves and others is the third response to the climate crisis that we will consider here.

The people who have done these things, and the communities that have formed around them, are commonly labeled climate activists. Their actions and their messages are sometimes so widely celebrated that it seems everyone in the world has taken notice. Other times, their actions have been criticized or ignored. Moreover, they can seemingly be as critical of the people who are doing something about climate change as those who are doing nothing, depending on what is being done. Their voices have helped people to understand that climate change interacts with human societies in a variety of ways, and they have emphasized the view that in addition to slowing the pace of climate change, our actions should be just and respectful of the dignity of humans everywhere.

Speaking Out for "Climate Justice"

Sometimes it may seem that the world can be divided into two groups: those who support climate action and those who oppose it. Climate activists can be seen as people who very enthusiastically want to do something about climate and try to get as many people as possible to cross over to their side as well. This picture might describe some climate activists very accurately, but for many climate activists it is an incomplete picture.

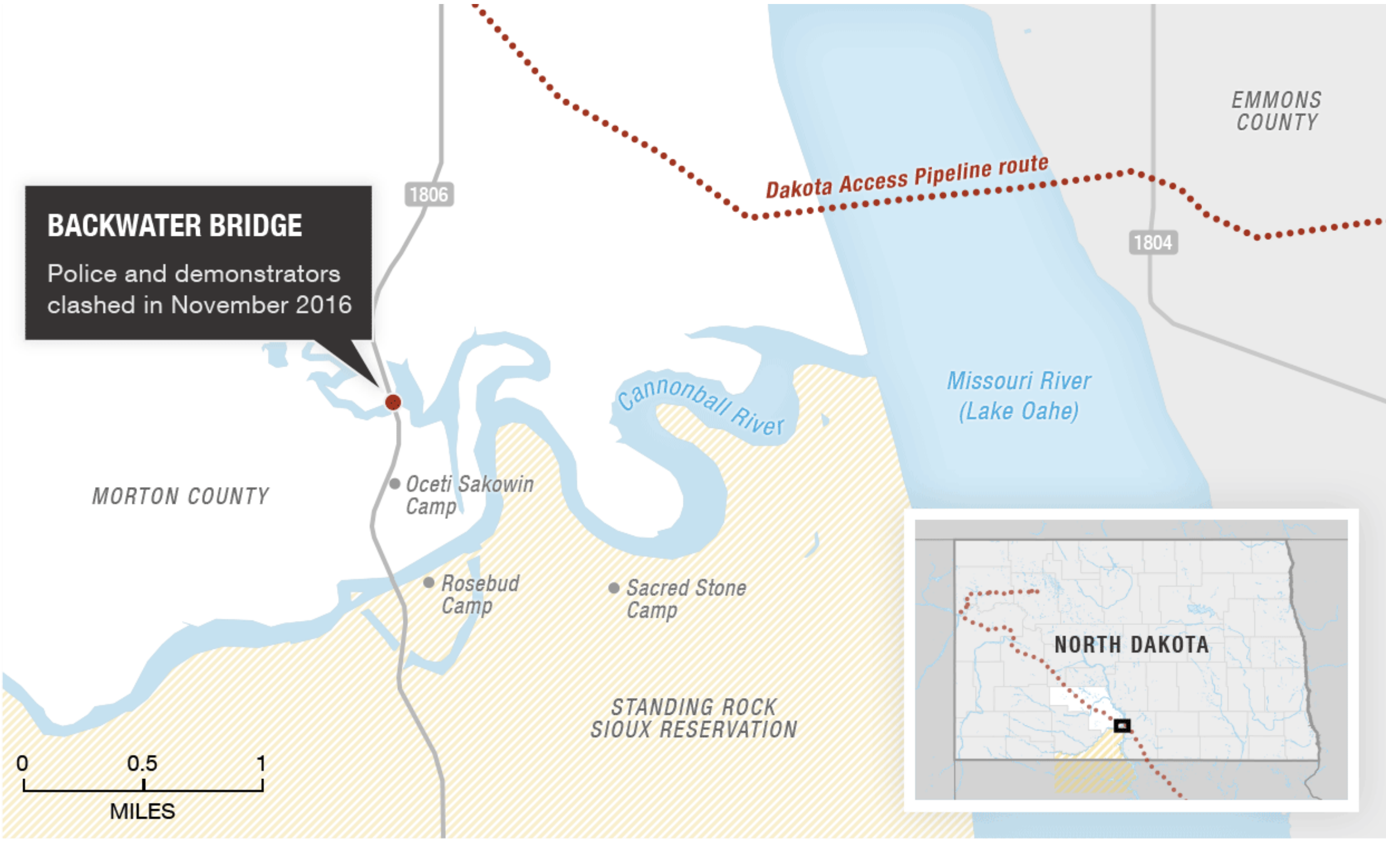
Often, climate activists are not merely trying to get people to act on climate change; they are engaged in the work of considering what actions we should take in the first place. And their concern is not merely to stop climate change, but more broadly to promote justice across the globe as humanity potentially faces a monumental disruption to our societies as a result of climate change. These individuals point out that not all actions that might slow climate change are just. To take one example, funding for climate action provided through the IRA created the opportunity to build carbon capture infrastructure in Louisiana, and the state is poised to move forward with this endeavor that could play a role in lowering greenhouse gas concentrations in the atmosphere.

However, local climate activists are raising the issue of where the captured carbon is going to be stored underground and if there are adequate safeguards in place to ensure that stored carbon is not going to pollute water and local ecosystems in under-resourced communities.[168] Advocates for climate justice acknowledge that climate change is impacting poor populations around the globe more severely than wealthy populations and maintain that this reality does not give a free pass for climate action that also comes at the expense of vulnerable communities. Raising these issues, as in the Louisiana example, complicates our view of the situation and complicates the question of what actions we should be taking to limit climate change.

Rather than viewing a concern for justice as a roadblock to quick and effective action against climate change, advocates for climate justice maintain a hopeful outlook. They argue that a shared concern for justice actually opens a path to global climate solutions that work, unlike the measures that have been taken by governments so far, which promise more than they deliver. Philosopher Olúfẹ́mi O. Táíwò argues that climate justice should entail an awareness of the historical legacies that have created today's global wealth and power disparities. He warned that the Green New Deal, if it were enacted, could problematically bolster what he calls "climate colonialism"—practices of extracting resources or producing clean energy in less-powerful countries and utilizing the benefits in wealthier nations.[169] In essence, the same types of practices from the past—remember the extraction of oil by Western nations from the Middle East in the twentieth century—could be reduplicated as part of the new global infrastructure for combating climate change, and inequalities could be exacerbated through the very process of taking climate action.

Arguably, if the primary concern is for justice and global equality, climate solutions could come much more easily. Some climate activists note that if wealthy Western countries were to view the strength they hold in wealth and power as something they achieved as a result of the natural resources of the globe, they might be more willing to use that money to give back to countries who are vulnerable to climate change, but do not have as many financial resources for transitioning to new forms of energy and infrastructure. Climate activists such as Táíwò invite citizens to imagine a world where different priorities guide the actions of the world's most powerful governments, many of which are democracies.

In the United States, the Biden administration has



Map of the Standing Rock protests and the proposed Dakota Access oil pipeline.

Source: NPR

attempted to incorporate a concern for justice into policy with the **Justice40 Initiative**. This initiative set the goal that 40 percent of overall benefits from Federal investment in certain categories flow to underserved communities. The categories of investment to which the initiative applies are climate change, clean energy and energy efficiency, clean transit, affordable and sustainable housing, training and workforce development, remediation and reduction of legacy pollution,[170] and the development of critical clean water and wastewater infrastructure.[171]

While climate activists try to inspire the public to think creatively about justice on massive, global scales, they can also provide a sense of community and belonging on a personal level and give concerned citizens opportunities to make a difference in their own lives. Climate scientist Katherine Hayhoe has contributed to IPCC reports, but also works on communicating the science of climate change in a way that resonates with her own faith community: Evangelical Christians.[172] In a TED Talk that has been viewed over 4 million times, Hayhoe offers a very simple way to fight climate change, just talk about it.[173] Rather than putting the pressure on any one person to solve the world's problems, climate activists who focus on climate justice are more likely to encourage anyone who is concerned to find a community—or even one other person—who feels the same way and work together to spread the word about climate change. Some of the most inspiring, powerful, and controversial climate activist move ments, which are discussed below, achieved recognition and success precisely because they led to cooperation be tween people.

Standing Rock and Indigenous Voices

In 2016 and 2017, protests led by the **Standing Rock Sioux Tribe** in North Dakota halted construction of the Dakota Access Pipeline, a portion of the Keystone Pipeline System that transports oil from Alberta, Canada, to the shores of the Gulf of Mexico in the U.S. state of Texas. Those supporting the pipeline contended that it would create new jobs and contribute significantly to the U.S. economy. Others opposed the pipeline, as they were concerned about its environmental impact and its impact on the local community. For several months, protesters camped near the construction site and waged a legal battle

Lauren Howland (Jicarilla Apache) protests the Dakota Access Pipeline and is met by law enforcement.

Source: PBS NewsHour

Alessandra Korap Munduruku.

Source: The Goldman Environmental Prize

to stop the proposed construction. The issues at stake were multifaceted and included the matter of sovereignty. At the local level, the route of the pipeline passed under the Missouri River at a location that was vital for protecting the drinking water of the Standing Rock Sioux Reservation. Intertwined with the local environmental and ecological impacts were the cultural and religious sites damaged by construction and threatened by future oil leaks.[174] Attention to the local threats to people's physical and spiritual health became a way for people to consider the human impact of global issues.

The Standing Rock protests seemed to meet at the intersection of many global justice concerns. The protesters targeted a pipeline that was transporting fossil fuels from underground in Canada to Texas, where it would be prepared for use that would release greenhouse gases out of the ground and into the atmosphere. In addition to the environmental harm of that process, the infrastructure for transporting the oil was damaging local ecosystems and sites of cultural significance across Canada and the U.S. The protesters raised the question of who should be allowed to have the final say over what humanity does to the Earth's land. Law enforcement and government representatives who favored shutting down the protests contended that the protesters were trespassing on, and in some cases vandalizing private property and argued that public roadways were not appropriate locations for acts of civil disobedience.[175]

On one level, the battle was a legal one, in which various advocates for and against the pipeline waged a long series of arguments in numerous courts. But on a more immediate level, the protests reminded some of many historical events from U.S. history when the government wanted to use land for purposes that Native Americans oppose and used violence and the threat of violence to achieve their aims. In this case, law enforcement made hundreds of arrests over the course of the protests and eventually used force to end the protests when newly elected President Donald Trump ordered that the construction continue in early 2017.

Similar struggles to that which took place at Standing Rock are happening around the globe. In Brazil, **Alessandra Korap Munduruku**, a member of the Munduruku Indigenous group of Sawré Muybu, took action when her home was threatened by one of the world's largest mining companies. The British company Anglo American planned to expand mining operations in Munduruku regions of the Amazon rainforest, where illegal mining was already polluting the water. During the Brazilian presidency of Jair Bolsonaro, when rates of deforestation in Brazilian rainforests were rising, Alessandra Korap Munduruku organized a coalition of Indigenous communities and publicly demanded that Anglo American withdraw from their land.

After initially denying that they had plans to mine in the area, Anglo American decided to withdraw its twenty-seven applications to mine inside Indigenous territories, formally notifying the Brazilian government of its decision in May 2021. Alessandra Korap Munduruku's victory has helped protect more than 400,000 acres of rainforest located in Sawré Muybu Indigenous Territory.[176] Indigenous leaders from around the world offer a vision for the relationship between nature and human societies in which local needs take priority over global profits.[177]

SECTION IV SUMMARY

- From the late 1950s through the late 1980s, scientists developed a robust understanding that greenhouse gas emissions were causing global warming and established the World Climate Research Programme (WCRP) and the International Geosphere-Biosphere Programme (IGBP) to study the phenomenon.
- In the 1980s, climate scientists realized that government officials needed to know about warming caused by greenhouse gases and take action on the matter. To better facilitate cooperation between scientists and governments, the Intergovernmental Panel on Climate Change (IPCC) was formed in 1988.
- The oil industry has felt threatened by government action to slow climate change, and it has funded misinformation about scientific views on the climate while continuing to make enormous profits.
- Big business, politicians, and news media have occasionally combined to advance messaging that casts doubt on scientific views about the climate.
- Today in the United States, many people's views about climate change and climate science align with their identification as a Democrat or a Republican.
- Governments around the globe have set targets for actions that would limit global warming to 1.5°C by the end of the twenty-first century, but they are not on pace to reach those goals.
- Investment in revolutionizing current industries and creating new ones could lead to novel methods of reducing greenhouse gas concentrations in the atmosphere or counteracting their impact on the climate.
- Climate activists have raised awareness that not all actions that combat climate change are just and that how we fight climate change matters.
- Indigenous communities around the world are advocating for environmental practices that value local populations over the profits of large companies; and those practices also may help address global climate problems.

Conclusion

Those living today have a unique opportunity to reflect upon the relationship between humanity and the Earth's climate. Never before have humans known about the Earth's climate history as extensively as we do now. We can see that our climate has always been changing, but we can also see that the speed at which it is currently changing is abnormal in human history. The last time humans lived through a shift in a climate epoch, from the Pleistocene to the Holocene around 11,700 years ago, the shift was followed by some stark changes in the global human population. The social world that humanity lives in today was built during and for the environmental conditions of Holocene.

It is difficult to predict how humans will respond as we now live in an era in which we as a species are significantly impacting the Earth System, an era often referred to as the Anthropocene. Already, people have reacted in different ways. Some deny it is happening. Others advocate for changing the behaviors that are altering our climate by reducing greenhouse gas emissions. Still others are thinking broadly about taking this moment as an opportunity to correct global inequalities that have developed over the past few centuries. A basic scientific knowledge of climate change and an awareness of the historical relationship between human societies and climate are important resources for considering our priorities for life in the Anthropocene.

Timeline

c. 2.6 million YA *(YA=Years Ago, with 2000 CE as Present)*	Start of the Pleistocene
c. 300,000 YA	Date of oldest known fossil record of *Homo sapiens*
c. 20,000 YA	Last Glacial Maximum (LGM)
c. 12,900–11,700 YA	Younger Dryas period
c. 11,700 YA	Start of the Holocene
c. 11,700–8,236 YA	Early Holocene Subseries/Subepoch or the Greenlandian Stage/Age
c. 8,236 YA	Final collapse of the glacial Laurentide Ice Sheet
c. 8,236–4,250 YA	Middle Holocene Subseries/Subepoch or the Northgrippian Stage/Age
c. 5,500 YA/3500 BCE	Emergence of the Mesopotamian civilization
c. 3200 BCE	Writing used in Egypt
c. 3200–2600 BCE	Early Harappan phase of the Indus Civilization
c. 2700–2200 BCE	Old Kingdom in Egypt
c. 2600–1900 BCE	Mature Harappan phase of the Indus Civilization
c. 2334–2218 BCE	Akkadian Empire in Mesopotamia
c. 4,250 YA/2250 BCE	Severe climate event, possibly resulting in Eurasian megadrought
c. 4,250 YA–present	Late Holocene Subseries/Subepoch or the Meghalayan Stage/Age
c. 1900–1000 BCE	Late Harappan phase of Indus Civilization
c. 1600–1050 BCE	Shang Dynasty in China
c. 1200–400 BCE	Olmec rule in Mesoamerica

c.1046–256 BCE	Zhou Dynasty in China
c. 900–200 BCE	Chavín Andean civilization
356–323 BCE	Lifespan of Alexander the Great
c. 150 BCE –250 CE	Peak of the Roman Empire
c. 476 CE	Collapse of the Western Roman Empire
536 CE	A cold year in Europe regarded by some historians as the worst year to be alive
c. 1280–1350	Wolf solar minimum
c. 1300–1850	The Little Ice Age (LIA)
c. 1400s	The Inca Empire consolidates control in the Andean region.
c. 1500s	European states establish colonies around the globe.
c. 1645–1715 CE	Maunder solar minimum
1714	Invention of the mercury thermometer
c. 1780s	Start of the Industrial Revolution
c. 1790–1820	Dalton solar minimum
1812–70	Lifespan of Charles Dickens
1931	Founding of the International Council for Science (ICSU)
c. 1946–91	The Cold War
1950	Start of the Anthropocene according to a proposal by the Anthropocene Working Group (AWG); start of the "Great Acceleration"; the World Meteorological Association is founded.
1957–58	International GeoPhysical Year (IGY)
1957	The Soviet Union launches the Sputnik satellite into orbit.
1958	The National Aeronautics and Space Administration (NASA) is founded.
1959	Bert Bolin alerts the National Academy of Sciences to rising CO_2 concentrations in the atmosphere.
c. 1960s	Earth System Science emerges as way to study the natural world.
1961	Charles David Keeling releases measurements of the CO_2 concentration in the atmosphere.

1962	Rachel Carson publishes *Silent Spring*.
1967	The Global Atmospheric Research Program (GARP) is founded.
1978	The International Council for Science and the United Nations Environment Programme (UNEP) organize an International Workshop on Climate Issues in Austria.
1979	The World Climate Programme, including the successor to GARP, the World Climate Research Programme (WCRP), is founded.
1985	A conference titled "Assessment of the Role of Carbon Dioxide and of Other Greenhouse Gases in Climate Variations and Associated Impacts" is held in Villach, Austria.
1986	The Advisory Group on Greenhouse Gases (AGGG) is founded by the International Council of Scientific Unions, the UNEP, and the World Meteorological Organization to follow up on the recommendations of the 1985 conference.
1987	The International Geosphere-Biosphere Programme (IGBP) is founded.
1988	A conference titled "The Changing Atmosphere: Implications for Global Security: Founding of Intergovernmental Panel on Climate Change (IPCC)" is held in Toronto, Canada.
1989	The fall of the Berlin Wall
1992	The United Nations Conference on Environment and Development is held in Rio de Janeiro, Brazil; the United Nations Framework Convention on Climate Change (UNFCCC) is established.
July 25, 1997	The U.S. Senate passes the Byrd-Hagel resolution 95–0, signaling the Senate's unwillingness to ratify the proposed Kyoto Protocol. The resolution is passed several months before negotiations on the Protocol conclude on December 11, 1997.
2000	Paul J. Crutzen and Eugene F. Stoermer coin the term "Anthropocene"; the U.S. presidential election is held between candidates George W. Bush and Al Gore.
2015	The Paris Agreement is adopted by 196 nations, including United States, at the UN Climate Change Conference (COP21) in Paris, France.
2016–17	The Standing Rock Sioux Tribe protests the construction of the Dakota Access Pipeline.
2020	The United States withdraws from the Paris Agreement during the Trump presidency.
2021	The United States rejoins the Paris Agreement during the Biden presidency; Alessandra Korap Munduruku leads a successful protest to block mining expansion in part of the Brazilian rainforests.
2022	Hurricane Ian; The Inflation Reduction Act (IRA) is signed into law by President Biden.

Glossary

1.5°C – a goal for limiting the amount of global warming above the pre-industrial average to 1.5°C by the end of the century

536 CE – an abnormally cold year in Europe as a result of volcanic eruptions; historian Michael McCormick recently named it the worst year to be alive.

4,250 Years Ago/2250 BCE – the starting date of the Late Holocene and the date of a severe climate event that may have included droughts that strained early human civilizations across Eurasia

Akkadian Empire – an early Mesopotamian empire that lasted from around 2334 to 2218 BCE

Alexander the Great (356–323 BCE) – Greek king from Macedonia whose vast conquests created cultural links throughout the Mediterranean region and east as far as the Indus Valley

Andes Mountains – a long mountain range that stretches from north to south along the Pacific Rim in South America that was home to early agrarian societies

Anthropocene – according to the IPCC, "a proposed new geological epoch resulting from significant human-driven changes to the structure and functioning of the Earth System, including the climate system"[182]; in March of 2024, the International Union of Geological Sciences rejected a proposal to formally name the Anthropocene as a new geological epoch, but noted that the term will "continue to be used not only by Earth and environmental scientists but also by social scientists, politicians and economists as well as by the public at large" and "will remain an invaluable descriptor of human impact on the Earth system."[183]

Archive – traditionally a storehouse of historical documents, but now also a figurative way of naming something as containing information about the past, such as the archives of nature

Atmosphere – one of the subsystems in the Earth System; the layers of gases encircling the Earth

Biosphere – one of the subsystems in the Earth System; all living organisms in and on the Earth

Black gold – a popular term that references oil, its immense value, and the fact that it, like gold, must be extracted from the earth

Bolin, Bert (1925–2007) – a Swedish meteorologist who led numerous scientific communities studying climate and was the first Chair of the Intergovernmental Panel on Climate Change (IPCC)

"Carbon footprint" – a term that references a person or organization's personal contribution to global carbon emissions

Carbon sequestration – also known as carbon capture; the act of taking carbon dioxide out of the atmosphere and storing it

Causal mechanism – something that causes something else to occur

Climate crisis – a term summarizing the dangerous impacts of climate change

Climate determinism – a method of telling historical narratives in which climate drives social and environmental changes over time

Climate history – an academic field that uses the methods of historians and studies sources produced by human societies to reconstruct past climatic conditions

Cold War – the global political conflict between the United States and the Soviet Union, and their allied nations, from around 1946 to 1991

Colonialism – a political system in which states or companies establish control over natural resources and people in distant lands

Crutzen, Paul J. (1933–2021) – a Dutch meteorologist who helped found the IPCC in 1988, won the Nobel Prize in Chemistry in 1995, and coined the term "Anthropocene" in 2000

Cryosphere – a part of the hydrosphere subsystem; all the ice in the Earth System

Early Holocene Subseries/Subepoch or the **Greenlandian Stage/Age** – the first of three stages of the Holocene that lasted from around 11,700 to 8,236 years ago and was characterized by warmer conditions

Earth System Science (ESS) – a relatively new approach to studying the natural world as a connected whole with a focus on the interactions between the Earth System's subsystems: the geosphere, hydrosphere, atmosphere, and biosphere

Foote, Eunice Newton (1819−1888) – an American scientist and women's rights advocate who recognized in the mid-nineteenth century that changing amounts of carbon dioxide in the atmosphere could impact the climate

Forcings – factors that are external to a climate system and influence climate change, such as volcanic activity, solar variations, and greenhouse gases

Fossil fuels – matter—including coal, oil, and natural gas—left behind from formerly living organisms that can be burned to create energy but when burned, releases carbon into the atmosphere

Geological time scale – a measure of time based on the record of rocks in which change is sometimes measured at the pace of millions or billions of years

Geosphere (Lithosphere) – one of the subsystems in the Earth System; the earth and rock that comprise the Earth

"Great Acceleration" – a term coined by environmental historian J. R. McNeill and Peter Engelke that identifies 1950 as a date when, partly due to a rapid growth in the human population, humanity's impact on the natural world rapidly increased, including a steep rise in fossil fuel emissions

Green New Deal – a policy proposal championed by U.S. Congressional Representative Alexandria Ocasio-Cortez, Senator Ed Markey, and others that would involve government investment in building infrastructure and incentivizing industries to help address the climate crisis

Greenhouse gas effect – a term that describes how higher concentrations of carbon dioxide and certain other gases in the atmosphere are warming temperatures in the lower levels of the atmosphere by trapping heat

Historical climatology – also known as paleoclimatology; this field reconstructs past climates utilizing methods for studying sources in nature, such as ice core sampling and dendrochronology.

History of climate and society (HCS) – a new interdisciplinary field, suggested by environmental historian Dagomar Degroot, focused on the history of the relationship between climate and human societies

Holocene – a geological interglacial period that began around 11,700 years ago, at the end of the last ice age

Hydrosphere – one of the subsystems in the Earth System; all the water in, on, and around the Earth in various forms

Indus Valley – a valley in present-day Pakistan that is home to the Indus River, was the location of some of the world's earliest agrarian societies, and marks the eastern extent of Alexander the Great's conquests

Industrial Revolution – a development starting in the late eighteenth century that expanded manufacturing by using machines

Inflation Reduction Act (IRA) – a law passed in the United States in 2022 during the presidency of Joe Biden that provides unprecedented government funding for addressing the challenges posed by climate change

Interglacial – a geological period of warmer conditions between ice ages

Intergovernmental Panel on Climate Change (IPCC) – a body of the United Nations (UN) that was formed in 1988 to offer an independent assessment of climate science and mediate between scientists and policymakers

International Geosphere-Biosphere Programme (IGBP) – established in 1987 to study global change in the context of Earth System Science

Justice40 Initiative – a policy commitment from the Biden administration that aims for 40 percent of overall benefits from Federal investments related to climate change and other categories to flow to underserved communities

Keeling, Charles David (1928–2005) – an American scientist who in 1958 began systematic record keeping of atmospheric carbon dioxide concentrations from the Mauna Loa Observatory in Hawaii

"Keeling Curve" – a graph that represents atmospheric carbon dioxide concentrations based on the records originally compiled by Charles David Keeling at the Mauna Loa Observatory in Hawaii

Kyoto Protocol – a non-binding international agreement, which was adopted in 1997 and entered into force in 2005, extending the 1992 United Nations Framework Convention on Climate Change (UNFCCC)

Late Holocene Subseries/Subepoch or the **Meghalayan Stage/Age** – the third of three stages of the Holocene that began around 4,250 years ago and continues to the present

Laurentide Ice Sheet – a vast ice sheet that covered much of North America during ice ages throughout the Pleistocene and which advanced most recently from c. 95,000 to c. 20,000 years ago

Lithosphere – see *geosphere*

Little Ice Age (LIA) – a period from around 1300 to 1850 during which cooler temperatures at various places and times resulted in cooler global temperatures on average

Mandate of heaven – an idea originating in ancient China that suggests an emperor has a mandate to rule, which is evidenced by order in the universe

McNeill, J. R. (b. 1954) – an American scholar and pioneering environmental historian who popularized the idea of the "Great Acceleration"

Mesoamerica – a region spanning North and Central America that was home to early agrarian societies, including the Olmec, and later societies, including the Maya and the Aztec

Mesopotamia – a region in present-day Iraq that was home to some of the earliest agrarian societies, including the Akkadian Empire

Middle Ages – a period in Europe from the collapse of the western part of the Roman Empire in the fifth century CE until around 1500 CE

Middle Holocene Subseries/Subepoch or the **Northgrippian Stage/Age** – the second of three stages of the Holocene that began around 8,236 years ago and ended 4,250 years ago and saw cooling conditions compared to the Early Holocene

Milankovitch cycles – patterns of the Earth's movement in relation to the Sun that influence the Sun's impact on climate over time

Modernization – a narrative of the process by which humanity or a specific group of people move from pre-modern conditions to modern conditions

Munduruku, Alessandra Korap (b. 1985) – a Brazilian environmental activist and member of the Munduruku people of the Tapajós River Middle Course who has defended portions of the Brazilian rainforest in various ways

National Aeronautics and Space Administration (NASA) – an agency of the United States government founded in 1958 that is a global leader in space exploration and the study of the Earth System

Negative feedback – in the context of Earth System Science, a phenomenon where climate change causes a natural reaction that acts against the initial direction of climate change

Nile Delta – a delta spreading over hundreds of miles in the region where the Nile River reaches the Mediterranean Sea that was home to early Egyptian agrarian societies

Paleoclimatology – see *historical climatology*

Paris Agreement – a 2015 intergovernmental agreement with the goal of limiting the increase in global temperatures to well below 2°C above pre-industrial levels

Pleistocene – a geological epoch that started around 2.58 million years ago and was characterized by cycles of ice ages and interglacials

Positive feedback – in the context of Earth System Science, a phenomenon where climate change causes a natural reaction that amplifies the initial direction of climate change

Precipitation – water released from clouds, including rain, snow, sleet, and hail

Proxy – something observable in nature that gives an indication of past climate conditions, such as tree rings, ice cores, or human-made records

Scale – the scope of an investigation; it can be large or small, either geographically or chronologically.

Scholarly field – a group of scholars who generally share an object of study, the type of evidence used to study the object, and methods used for analyzing that evidence

Solar minima – periods of decreased sunspot activity that follow an eleven-year cycle, but can also fall over longer stretches of time, such as occurred during certain phases of the Little Ice Age

Space Race – a competition of space exploration and technology between the United States and the Soviet Union during the Cold War

Standing Rock Sioux Tribe – a nation whose members led a protest that delayed the construction of an oil pipeline in North America

Stoermer, Eugene F. (1934–2012) – a biologist who, along with Paul J. Crutzen, coined the term "Anthropocene"

Subsystems – a classification within Earth System Science consisting of the geosphere, hydrosphere, atmosphere, and biosphere

Tipping point – a point of no return when changes in a climate system become irreversible

Triple planetary crisis – the ongoing and looming global risks related to climate change, air pollution, and biodiversity loss

United Nations (UN) – an intergovernmental organization founded in 1945 in the wake of World War II with a continued aim of promoting peace, dignity, equality, and a healthy planet

United Nations Framework Convention on Climate Change (UNFCCC, also known as UN Climate Change) – established in 1992 as a foundation for intergovernmental action on climate change that led to the Kyoto Protocol (1997) and the Paris Agreement (2015)

World Climate Research Programme (WCRP) – a program established in 1979 for the study of the Earth System and climate change

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